Basics of Environment Energy Ecology

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1 Environment and Ecology

Introduction

- The word 'environment' is derived from the French word 'environner,' which means to encircle.
- The word 'environment' is defined as the sum of all biotic and abiotic components that surround an individual and, in turn, affect the evolution and development of life on Earth's surface.
- On Earth's surface, biotic components, such as plants, animals, microorganisms, etc., and abiotic (devoid of life) components, such as soil, water, air, etc., exist and are in continuous interaction with each other. Thus, we can say that environment is not static.
- 'Nature' refers to the 'characteristics and features of plants, animals, and other creatures'.
- According to the Environment (protection) Act, 1986, Environment includes water, air, land, and the inter-relationship that exist among and between water, air, land, and human beings, plants, microorganisms, and other living creatures.

Weather

Short-term properties of the atmosphere (cloud cover, wind, etc.) at a given time and place are called weather.

Climate

The average weather forecast (atmospheric conditions, seasonal variations, weather extremes) of an area over a long period of time is called climate.

Environmental Factors

Biosphere

Biosphere is a part of the Earth, where life exists. Biosphere is formed by the interactions among hydrosphere (water), atmosphere (air), and lithosphere (land) on Earth's surface.

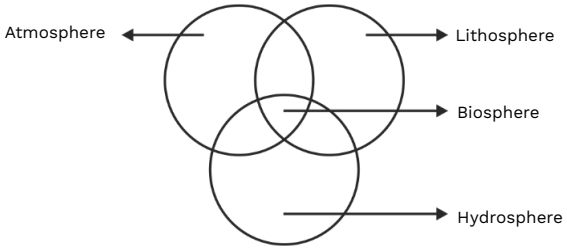


Fig. 1.1

Atmosphere

- It is the body of air that surrounds the Earth in the form of concentric shells of a sphere. The atmosphere is divided into concentric shells due to variations in temperature and pressure at various altitudes.
- It is a vast expense of air or a mixture of various gases, which surrounds the Earth's surface and is held due to the gravitational force or pull of the Earth.
- Most of the atmosphere is located close to the Earth's surface and is the densest at that part.

Structure of Atmosphere

Troposphere

- It is the first layer of the atmosphere extending up to 12 km in height, the one we live on.
- The heat that is generated from the Earth's surface warms this layer, and therefore, the temperature is reduced as it moves upwards to the tropopause, which represents the passage to the stratosphere.
- The tropopause is the maximum limit where water vapours can rise and convert to ice. It is also called a cold trap or thermal layer. In total 90 percent of the atmospheric gases exist in the troposphere.

Stratosphere

- It extends up to 50 km. Temperature increases as the height increases due to the presence of ozone layer, which absorbs a part of solar radiation.
- Stratopause is the transition layer or border between stratosphere and mesosphere.

- There are no phenomena determined by atmospheric turbulences, which happen due to water vapor and dust particles. The absence of this turbulence makes visibility high in the stratosphere, a perfect layer for flying jet aircraft.

Mesosphere

- In mesosphere the temperature reduces as the height increases because the heat that is radiated from the Earth's surface is far away. The mesosphere extends up to an altitude of 80 km.
- Meteors make it through the exosphere and thermosphere without much trouble because these layers do not have much air. There are enough gases in the mesosphere to cause friction and heat, therefore meteors burn up in the mesosphere.

Thermosphere

- Temperature increases as the height increases in the thermosphere, which extends up to 700 km. The density of gases diminishes as height increases.
- Artificial satellites and spacecraft orbit within the thermosphere and exosphere.
- The exosphere has no clear upper boundary since it gradually fades into outer space.

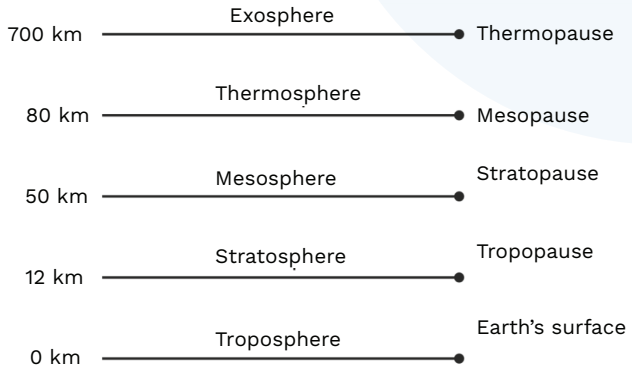


Fig. 1.2

Air Composition

- As we discussed earlier, 90% of the atmospheric gases exist in the Troposphere. Nitrogen and oxygen constitute 99% of the atmospheric gases. The remaining gases have only 1% share of Atmospheric Gases.

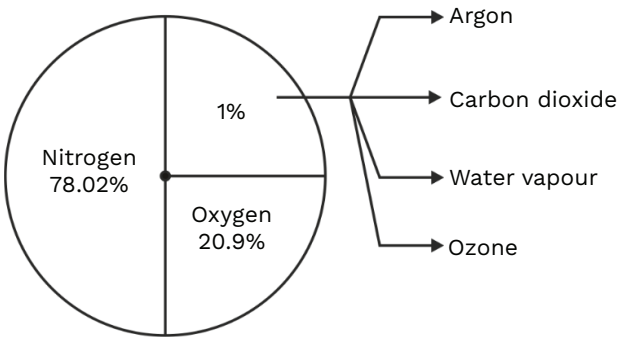


Fig. 1.3

- Carbon dioxide, water vapor, and ozone play an important role in maintaining the heat balance of the Earth.
- Air consists traces of neon, helium, krypton, and methane.

Lithosphere

- The word 'lithosphere' is derived from the word 'lithos', which means rock and sphere means zone. It is a solid layer of the Earth that consists of rocks. The crust is inorganic and is composed of various minerals.
- The lithosphere provides a platform and habitat to biotic components of the environment and is a source of nutrients. The lithosphere includes crust and asthenosphere, which is the upper portion of the mantle.

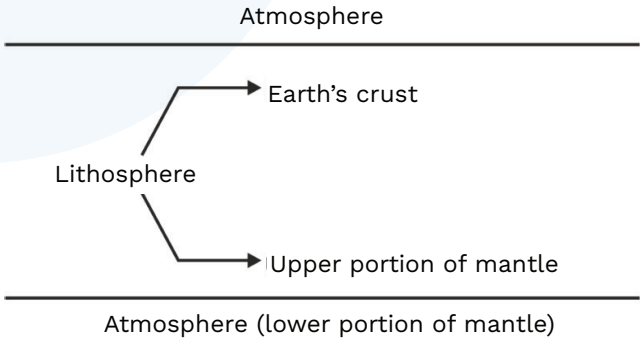


Fig. 1.4

- Tectonic activity is the most prominent characteristic of the Earth's lithosphere. The interaction of massive lithosphere slabs (tectonic plates) by collision, tearing apart, or slipping against one another is known as tectonic activity.
- Tectonic activity mainly occurs at the boundary of these tectonic plates. The

movement of tectonic plates is caused by thermal energy from the mantle of the lithosphere. Tectonic activity is responsible for earthquakes, volcanoes, mountain formation (orogeny), etc.

Hydrosphere

- The hydrosphere is composed of the total amount of water on the planet. It includes the oceans, rivers, lakes, and moisture in the air.
- In total, 97% of Earth's water is in the oceans and the remaining 3% is freshwater. A total of 75% of the freshwater is present in the form of ice on Earth's surface.
- The frozen part of the hydrosphere is known as the cryosphere.
- Water moves through the hydrosphere in a cyclic manner, which is called as Hydrological Cycle.
- In the hydrological cycle, water collects in clouds and then falls to the Earth in the form of rain or snow. This water flows in rivers, collects in lakes and oceans, evaporates again, forming clouds, and falls on the Earth's surface in the form of rain or snow.

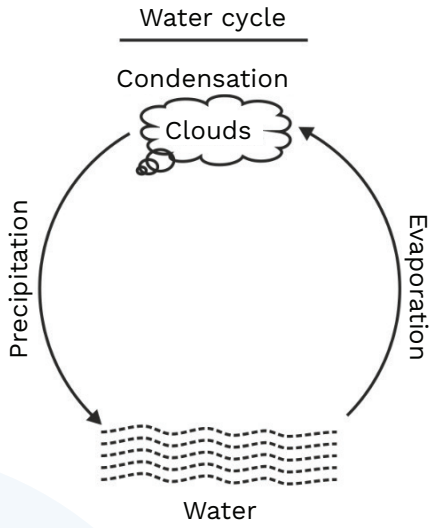


Fig. 1.5

Light

- Sun is the ultimate source of energy and emits light as electromagnetic radiation, which is visible within a certain portion of the electromagnetic spectrum.

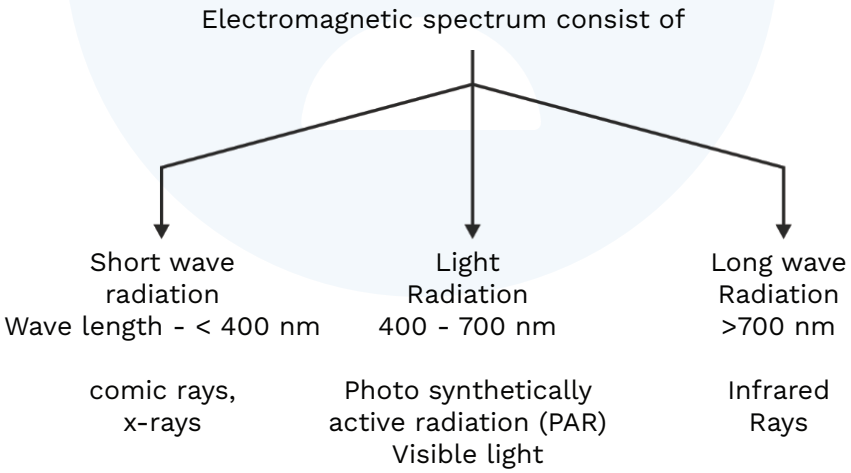


Fig. 1.6

- Light or visible spectrum having a wavelength of 400-700nm is also called photosynthetically active radiation (PAR).
- Depending upon wavelength three types of ultraviolet radiations are distinguished.

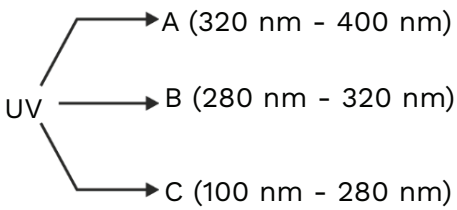


Fig. 1.7

- The shorter the wavelength, the more harmful the UV radiation
- However, UV-C cannot penetrate the Earth's atmosphere because UV radiation of wavelength 100–400 nm is mostly absorbed by the ozone layer, which is present in the stratosphere, and only a small fraction of it reaches the Earth's surface.

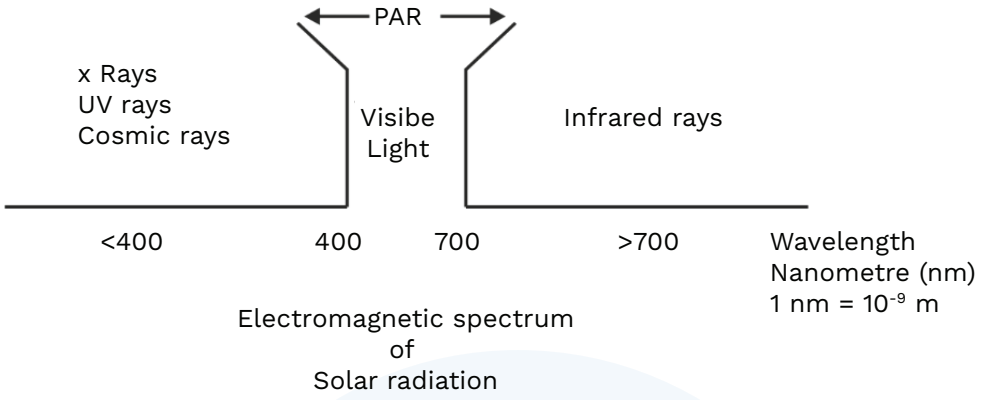


Fig. 1.8

Ecology

- The word 'ecology' is derived from the Greek word 'oikos,' which means home and 'logos,' which means to study.
- It is a systematic study of the relationship between biotic components and their environment.
- The term 'ecology' was first used by German zoologist Ernst Haeckel in 1869.

Autecology

The study of the relationship between individual species and the environment.

Synecology

The analysis of floral communities in relation to the habitats of a particular ecosystem.

Gaia Hypothesis

- The Gaia hypothesis is a scientific idea stating that the Earth is a complex living entity dependent on the self-regulating interactions between species and their inorganic environment.
- For example, Climatic conditions depend on the interactions between Biotic communities and their abiotic environment, all of which regulate each other continuously.
- The Gaia hypothesis is named after the mythical Greek goddess Gaia who personifies the Earth. It was first proposed by the British scientist James Lovelock in his 1972 paper 'Gaia as Seen through the Atmosphere'.

Solved Examples

1. The Gaia hypothesis, which suggested that the Earth should be seen as a single organism was devised by: **[UPSC ESE 2022]**

(A) James Lovelock
(B) Francoise d'Eaubonne
(C) Earnest Haeckel
(D) Paul Ehrlich

Solution: (A)

Ecological Principles

Ecology is mainly based on three principles.

Homeostasis

The tendency of the body to seek and maintain a condition of balance within its internal environment even when it undergoes external changes.

Limiting Factors

Limiting factors are environmental influences that constrain the productivity of organisms, population in communities.

Law of Thermodynamics

Energy can neither be created nor be destroyed, it can be only be changed from one form to another.

Levels of Organisation of Ecology

The seven major ecological levels of an organisation are as follows.

Organism

They represent the main unit of study in ecology. The biological unit possesses a distinct structure and function at each level. Form, physiology, behaviour, distribution, and environmental adaptations are investigated at this level.

Population

Population is a group of individuals of the same species, inhabiting the same area and functioning as a unit of a biotic community.

Biological Community

- A biological community is the collection of various species of the population (flora and fauna) in a certain location at a particular period. These species can vary greatly in size and number.
- Interdependence and interactions between the populations of several species within a habitat provide biological or biotic community structure.

Ecosystem

- An ecosystem is defined as a community of life forms in concurrence with non-living components, interacting with each other. The term 'ecosystem' was first coined by AG Tansley.
- The concept of 'ecosystem' was initially given by EP Odum, who is considered as the father of ecosystem.

Biome

- Biome is a huge geographic unit characterised by a dominant flora type and fauna linked with a particular climate zone. For example, forest, desert, etc.
- The climate establishes the limits of a biome. Temperature and rainfall are the most significant climate characteristics.
- Aquatic ecosystems are not considered biomes. They are separated into several life zones, each of which has regions with distinct fauna and flora.
- The biosphere consists of all terrestrial biomes and bodies of water on the Earth.

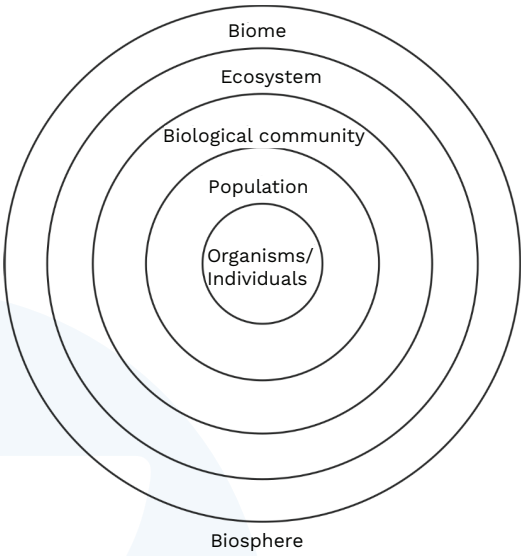


Fig. 1.9

Biotic Potential

- The maximum rate at which a population can increase when resources are unlimited and environmental conditions are ideal is termed the population's biotic potential.
- Each species has a unique biotic potential owing to variation in the:
  - Reproductive period, capacity, and frequency of the species
  - Litter size (how many were born each time) and survival rate (how many survive up to reproductive age)

Carrying Capacity

For a given region, carrying capacity is the maximum number of individuals of a given species that an area's resources can sustain indefinitely without significantly degrading those resources.

Species

A 'species' is defined as 'a group of similar species which can interbreed and produce fruitful offspring.'

Classification of Species

Keystone Species

Keystone species play a critical role in maintaining the structure, functioning, or productivity of an ecosystem at a defined level. For example, hummingbird, sharks (called ocean cleaners).

Indicator Species

A group of species is chosen as an indicator or proxy to represent the state of the ecosystem and are sensitive to the environment. For example, cray fishes, tubifex larva.

Umbrella Species

Umbrella species are species that are selected as representatives of their ecosystem. By protecting these species, other species that are a part of their ecosystem will also benefit. under the same umbrella. Umbrella species are large animals or organisms, for example, giant pandas, spotted owls, and red wolves.

Foundation Species

These are the species that have a strong role in creating, structuring, and maintaining habitat. Foundation species can occupy any trophic level in a food web.

Ecological Niche

- A niche refers to the unique functional role and position of a species in its habitat or ecosystem. The functional characteristic of a species in its habitat is referred to as 'niche' in that common habitat.
- In nature, many species occupy the same habitat, but they perform different functions.
- **Habitat niche** - where it lives, food niche - what it eats or decomposes and what species it competes with.
- **Reproductive niche** - how and when it reproduces.
- **Physical and chemical niche** - temperature, land shape, land slope, humidity and another requirement.
- Niche plays an important role in the conservation of organisms. If we have to conserve species in their native habitat, we should know the niche requirements of the species.

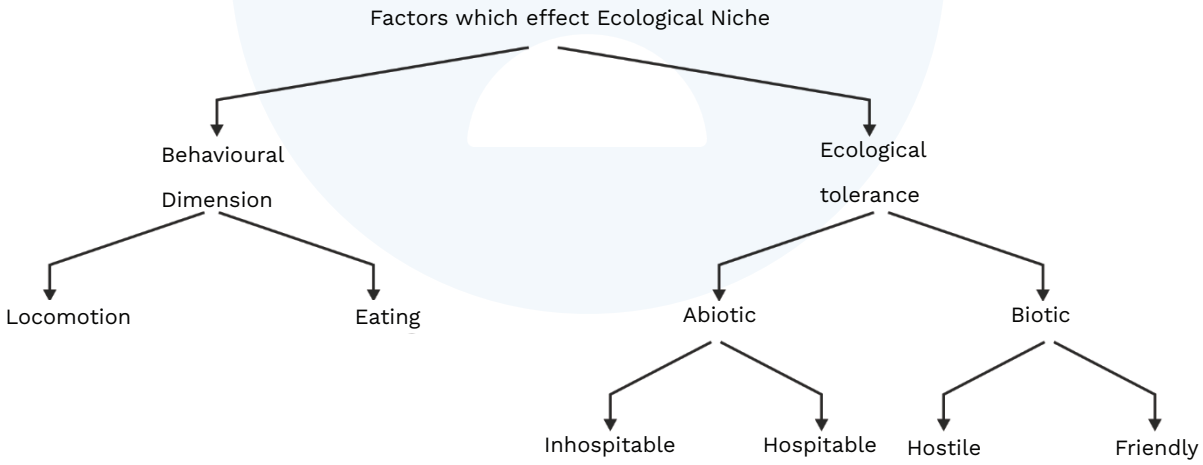


Fig. 1.10

Niche Partitioning or Resource Partitioning

- Species divided limited resources to help avoid competition in an ecological niche it is called Niche partitioning
- Intraspecific competition denotes competition for resources by individuals of the same species
- Interspecific competition denotes competition for resources by individuals of different species

Biological Interaction

These are the way species interact with each other. These interactions can be interspecific (interactions with different species) or intra-specific (interactions between the same species).

Type Interaction	Species		Effect	Comment
	1	2		
Negative Interactions				
Amensalism	–	0	One species is inhibited while the other species is unaffected.	• Penicillium generates the antibiotic penicillin, which inhibits the growth of microorganisms. • A huge tree gives shade to a small plant and limits its growth in turn small plant has no effect on the large tree.
Predation	+	–	One species (predator) benefits while the second species (prey) is harmed and inhibited.	• Predators lower the competitive pressure among species by hunting and killing the prey and helps in maintaining species diversity. For example, tiger, cheetah.
Parasitism	+	–	Beneficial to one species (parasite) and harmful to the other species (host).	• Parasitism implies a parasite residing in or on another living species called the host and obtains feed and shelter from the host. Examples of parasites are tapeworm, roundworm, malaria parasite.
Competition	–	–	Adversely affects both species.	• Competition exists when two populations or species rely on a limited resource for survival. For e.g., woodpeckers and squirrels usually fight for nesting rights.
Positive Associations				
Commensalism	–	0	One species (the commensal) benefits, while the other species the host) is neither harmed nor inhibited.	• Suckerfish attach to sharks which provides protection, transport, and food. However, the shark has neither profit nor damage with this connection.
Mutualism	+	+	Interaction is favourable to both species.	• Sea anemone attaches to the hermit crab shell for transport and food, while the anemone offers camouflage and protection using its stinging cells to the hermit crab. • Some mutualisms are highly interdependent on one another for survival such organisms associated could no longer survive without one another. Such intimate relationships are known as symbiosis (symbiosis is intense mutualism (e.g., coral and zooxanthellae)).
Neutral Interactions				
Neutralism	0	0		• True neutralism is extremely unlikely.

Table 1.1

Ecological Footprint

An ecological footprint is a measure of human impact on the ecosystems. It is typically measured in the area of wilderness or the amount of natural capital consumed each year. It refers demand for resources by the human population from nature to survive.

Carbon Footprint

- A carbon footprint is defined as the total amount of greenhouse gases produced directly or indirectly to support human activities, usually expressed in equivalent tons of carbon dioxide (CO_2). It is a measure of the impact of our activities on the environment in general and on climate change.
- It refers to the number of greenhouse gases produced in our day-to-day life through fossil fuels for electricity, heating, and transportation.
- The carbon footprint is a highly powerful tool to understand the impacts of personal behaviour on global warming. It shows the amount of CO_2 released by their activities.

Ecotone

An ecotone is a zone of junction or a transition area between two biomes. It is the zone where two communities meet and integrate. For example, the mangrove forests represent an ecotone between marine and terrestrial ecosystems. Other grasslands (between forest and desert), estuary (between fresh water and saltwater), and riverbank or marshland (between dry and wet).

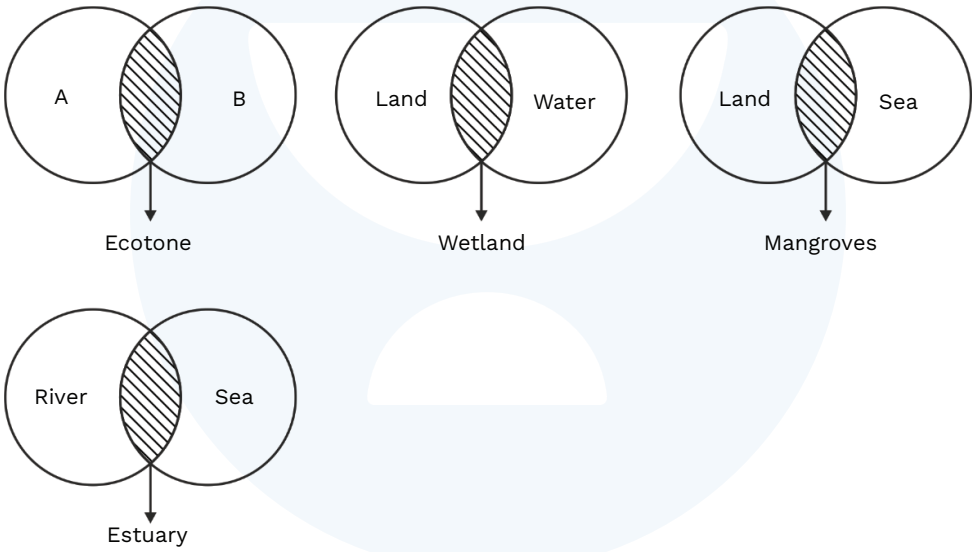


Fig. 1.11

Ecocline

- Ecocline is a zone of gradual but continuous change from one ecosystem to another when there is no sharp boundary between the two in terms of species composition. Ecocline occurs across the environmental gradient (gradual change in abiotic factors such as altitude, temperature (thermocline), salinity (halocline), depth, etc.)

Ecotone	Ecocline
Describes variation in species of flora and fauna.	Describes variation in forms of species based on genetic differences.
Harder of different two ecosystems in transition.	Based on technical criteria, it is easy to differentiate.

Table 1.2

Edge Effect - Edge Species

- The edge effect refers to the changes in population or community structures that occur at the boundary of two habitats (ecotone). Sometimes the number of species and the population density of some of the species in the ecotone is much greater than either community. This is called the edge effect.
- The organisms that exist primarily or most abundantly in this zone are known as edge species.
- In terrestrial ecosystems, edge effect is especially applicable to birds. The density of birds is greater in the ecotone between the forest and the desert.

Biological Spectrum

- The biological spectrum is comprised of all living species categorised into three areas that coexist at differing levels of biological organisation. The three domains, bacteria, archaea, and eukaryotes, are further split into kingdoms. All three domains exist at different levels of biological organisation, from the cell level to the biosphere.
- These are categorised according to their capacity to adapt to adverse situations. A plant's lifeform is the sum of all of its life processes and has evolved in direct reaction to its environment. The biological spectrum denotes the extent of vivacity/variety found in an ecosystem.

Solved Examples

1. Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms? **[UPSC 2013]**

(A) Ecotone
(B) Ecological niche
(C) Habitat
(D) Home range

Solution: (B)

Ecological/Biotic Succession or Ecological Development

Ecological successions are the entire process of directional and sequential change of either plant community or the whole ecosystem over a period.

Types of Succession

Primary Succession

Primary succession occurs in essentially lifeless areas, regions in which the soil is incapable of sustaining life as a result of such factors as lava flows, newly formed dunes, or rocks left from a retreating glacier.

Pioneer Community

Group of organisms that first invade a new area in the process of ecological succession and often make the habitat hospitable. Organisms such as algae, fungi, and lichens are called pioneer communities.

Secondary Succession

Secondary succession occurs in areas where a community that previously existed has been removed; it is typified by smaller-scale disturbances that do not eliminate all life and nutrients from the environment.

Climax Community

- A climax community is the final stage of succession, remains relatively unchanged helps in the establishment of a stable community until destroyed by an event such as fire or human interference.
- In ecological terminology, the developmental stages of a community are known as seral stages and the final stage as the climax community. The entire series of communities that is characteristic of a given site is called a sere.
- **Prisere:** A collection of sere making up development of an area from a pioneer community to climax community.

Other Prominent Types of Ecological Succession

Autotrophic Succession

Autotrophic succession is characterised by the early and continued dominance of autotrophic organisms like green plants. It begins in a predominantly inorganic environment and the energy flow is maintained indefinitely.

Allogeneic Succession

In Allogeneic succession, the replacement of the existing community, is caused largely by any other external condition and not by the existing organisms.

Autogenic Succession

Autogenic succession refers to that type when the community itself because of its reactions to the environment, modifies its environment and thus causes its replacement by new communities.

Hydrarch Succession

Plant succession starts in relatively shallow water, such as ponds and lakes, and culminates in a mature forest.

Xerarch Succession

Plant succession starts on bare ground or rock and culminates in a mature climax forest. The pioneer species, such as lichens and mosses, result in the gradual accumulation of soil.

Process of Ecological Succession

Nudation

It is the development of a bare site uninhabited by any organisms. Nudation is usually caused by disturbances. These can be topographic (soil erosion, wind, etc.), climatic (hails, storms, fire, etc.), and biotic (human activities).

Invasion or Migration

This process helps the arrival of seeds, spores, or other reproductive propagules for the establishment of species. Invasive species are threateners or agents of community disturbance.

Ecesis

Ecesis is also called colonisation. It is the initial establishment of a plant community it is dependent on soil structures.

Aggregation

An increase in the population of species that have become established in the area is called aggregation.

Competition

Once a few initial species have become established, the intra, as well as interspecific competition, starts. This is called competition. It is usually for resources such as food, water, etc. Competition is found in both plants as well as animal species.

Reaction

Environmental conditions get modified by the action of species occupying the habitat. These changes subsequently trigger displacement and replacement of one species by another.

Stabilisation

The process by which the climax community gets stabilised is called stabilisation. A climax community is mature self-sustaining, and stable, and is the final stage of succession.

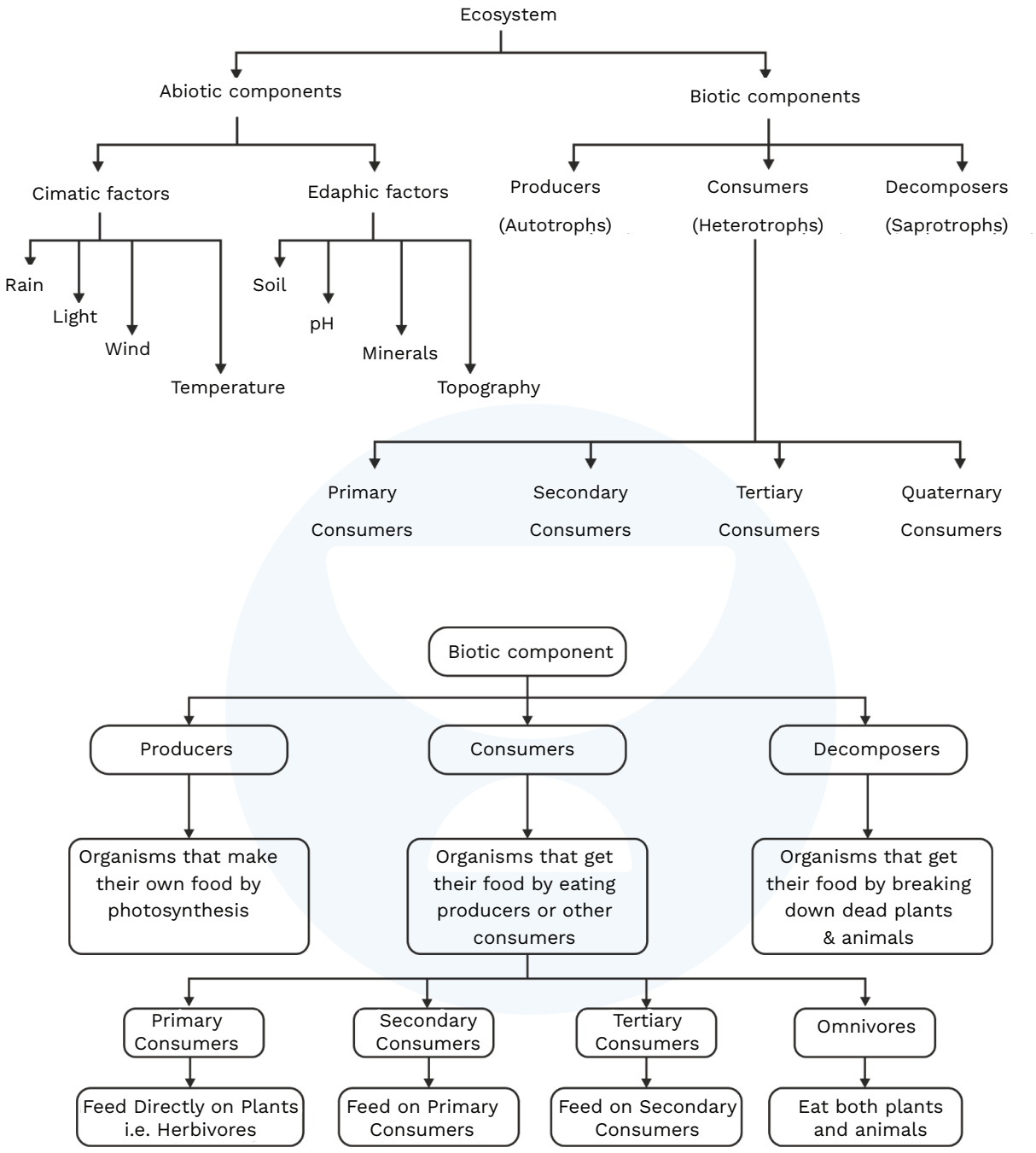

Ecosystem
Abiotic components
Biotic components
Cimatic factors
Edaphic factors
Producers
(Autotrophs)
Consumers
(Heterotrophs)
Decomposers
(Saprotrophs)
Rain
Light
Wind
Temperature
Soil
pH
Minerals
Topography
Primary
Consumers
Secondary
Consumers
Tertiary
Consumers
Quaternary
Consumers
Biotic component
Producers
Consumers
Decomposers
Organisms that make their own food by photosynthesis
Organisms that get their food by eating producers or other consumers
Organisms that get their food by breaking down dead plants & animals
Primary Consumers
Secondary Consumers
Tertiary Consumers
Omnivores
Feed Directly on Plants i.e. Herbivores
Feed on Primary Consumers
Feed on Secondary Consumers
Eat both plants and animals

Fig. 1.12

Solved Examples

1. With reference to the food chains in ecosystems, which of the following kinds of organism is/are known as decomposer organism/organisms?

[UPSC 2013]

I. Virus
II. Fungi
III. Bacteria

Select the correct answer using the codes given below.

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

Solution: (B)

Types of Ecosystem

Ecosystems can generally be classified into two classes such as natural and artificial. Artificial ecosystems are natural regions affected by man's interferences. They are artificial lakes, reservoirs, townships, and cities.

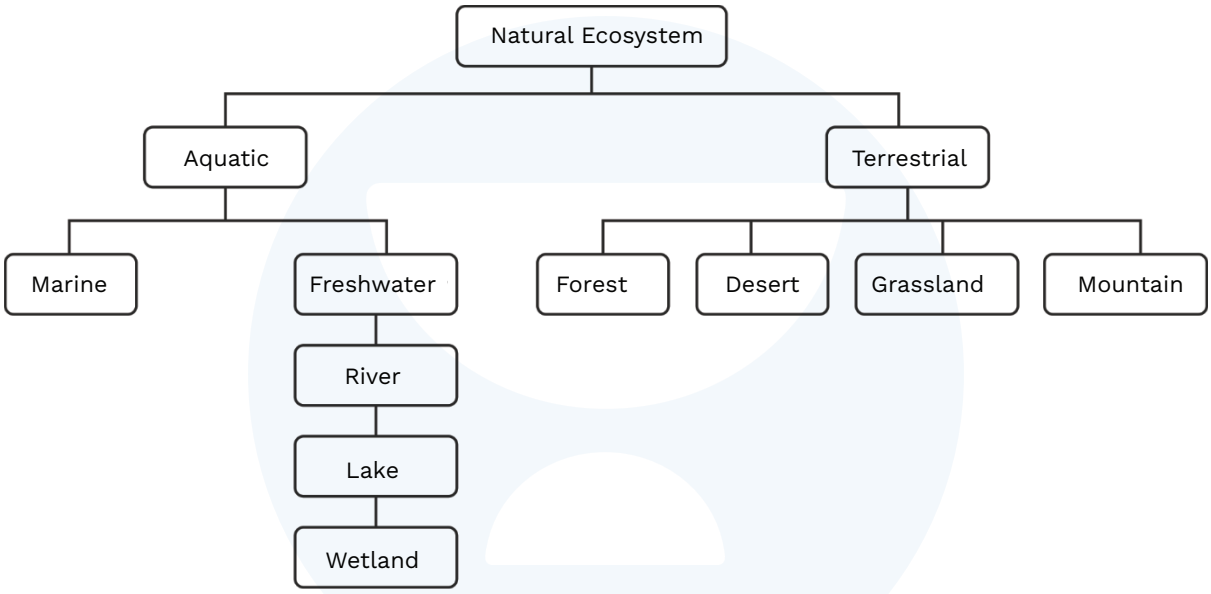


Fig. 1.13

Terrestrial Ecosystems

- Terrestrial and aquatic ecosystems are differentiated by taking water as a constraint, its availability, fluctuations in temperature, and light availability.
- The terrestrial atmosphere is more clear than the aquatic environment due to greater access to light than in aquatic ones

Biome

The terrestrial portion of the biosphere is subdivided into vast sections known as biomes. No two biomes are identical. They are distinguished by their climate, flora, fauna, and soil.

Types of Biomes

Tundra Biome

- There are two types of tundra: arctic and alpine.
- The alpine tundra is found in the high mountains, above the upper slopes like the Himalayas. Due to permafrost, there is minimal vegetation like mosses and lichens found on bare rocks.

- To prevent heat loss and from cold, animals animals have big bodies with short tails and ears, thick cuticles, and fur. For example, reindeer, Arctic fox, wolves, muskox, polar bear, Arctic hare.

Taiga or Boreal Biome

- The vegetation in this biome is an evergreen coniferous forest with less humous content and least productive, for example, spruce, fir, and pine.
- Siberian tiger, wolverine, bear, red fox, Hyla, and Rana are the animal species found in the boreal biome.

Podzols

Podzols are an infertile thin layer of soil, which lies on sandy or loamy subsoil surfaces with no organic matter used for grazing purposes. These are found in coniferous or boreal environment.

Temperate Deciduous Biome

The vegetation is deciduous, plants survive winter by shedding leaves to safeguard from snow and soils are podzolic. For example, oak, elm, ash, birch, beech, and so on.

Temperate Rainforest Biome

- Vegetation occupies a small area that exists in North America Coat with trees, mosses, and lichens grown as epiphytes (a plant that grows on another plant, especially one that is not parasitic).
- Epiphytes are known as 'air plants' because they are not rooted in the soil. Epiphytes obtain nutrients from rainwater, air.

Temperate Deciduous Biome (Mediterranean Climate)

- Vegetation is drought resistant as they fight excessive heat and evaporation. Plants adapt to fire effectively and grow rapidly after being burnt.
- Low, broad-leafed evergreen oak trees are occupied in these regions with adequate rainfall.

Sub-Tropical Deciduous Biome

Lush vegetation with broadleaf evergreen forests and deciduous trees. Neither a dry season nor a cold season can stop the growth of perennial plants.

Steppe or Temperate Grassland Biome

This biome is Treeless and the grass available is fresh, healthy, and short. Towards the Earth's pole, an increase in rainfall helps in forming a transitional zone of wooded steppes where conifers appear.

Tropical Deciduous Biome (Monsoon Climate)

Tropical Monsoon Forests are referred to as drought-deciduous forests/dry forests/dry deciduous forests/tropical deciduous forests. For example, teak, neem, bamboos, sal, shisham, sandalwood, khair, and mulberry.

Savannah or Tropical Wet and Dry Biome

- Savannah Vegetation typically survives drought with plants shedding leaves to reduce water loss through transpiration, thick trunks which store water.
- The Savannah ecosystem is characterised by tall grass and short trees and it has a variety of mammals, birds, and reptiles.

Tropical Rain Forest Biome

- Tropical rain forests are characterised by high temperatures and huge rainfall.
- All plants plants, particularly epiphytes, strive for sunlight, leading to the formation of a unique layered structure called the canopy. The tree canopy comprises two layers which have dense underlying vegetation, which allows no light to reach the ground.

Difference between Evergreen and Deciduous Forests

Evergreen Forests	Deciduous Forests
The evergreen forests are found in areas of heavy rainfall and abundant sunshine.	Deciduous forests are found in a moderately cool climate with moderate rainfall.
Trees in these forests do not shed their leaves simultaneously during any part of the year.	Trees in these forests shed their leaves in a particular season to conserve loos of moisture through transpiration.
These forests are found on the eastern margins of continents like South China, South Brazil, etc.	These forests are found in subtropical regions of Asia, Central America, Brazil, etc., with a distinct dry season like monsoons.
These forests consist of hardwood species like bamboos, rubber, sandalwood, rosewood, etc.	In these forests, trees are generally deciduous and have broad leaves like oak, ash, cedar, fir, etc.

Table 1.3

Desert Biome

- Xerophytic or drought-resistant vegetation with plants having no or few leaves with a needle or hairy foliage, which helps in reducing water loss through transpiration and wide-spaced roots to absorb groundwater. For example, cacti, thorny bushes, and scattered dwarf acacias.
- The seeds of these plant species possess thick coverings to protect them during their dormancy.

Forest Vegetation in India

Tropical Evergreen Forests

- This vegetation is found in areas with greater than 200 cm rainfall and a 15-30°C temperature range. Climate is warm and humid throughout the entire year.
- They are primarily found near the equator. Rosewood, mahogany, ebony, bamboo, etc., are the observed plant species.
- Elephants, monkeys, lemurs, and deer are common animals observed. The one-horned rhinoceros inhabits the Assam and West Bengal forests.

Tropical Deciduous Forests

- These forests also known as monsoon forests are found in areas with rainfall between 70-200 cm. During the dry summer, trees lose their leaves for around six to eight weeks.
- These deciduous forests are further classified into moist or dry based on water availability.
- Moist deciduous dorests are found in areas with rainfall between 100 to 200 cm. Teak, bamboo, sal, shisham, sandalwood, khair, Kusum, Arjun, etc., are the plant species found in these forests.
- Dry deciduous woods are prevalent in regions with precipitation between 70 and 100 cm. Teak, sal, peepal, neem, etc., are the plant species found in these forests.

Thorn Forests and Scrubs

- This vegetation is found in areas with less than 70 cm of rainfall. Acacias, euphorbias, cacti, etc., are found in these forests.
- Trees are dispersed with deep roots into the soil in search of water. Succulent stems help to preserve water. Leaves are thick and tiny to reduce evaporation.

Montane Forests

- Vegetation undergoes changes as temperature decreases with an increase in height in mountainous regions.
- The wet temperate forest type exists between 1000 and 2000 metres in altitude.

- Between 1500 and 3 m, temperate forests are composed of coniferous trees such as pine, deodar, silver fir, etc.
- At altitudes greater than 3600 m, alpine vegetation replaces temperate forests and grasslands.
- As they reach the snowline, they become increasingly stunted and are heavily grazed by nomadic tribes like the Gujjars and Bakarwal.
- Spotted deer, wild sheep, Tibetan antelope, yak, snow leopard, bear, rare red panda, etc., are commonly found in these areas.

Mangrove Forests

- Mangrove forests are spread in coastal areas piled up with mud, and silt and affected by tides. Mangroves are found in Ganga, Mahanadi, Krishna, Godavari, and Kaveri deltas.
- Mangrove plant species have their roots submerged in water and are dense. The Ganga-Brahmaputra delta contains sundari trees. Palm, coconut, and keora, are also found in the delta. The famous Royal Bengal Tiger is found in these forests.

Aquatic Ecosystems

- Group of interacting organisms dependent on one another and their water environment for nutrients (nitrogen and phosphorus). Water provides shelter.
- Aquatic ecosystems are classified into distinct zones based on depth, tides, temperature, light, and proximity to land.
- Freshwater ecosystems: fresh water is land water with a low salt concentration (< 5 ppt) and continuous cycling.
- Freshwater ecosystems are classified into two types Lentic and lotic.

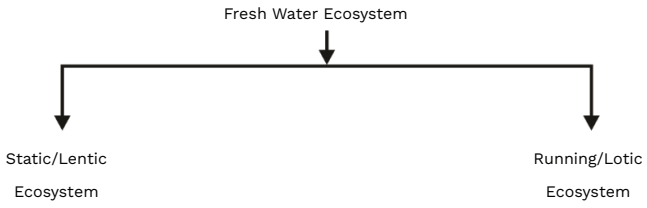


Fig. 1.14

- Marine ecosystems: the water bodies with a salt concentration equal to or above 35 ppt.
- Brackish water ecosystems: the water with a salt concentration of 5–35 ppt.
- Lakes and ponds are classified based on organic matter production (nutrients) or productivity.
- Oligotrophic: These are newly formed lakes' nutrients and productivity is less.
- Eutrophic: These have a large or excess supply of nutrients. They are shallow and murky.
- Mesotrophic: Moderate nutrients and productivity is observed in mesotrophic lakes.

Factors Limiting the Productivity of Aquatic Habitats

Sunlight

- The amount of sunlight reaching decreases with an increase in depth of the waterbody. Due to this, proliferation of plants and their photosynthetic activity diminishes.
- Turbidity of the water is also responsible for diminishing light penetration and photosynthetic activity. Suspended particle debris, such as clay, silt, phytoplankton, etc., contribute to the turbidity of the water.

Dissolved Oxygen

- The average dissolved oxygen concentration in freshwater is 10 ppm by weight. This is 150 times less oxygen than would be present in an equal volume of air.
- Oxygen enters the water bodies via air-water contact and through the photosynthetic processes of aquatic plants. DO diminishes in the water body due to e]respiration of organisms and air-water contact.
- Warm water decreases oxygen's solubility and boosts decomposer activity. So increase in the temperature of water accelerates the oxygen depletion rate.

Temperature

- Normally, temperature of the water has less fluctuations, so aquatic species have a limited temperature tolerance range. Even slight variations in water temperature pose a greater threat to the existence of aquatic organisms than variations in air temperature do to the life of terrestrial organisms.

- A sheet of ice over a body of water can effectively block out sunlight. In such an aquatic environment, photosynthesis ceases while respiration continues.
- Winterkill is the condition where aquatic species die in winter due to depleted oxygen levels in shall water bodies.

Based on the above factors, light penetration, dissolved oxygen, and temperature, proximity to land, lakes, and ponds is divided into different zones.

Littoral Zone

The warmest zone absorbs more heat. It is the topmost or near to share. Biodiversity is rich. Algae, rooted floating plants can be observed in the littoral zone.

Limonitic Zone

Near-surface open water surrounded by a littoral zone. Light penetrates this zone.

Profundal Zone

Light cannot be penetrated. It is also known as the Aphotic zone. The lower layers of the aquatic ecosystems, where light penetration and plant growth are restricted forms the aphotic zone. Only respiration activity takes place in this zone. The profundal zone lacks oxygen.

Benthic Zone

Bottom of the lake. The temperature will be cool as light does not penetrate. Decomposers live in this zone.

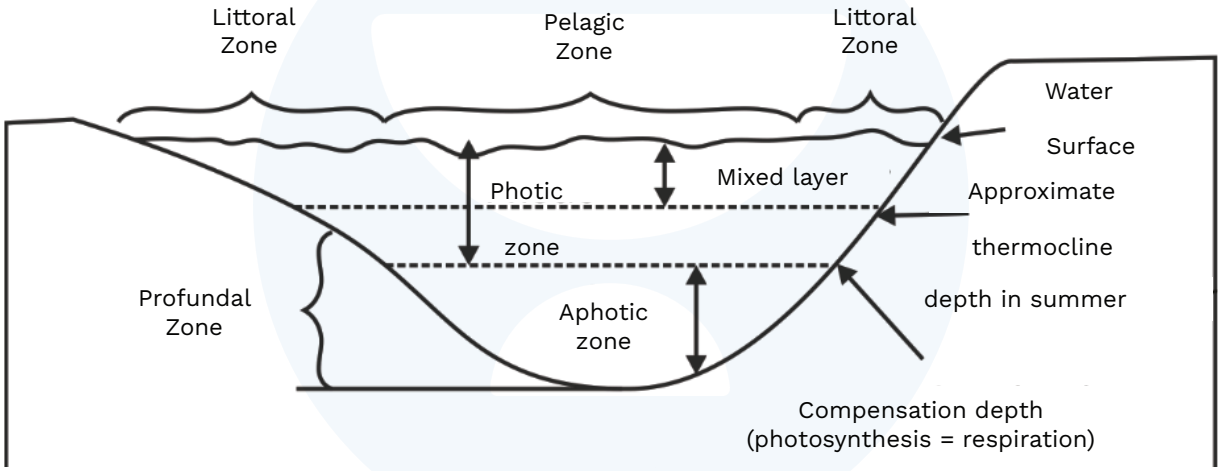


Fig. 1.15

Eutrophication

The natural or artificial addition of nutrients in the bodies of water is called eutrophication. For example, Natural runoff nutrients, weathering of rocks (erosion, mining), fertilizer runoff, and sewage discharge.

Effects

- Algal blooms increase rapidly due to eutrophication. Some algal blooms (phytoplankton) release biotoxins that kill aquatic life. These are called harmful algal blooms (HAB). When HABs are grown in large numbers they are called red tide.
- Hypoxia: Reduced dissolved oxygen content in the water body is called hypoxia.
- If algae die, they are converted to dead organic matter by decomposers, which consume oxygen to break down, leading to hypoxia.

Moving Bodies of Water

Moving bodies of water will move in one direction. Different zones of moving bodies of water are.

Source Zone

- It is narrow and contains headwater streams often begin as snow melts or springs. Water

will be cold and clear with little sediments. The water flow is swift.

- It dissolves a large amount of oxygen from the air, and most plants are attached to rocks.

Transition Zone

The transition zone forms wider and deeper streams that flow down gentler slopes. This wider stream joins tributaries. Water is warmer, with more nutrients, which supports more producers.

Flood Plain Zone

Water is murky warmer contains the highest nutrients and the flow is slow. River fragments from distributaries, empty into oceans at the river mouth (estuaries).

Marine Ecosystems

- Marine ecosystems are classified into different zones based on the environment.
  - Pelagic environment
  - Benthic environment
- In pelagic environment, the waters are divided into the neritic province, which includes the water above the continental shelf and the oceanic province, which includes all the open waters above the continental shelf.
- The neritic province has high nutrient levels due to dissolved materials in riverine runoff which distinguishes it from the oceanic province.
- The upper portion of both neritic and oceanic waters is called the epipelagic zone where photosynthesis occurs. The Epipelagic zone is equivalent to the photic zone.
- Below the epipelagic zone, the mesopelagic zone occurs ranges between 200–1000 m, the bathypelagic zone from 1000–4000 m, and abyssalpelagic the deepest parts of the oceans from 4000 m to deep-sea trenches.
- The benthic environment or zone is on the ocean floor. The ocean floor drops as you move away from the continents. There are fewer living things on the ocean floor where the water is very deep.
- Based on light penetration or the amount of light they receive, marine ecosystems are classified into the euphotic zone, which extends up to 200 m below the surface. In this zone, sufficient light penetrates which results in good photosynthetic activity. Most marine life inhabitants live in this zone.
- Below the euphotic zone is a dysphotic zone that extends from 200–1000 m below the surface. Sunlight is still available but low photosynthetic activity in this zone.
- Below the dysphotic zone lies the aphotic zone which does not receive any sunlight.

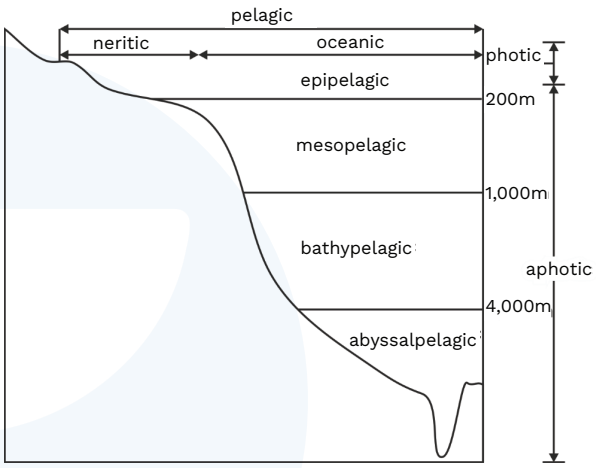


Fig. 1.16

Biodiversity of Aquatic Ecosystems

The aquatic animals are categorised according to their occurrence zone.

- **Neuston** are the species which inhabits the air-water assemblage. For example, floating plants.
- **Periphyton** are the species clinging to the leaves and stems of rooted plants or protruding above the bottom mud. For example, sessile algae.
- **Plankton** are microscopic floating species which include phytoplankton, diatoms, zooplankton, and larval forms. The poor locomotory capacity of plankton means that their dispersal in aquatic ecosystems is mostly determined by currents.
- **Nekton** are the species which consists of strong swimmers capable of overcoming water currents.
- **Benthic** species are those that reside at the bottom of a body of water.

Productivity

Rate of formation of energy and matter which is stored in different species of different Trophic levels.

Trophic Level

- The trophic level of an organism is the position it occupies in a food chain.
- A species can concurrently inhabit multiple trophic levels. The trophic level denotes a functional level, not a species. For example, a sparrow is a major consumer when it consumes seeds, fruits, and peas, but a secondary consumer when it consumes insects and worms.
- The rate at which chemical energy is formed from solar energy is known as primary productivity.
- Primary productivity is measured in dry weight (dry gram calories/unit area/unit time.).
- The rate at which energy is stored at different levels in consumers is called secondary productivity.

Gross Primary Productivity (GPP)

The total amount of organic matter produced through photosynthesis is called GPP.

Net Primary Productivity (NPP)

The amount of energy that remains available for plant growth after subtracting the fraction plants uses for respiration.

NPP = GPP – Fraction used for respiration

Functions of an Ecosystem

Ecosystems are complex dynamic systems, they perform the following functions

- Energy flow through the food chain
- Nutrient cycling (biogeochemical cycles)
- Ecological succession or ecosystem development
- Homeostasis or feedback control mechanisms

Energy Flow in an Ecosystem

- The flow of energy and nutrients across various trophic levels derived from various external sources maintains an ecosystem. All ecosystems on the Earth get their energy mostly from the sun.
- 'Autotrophs' or 'primary producers' use photosynthesis to convert solar energy into organic matter at the first trophic level.

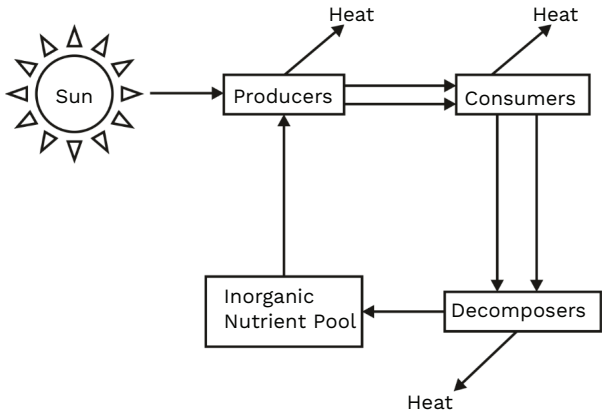


Fig. 1.17

- 'Herbivores' or 'primary consumers' obtain energy from the autotrophs they consume at the second trophic level. Most of this energy is used for metabolic processes like respiration, digestion, and maintenance of body temperature.
- 'Carnivores' or 'secondary consumers' acquire energy for their growth and survival from herbivores at this trophic level. Huge predators represent higher trophic levels as they derive their energy from carnivores.

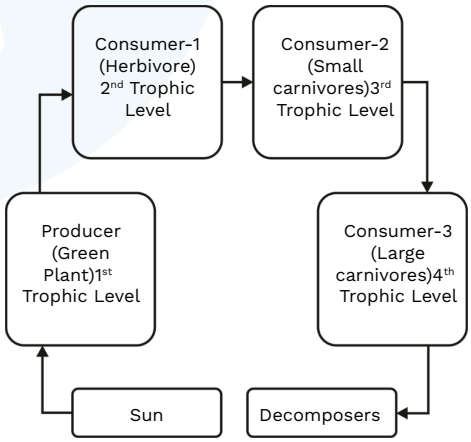


Fig. 1.18

Food Chain

A food chain is defined as the unidirectional transfer of food energy from the producer to the consumer through a series of organisms (herbivores to carnivores to omnivores to decomposers) with repeated eating and being eaten process.

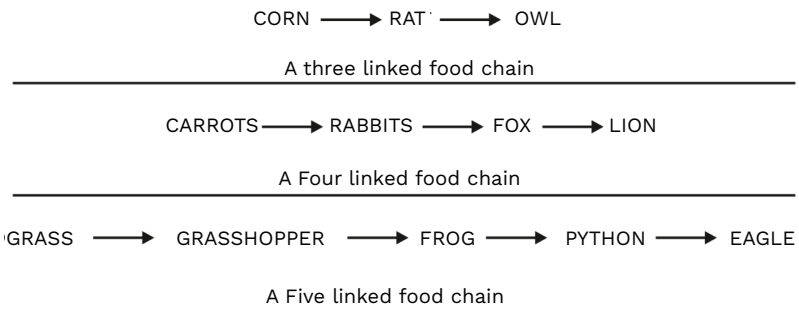


Fig. 1.19

Solved Examples

1. With reference to food chains in ecosystems, consider the following statements. **[UPSC 2013]**

I. A food chain illustrates the order in which a chain of organisms feed upon each other.
II. Food chains are found within the populations of a species.
III. A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is / are correct?

(A) I only
(B) I and II only
(C) I, II, and III
(D) None of the above

Solution: (A)

2. Which one of the following is the correct sequence of a food chain? **[UPSC 2014]**

(A) Diatoms-Crustaceans-Herrings
(B) Crustaceans-Diatoms-Herrings
(C) Diatoms-Herrings-Crustaceans
(D) Crustaceans-Herrings-Diatoms

Solution: (A)

Food Web

- 'Food web' is a graphical description of feeding relationships among species in an ecological community i.e., of who eats whom. It is also means of showing how energy material flows through a community of species as a result of these feeding relationships.
- In the food web, species are connected by a line/arrow which is known as a link, and species are referred to as Node. The concept of the food chain and food web was given by Charles Elton.

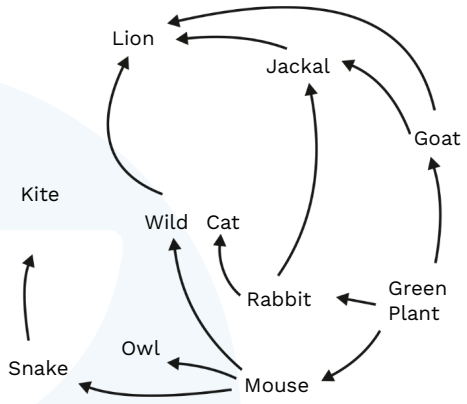


Fig. 1.20

Food Chain	Food Web
It is the single pathway through which food and energy travel in the ecosystem.	It consists of several interconnected food chains through which food and energy travel in an ecosystem.
Usually, members of higher trophic level feed upon a single type of organism of lower trophic level.	Usually, members of higher trophic levels feed upon many organisms of lower trophic level.
Isolated or separate food chains increase the instability of the ecosystem.	The presence of complex food webs increases the stability of the ecosystem.
It does not have any effect on improving the adaptability and competitiveness of the organisms.	More complex food webs improve the adaptability and competitiveness of the organisms.

Table 1.4

Ecological Pyramids/Eltonian Pyramids

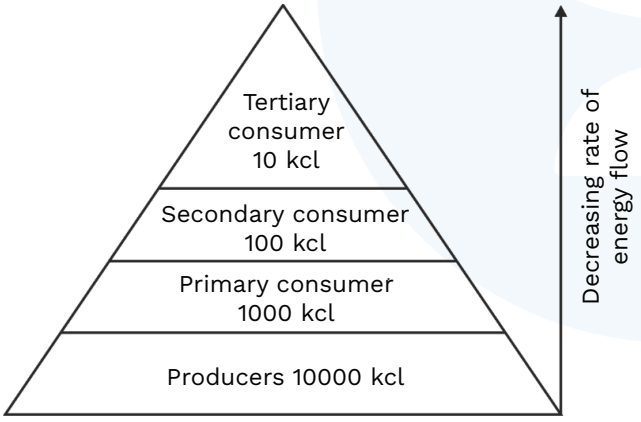
A diagram that explains how ecologically important factors (numbers, energy, biomass) vary between trophic levels.

Pyramid of Energy

- An Energy pyramid represents the amount of energy at each trophic level and the loss of energy at each is transferred to another trophic level.
- The amount of energy left at that trophic level cannot support another trophic level.

Lindeman's Law/10% Law

- When energy is transferred from one trophic level to another, 90% of the energy is used for metabolic processes (locomotion, digestion, respiration) and only 10% of the energy is transferred.
- Energy decreases as one moves through the trophic level from the bottom to the top of the pyramid. Thus, the energy pyramid is always upward.



Pyramid of Energy

Fig. 1.21

Ecological Efficiency

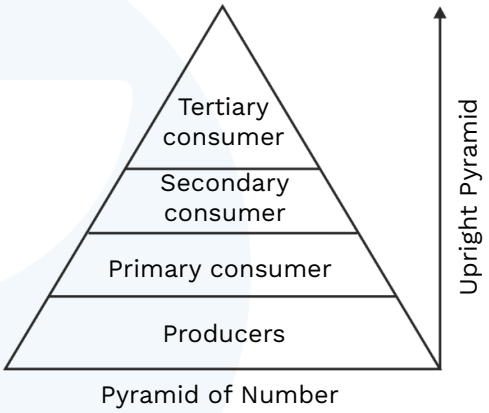
- Ecological Efficiency refers to the percentage of energy transferred from one trophic level to another.
- In a marine ecosystem, ecological efficiency is higher because of rich biodiversity (rich food chain) and that is why consumers do not move long distances for their food intake. Therefore, in this ecosystem, less energy is used by a consumer to maintain their metabolic processes. Due to this, more energy and matter are concentrated in the body of an organism.

Pyramid of Numbers

The pyramid of numbers represents the number of individuals per unit area at each trophic level. Different types of Pyramid of numbers exist, they are

Upright Pyramid of Numbers

From the bottom to the top of an upright pyramid of numbers, the number of individuals declines. Large numbers of producers are found in aquatic or grassland ecosystems. Fewer herbivores, in turn, feed fewer carnivores.



Pyramid of Number

Fig. 1.22

Inverted Pyramid of Numbers

From the lower to the higher trophic level of an inverted pyramid of numbers, the number of individuals increases.

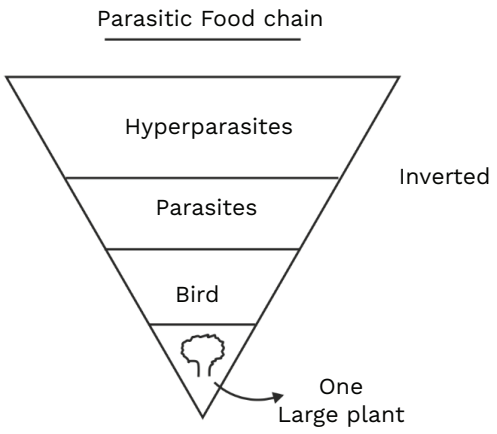


Fig. 1.23

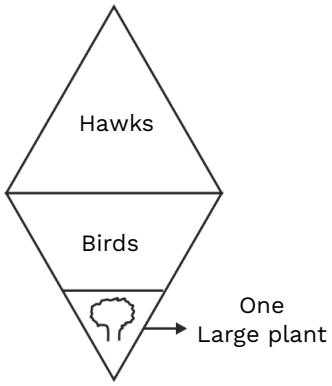


Fig. 1.24

Pyramid of Biomass

The pyramid of biomass represents the amount of biomass (dry weight) per unit area at each trophic level. Different types of pyramids of biomass exist; they are described below.

Upright Pyramid of Biomass

From the lower to the higher trophic level of an upright pyramid of biomass, the biomass of individuals decreases. For example, terrestrial ecosystem.

Inverted Pyramid of Biomass

- From the lower to the higher trophic level of an inverted pyramid of Biomass, the biomass of individuals increases. For example, aquatic ecosystem.
- The producers in an aquatic ecosystem are microscopic phytoplankton that has less biomass supporting the heavy consumer biomass and it adopts an inverted form.

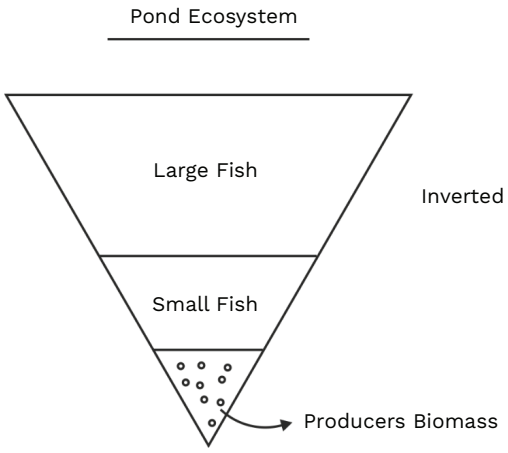


Fig. 1.25

Transfer of Biotoxins at Trophic Level

Bioaccumulation

- Accumulation of contaminants by species in concentrations that are orders of higher magnitude than the surrounding environment.
- Bioaccumulation can happen through breathing, eating, drinking, and skin contact. Polychlorinated biphenyls (PCB), dioxins, DDT, and mercury accumulate in living tissues.
- Drinking water with an excess of nitrates leads to blue baby syndrome/methemoglobinemia Haemoglobin will be non-functional as the body converts consumed nitrates into nitrites. These nitrites bind to the haemoglobin forming methaemoglobin which is unable to carry oxygen.

Bioconcentration

Direct uptake of a substance by a living organism through water and subsequent concentration of a substance within an organism will be more than the surrounding environment. It is also known as aquatic lower case.

Biomagnifications or Bioamplification

- The tendency of pollutants to concentrate as they move from one trophic level to another. An increase in the concentration of pollutants from one like in a food chain to another is called biomagnification. Let us take a look at some effects of biomagnification.
- Dichloro Diphenyl Trichloroethane (DDT) is an insecticide that is now banned in many countries. In 1992, DDT was sprayed in the lake of Michigan USA to kill mosquitoes and eliminate malaria After 20 years, it was found that the numbers of Pelican birds (which were specific to Michigan) were decreasing very rapidly.
- DDT, which was used for killing mosquitoes, entered the body of pelicans through the food chain. Due to the presence of DDT, the egg laid down by these birds had a very thin outer shell, even before eggs could hatch, the thin shell of the egg used to break off by itself which was responsible for the decrease in the bird population. It was found that the

population of Pelican birds was decreased due to the biomagnification of DDT.

- Diclofenac, which is a medicine for common inflammatory disease, fever, and pain associated with it was widely used in India. This drug proved to be fatal for vultures because vultures consume dead animals, and some amount of diclofenac is used to enter the food chain of vultures. It results in kidney failures in vultures. The number of vulture populations decreased significantly. Therefore, the Indian government banned Diclofenac in 2006.

Saving Asia's Vultures from Extinction (SAVE)

- It advocates for the effective removal of diclofenac from veterinary use, and the creation of vulture sale zone, breeding centres, and release programs.
- Endosulfan, which is a pesticide is being phased out globally. It is a 'xenoestrogen', a synthetic substance that imitates or enhances the effect of oestrogens and it can act as an endocrine disrupter causing reproductive and developmental damage in both animals and humans. A global ban on the manufacture and use of endosulfan was negotiated under Stockholm Convention in April 2011.

Bioremediation

- Bioremediation is a phenomenon in which living organisms are used to solve any degradation in the environment.
- In this process, microorganisms like bacteria, fungi, etc., are constantly at work and they break down complex organic matter into simpler matter in any given ecosystem. For example, leguminous crops or oilseeds, pulses, etc., are used to replenish the nitrogen content of soil, which is exhausted because of repeated monocropping of wheat and rice.

Features of Bioremediation

- Bioremediation is less expensive than other technologies. It is often used to clean up hazardous wastes.
- The bioremediation process is more effective and precise because it can be applied to the location (exact site) of the problem, without affecting the surrounding area.
- It is an ecologically sound and natural process. Instead of merely transferring contaminants from one environmental medium to another, it eliminates the target chemicals.

Biogeochemical Cycles

- All elements the Earth are recycled continuously. The main components of organisms are oxygen, carbon, nitrogen, phosphorous, and sulphur.
- The movement of C, H, O, N, S, Ph, and their compounds between organisms and their physical environment is known as the Biogeochemical cycle. Chemicals ingested by organisms are recycled back into the soil, air, and water through respiration, excretion, and decomposition.
- Biogeochemical cycles can be either gaseous (reservoir pool is the atmosphere) or sedimentary (reservoir pool is the Earth's crust).

Gaseous Biogeochemical Cycles

- Carbon cycle
- Nitrogen cycle
- Water cycle
- Oxygen cycle

Sedimentary Biogeochemical Cycle

- Phosphorus cycle
- Sulphur cycle

Carbon Cycle

Through photosynthesis, carbon enters nature as carbon dioxide in the form of carbohydrates. These carbohydrates are transferred from producers to consumers and these carbon components are returned to the environment through the respiration or decomposition of dead plants and animals. The combustion of fossil fuels also recycles carbon.

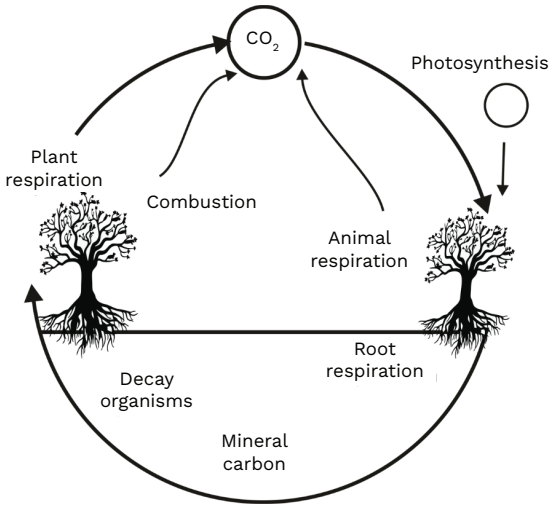


Fig. 1.26

- Nitrogen is frequently released into the atmosphere by denitrifying bacteria and restored to the cycle by lightening and electrification.

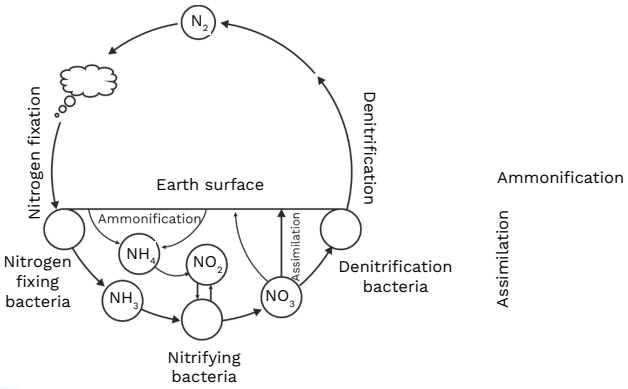


Fig. 1.27

Solved Examples

1. Which of the following adds/add carbon dioxide to the carbon cycle on the planet Earth? **[UPSC 2014]**

I. Volcanic action
II. Respiration
III. Photosynthesis
IV. Decay of organic matter

Select the correct answer using the codes given below.

(A) I and III only
(B) II only
(C) I, II, and IV only
(D) I, II, III, and IV

Solution: (C)

Nitrogen Cycle

- Nitrogen exists in its elemental form in the atmosphere and is made accessible to living creatures by combining this elemental form of nitrogen with hydrogen, carbon, and oxygen with help of nitrogen-fixing bacteria.
- Nitrogen-fixing bacteria: some bacteria can turn nitrogen into ammonia by the process known as nitrogen fixation. For example, azotobacter, achaea, etc.
- Denitrifying Bacteria: these bacteria metabolise nitrogenous compounds using various enzymes turning nitrogen oxides back to nitrogen gas or nitrous oxide.

Oxygen Cycle

- The oxygen cycle is the cycle that helps move oxygen through the three main regions of the Earth, the atmosphere, the biosphere, and the lithosphere. The atmosphere is the largest reservoir of free oxygen on the Earth. The biosphere also has some free oxygen produced from photosynthesis and other life processes.
- The largest reservoir of oxygen is the lithosphere. Most of this oxygen is not on its own or free moving but part of chemical compounds such as silicates and oxides.

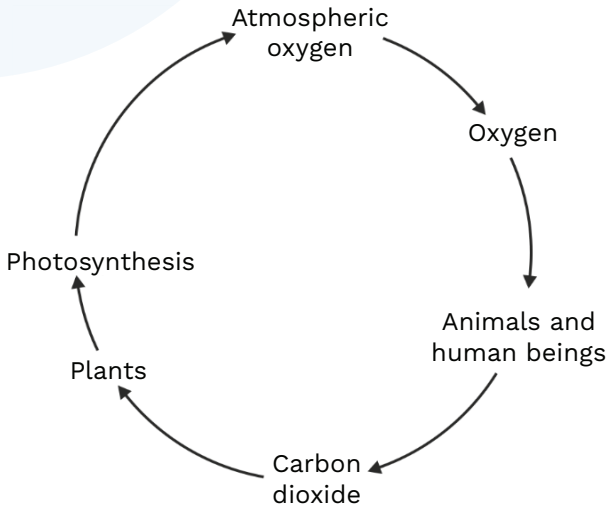


Fig. 1.28

Solved Examples

1. Consider the following statements.

I. Agricultural soils release nitrogen oxides into environment.

II. Cattle release ammonia into environment.

III. Poultry industry releases reactive nitrogen compounds into the environment.

Which of the statements given above is/are correct? **[UPSC 2019]**

(A) I and III only

(B) II and III only

(C) II only

(D) I, II, and III

Solution: (D)

Phosphorus Cycle

- The phosphorus cycle is initiated by the removal of phosphates from rocks due to weathering and they are dispersed across soil and water through the rain. The phosphate ions in soil are absorbed by plants. These phosphates are transferred from plants to animals through consumption.
- The phosphates consumed by animals and plants return to the soil via urine and faeces excretion and decomposition of dead plants and animals.
- Phosphorus is not very soluble in water and bonds strongly to molecules in the soil and reaches water by traveling with runoff soil particles. Phosphates also reach and settle on the floors of water bodies by fertiliser runoff, sewage leakages, and natural ores. These sediments are mixed up and phosphates re-enter the phosphorus cycle. Phosphates absorbed by aquatic plants are subsequently transported up the aquatic food chain through successive levels.

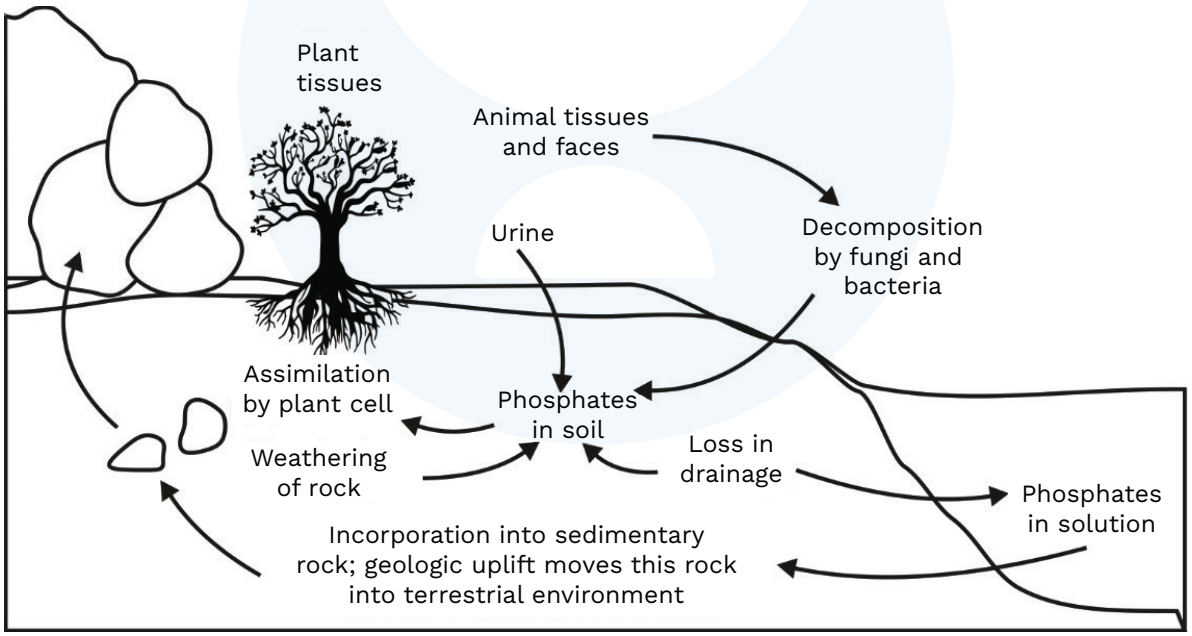


Fig. 1.29

Sulphur Cycle

- Sulphur is a component of amino acids, proteins, soil, etc.
- Various microbes present in the soil break down sulphur-containing proteins into their constituent amino acids. Sulphur from amino acids is turned into hydrogen sulphide (H_2S) by the second group of soil microbes. Sulphur bacteria convert H_2S to Sulphur and then to sulphate in the presence of oxygen. Ultimately, sulphate becomes H_2S.

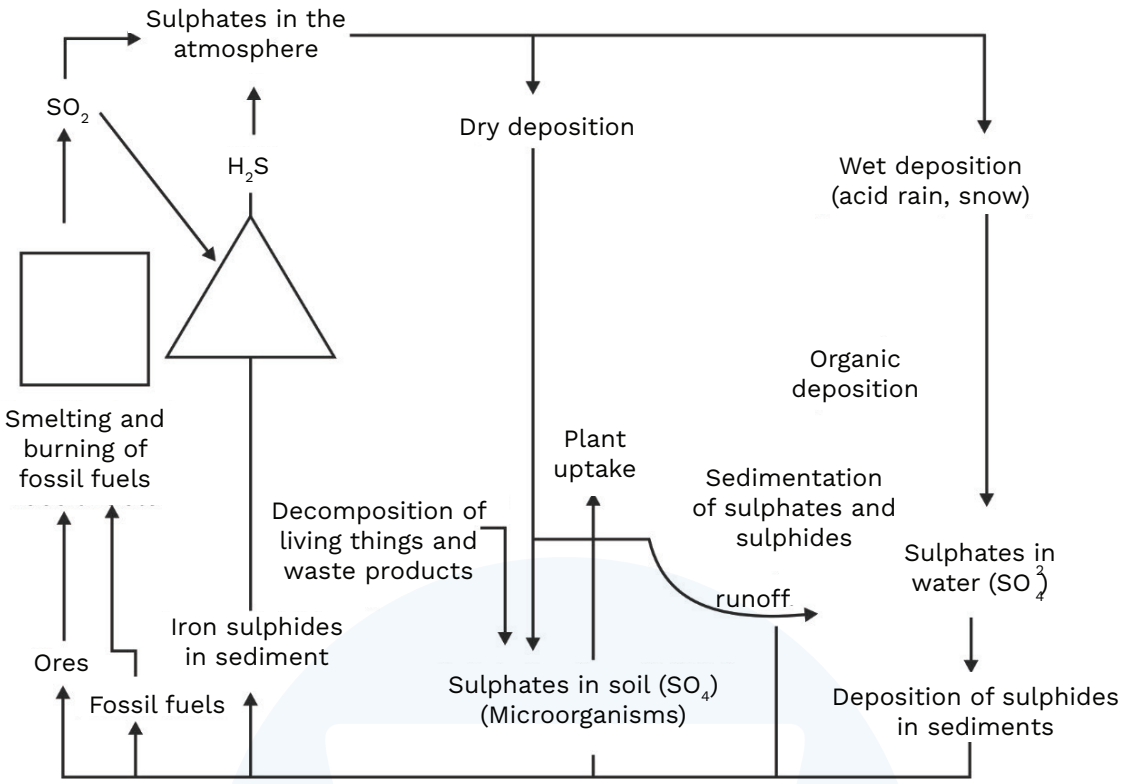


Fig. 1.30

Practice Exercise

1. Which of the following are the reasons for declining coral reefs around the world?

I. Rise in sea temperature
II. Ocean acidification
III. Overfishing
IV. Pollution from land

Select the correct answer using the codes given below.

(A) I, II, and III only
(B) II, III, and IV only
(C) I, II, and IV only
(D) All of the above

2. Consider the following statements regarding Wetlands International. **[UPSC 2019]**

I. It is a party in the implementation of the Ramsar convention.
II. It is an intergovernmental body that strives to protect wetlands and preserve their richness.
III. Wetlands International's work ranges from research and community-based field projects to advocacy and engagement with governments, and corporate and international conventions.

Select the correct answer using the codes given below.

(A) I and III only
(B) I and II only
(C) II and III only
(D) All of the above

3. Consider the following statements.

I. Biome is a large ecological area on Earth's surface in which flora and fauna adapts themselves in a particular way in a given environment.
II. Cold desert, savanna grasslands, hot desert, and temperate grasslands are some examples of terrestrial biome.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

4. With reference to wetlands, consider the following statements.

I. The Convention on Wetlands, signed in Ramsar, Iran, in 1971, is an intergovernmental treaty that provides the framework for national cooperation for the conservation and wise use of wetlands and their resources.

II. Second February of each year is observed as World Wetlands Day (WWD).

III. Indo-Gangetic wetlands are the largest wetland system in India, extending from the river Indus in the west to the Brahmaputra in the east.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

5. Ecological efficiency of an ecosystem refers to:

(A) Percentage of energy transferred from one trophic level to another.
(B) Percentage of mass transferred from one trophic level to another.
(C) Percentage of energy input to energy output ratio.
(D) Percentage of productivity input to productivity output.

6. Which portion of the constitution has components of environmental conservation explicitly?

I. Preamble
II. Fundamental rights
III. Directive principles of state policy
IV. Fundamental duties

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) III and IV only
(D) I, III, and IV only

7. Consider the following statements regarding mangroves.

I. Mangroves can tolerate excessive salt content and regular water-logging condition.

II. Mangrove forests stabilise the coastline and reduce erosion from storms, waves, tides, etc.

III. Sundari tree is an example of mangroves.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

8. Consider the following statements.

I. Aphotic zone is the upper layer of the aquatic ecosystems, up to which light penetrates and within which photosynthetic activity is confined.

II. With the increase in the temperature of a water body, the rate at which oxygen is depleted from water also increases.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

9. Consider the following scenarios.

I. Trees at trophic level one as primary producers.

II. Parasites at trophic level two as primary consumers.

III. Hyperparasites at trophic level three as secondary consumers.

The resultant pyramid of numbers would be:

(A) Erected
(B) Inverted
(C) Neither erected nor inverted
(D) Can be erected or inverted depending on other factors

10. The term 'biomagnification' is referred to as:

(A) Increase in body weight
(B) Uncontrolled growth of harmful organisms
(C) Accumulation of the increasing amount of non-degradable pollutants through the food chain
(D) Increase in the number of bacteria in the culture medium

11. Consider the following statements regarding ecological niche. **[UPSC 2015]**

I. Ecological niche is a description of all the biological, physical, and chemical factors that a species needs to survive, stay healthy, and reproduce.

II. Two species may or may not have an identical niche.

III. It plays an important role in the conservation of organisms.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

12. The Greek word 'oikos' means:

(A) plant, animals, and microbes
(B) man and animal
(C) habitation
(D) None of the above

13. Consider the following statements.

I. Sundarbans mangrove forest cover is fast depleting.

II. Sundarbans mangrove forest is located in Karnataka.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

14. If a wetland of international importance is brought under the 'Montreux Record', what does it imply? **[UPSC 2014]**

(A) Changes in ecological character have occurred, are occurring, or are likely to occur in the wetland as a result of human interference.
(B) The country in which the wetland is located should enact a law to prohibit any human activity within 5 km from the edge of the wetland.
(C) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity, and therefore, the cultural diversity therein should not be destroyed.
(D) It is given the status of 'World Heritage Site.'

15. Abiotic components include:

(A) Inorganic and non-living parts of the world.
(B) Chemicals like O_2, N_2, etc.
(C) Physical processes like volcanoes, floods, and forest fires.
(D) All of the above

16. Inverted ecological pyramid based on numbers of species is found in:

(A) Grassland ecosystem
(B) Pond ecosystem
(C) Parasitic food chain
(D) All of the above

17. Consider the following statements regarding coral reefs.

I. Coral reefs are diverse underwater ecosystems held together by $CaCO_3$ structures secreted by corals.

II. These are found around the world in tropical and subtropical oceans.

III. Coral reefs are built from story corals, which in turn consist of polyps that cluster in groups.

Which of the above statements is/are correct?

(A) I and II only
(B) II and III only
(C) III only
(D) I, II, and III

18. Which one of the following is not an abiotic component of the environment?

(A) Radiation
(B) Geologic substratum
(C) Symbionts
(D) Gravity

19. In the lake ecosystem, a pyramid of biomass is:

(A) Upright
(B) Inverted
(C) Anything is possible
(D) None of the above

20. Which one of the following is a top carnivore of the grassland food chain?

(A) Snake
(B) Deer

(C) Vulture
(D) Frog

21. Consider the following statements regarding productivity.
I. Primary productivity of the ecosystem is the rate at which plants and other photosynthetic organisms produce organic compounds.
II. Secondary productivity is the rate at which energy is stored at different levels in consumers.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

22. Which of the following is nature's cleaner?
(A) Producers
(B) Consumer
(C) Decomposers
(D) Humans

23. The succession as a result of changes in communities in the ecosystem is called:
(A) Biotic succession
(B) Climatic succession
(C) Geographical succession
(D) Physiographic succession

24. Pyramid of energy in a pond ecosystem is always:
(A) Inverted
(B) Upright
(C) Irregular
(D) Linear

25. Xerosere succession is related to:
(A) Land
(B) Rocks
(C) Pond
(D) Desert

26. A food web consists of:
(A) A portion of a food chain
(B) Interlocking food chain
(C) Producers, consumers, and decomposers
(D) A set of similar consumers

27. A trophic level refers to:
(A) A portion of a food chain.
(B) An organism's position in an ecosystem.
(C) An organism's position in a food chain.
(D) An organism's position in a biome.

28. A primary succession on bare rock starts with:
(A) Lichens
(B) Herbs
(C) Trees
(D) Animals

29. In terms of a food chain or food pyramid, organisms that trap energy, such as plants are known as:
(A) Primary consumers
(B) Secondary consumers
(C) Tertiary consumers
(D) Primary producers

30. Microbes are a:
(A) Biotic factor
(B) Abiotic factor
(C) Edaphic factor
(D) None of them

31. Which of the following is known as rainforests of the ocean?
(A) Seagrass
(B) Oysters
(C) Coral reefs
(D) Kelp

32. International Day for Biological Diversity is celebrated on:
(A) 12 May
(B) 22 May
(C) 23 May
(D) 24 May

33. Consider the following statements regarding primary productivity.
I. Primary productivity of the ocean is more than that of the land.
II. Net primary productivity is gross primary productivity minus respiration losses.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

34. Wetland sites are also known as:
(A) Rem sites
(B) Ramsar sites
(C) Ramsor sites
(D) Ramsey sites

35. Consider the following statements.
I. Coral reefs are diverse underwater ecosystems held together by CaCO3 structures secreted by corals.
II. Coral reef works as a sink for CO_2.
III. Global warming is one of the causes of coral bleaching.
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

36. Consider the following statements regarding the ecological pyramid.
I. The length of each horizontal bar of the ecological pyramid represents the total number of individuals at each trophic level in an ecosystem.
II. In the pyramid of biomass, the organisms in each trophic level are weighted instead of being counted.
III. The pyramid of energy is always inverted, with a large energy base at the top.
Which of the above statements is/are correct?
(A) I only
(B) I and II only
(C) II and III only
(D) I, II, and III

37. Consider the following statements regarding coral reefs.
I. Corals have a symbiotic relationship with zooxanthellae microscopic algae that live on coral.
II. Corals are the largest biogenic calcium carbonate provider.
III. Coral bleaching can occur if the concentration of photosynthesis pigments within the zooxanthellae falls.
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

38. The transition area between two biomes is known as:
(A) Ecotone
(B) Ecological niche
(C) Biosphere
(D) Succession

39. The environment which has been modified by human activities is called:
(A) Natural environment
(B) Anthropogenic environment
(C) Urban environment
(D) Modern environment

40. Chipko movement in Karnataka is known as:
(A) Appiko movement
(B) Sipko movement
(C) Assiko movement
(D) None of these

41. Amrita Devi is known for:
(A) Water conservation
(B) Soil conservation
(C) Wildlife conservation
(D) Tree conservation

42. Food levels of an ecosystem are called:
(A) Consumer levels
(B) Producer levels
(C) Trophic levels
(D) None of the above

43. Ecological footprint is:
(A) Measure of human impact on the ecosystem of earth.
(B) Measured in the amount of natural capital.
(C) Measured in area of wilderness and refers to the demand of resources by human population from nature to survive.
(D) All of the above

44. Consider the following statements.
I. An ecosystem is a biotic community together with its physical environment considered as an integrated unit.

II. All the living beings in a given spatial unit are biotic elements.

Which of the statement given above is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

45. Match the following.

List I

A. Herbivores
B. Carnivores
C. Omnivores
D. Detritivores

List II

I. Obtain their food only from animals
II. Obtain their food from both plants and animals
III. Obtain their nutrients by consuming detritus
IV. Obtain their food only from living plants

	A	B	C	D
(A)	I	IV	II	III
(B)	IV	I	II	III
(C)	IV	I	III	II
(D)	I	IV	III	II

46. Consider the following statements.

I. Autotrophs are those organisms that obtain their food by eating plants, other animals or their products.
II. Heterotrophs are those plants that synthesise their food themselves.

Which of the statements given above is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

47. Which one of the following terms describes not only the physical space occupied by an organism but also its functional role in the community of organisms?

(A) Ecotone
(B) Ecological niche
(C) Habitat
(D) Home range

48. Which of the following means progressing replacement of one vegetation community in a given habitat by the other?

(A) Ecological pyramid
(B) Ecological succession
(C) Ecological niche
(D) None of the above

49. Consider the following statements regarding ecocline.

I. Ecocline describes an ecotone in which a series of bio-communities display a continuous gradient.
II. Ecocline refers to variation in the physio-chemical environment of ecotone.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

50. Match the following.

List I

A. Producer
B. Primary consumer
C. Secondary consumer
D. Decomposers

List II

I. Bacteria
II. Phytoplankton
III. Herbivore
IV. Carnivore

	A	B	C	D
(A)	III	II	I	IV
(B)	III	II	IV	I
(C)	II	III	I	IV
(D)	II	III	IV	I

51. Consider the following statements.

I. Mangrove swamps are coastal wetlands in the tropics containing trees and shrubs.
II. Mangrove roots provide a habitat for oysters, crabs and other marine organisms.

Which of the statements given above is/are correct?

(A) I only
(B) II only

(C) Both I and II
(D) Neither I nor II

52. Consider the following statements.
I. Coral reefs usually do not form in the areas where there is a significant intrusion of fresh water and cold water.
II. Ocean acidification increases the calcifying ability of coral reefs.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

53. Consider the following statements regarding coral reefs.
I. Coral reefs are found all around the world in tropical and subtropical oceans.
II. Coral reefs are found at depths of less than 150 feet, but some corals can extend up to about 450 feet
III. Survival conditions for coral reef are temperature 20°C–21°C, salinity 27%–30%, depth (0–60), and zone 30°N–30°S.
Which of the above statements is/are correct?
(A) I only
(B) III only
(C) II and III only
(D) All of the above

54. Which of the following statements is not true about coral reefs?
(A) Coral reefs are increasing in areas everywhere.
(B) Corals are formed by polyps.
(C) Corals are the nurseries for hundreds of marine organisms.
(D) Coral bleaching is the biggest threat faced by reefs.

55. With reference to the distinction between lakes and wetlands, consider the following statements.
I. Wetlands are shallow water bodies whereas lakes can be deep or shallow.
II. The vertical mixing in the lake is thermally regulated whereas in wetlands it is wind regulated.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

56. Consider the following statements.
I. Pyramid of energy is always upright irrespective of the type of ecosystem.
II. Pyramid of numbers for both grasslands and forest ecosystems are similar.
Select the correct answer using the codes given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

57. Consider the following statements regarding niche.
I. It is the functional role played by a species in an ecosystem.
II. Two species can have identical niches.
Which of the statements given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

58. Consider the following statements regarding ecotone.
I. It is a transition area between two biomes where two communities meet and integrate.
II. It may be narrow or wide, local zone (between field and forest), or regional zone (between forest and grassland).
III. Ecotone may be a steep or gradual blending of communities.
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

59. Consider the following statements regarding ecological pyramid.

I. Ecological pyramid based on numbers of individual species is inverted in the marine food chain.

II. Ecological pyramid based on numbers of individual species is inverted in the parasitic food chain.

III. Ecological pyramid based on the weight of biota, is inverted in the marine food chain.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

60. Consider the following statements.

I. Bioremediation is the phenomenon in which living organisms are used for solving any degradation in the environment.

II. Leguminous crops like oilseeds, pulses, etc., are used to replenish the N_2 content of the soil, which is exhausted due to monocropping.

III. The rhizobium bacteria in roots of leguminous plants refixes N_2 in soil.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

61. Which one of the following is the best description of the term 'ecosystem'?

[UPSC 2015]

(A) A community of organisms interacting with one another.
(B) That part of the Earth is inhabited by living organisms.
(C) A community of organisms together with the environment in which they live.
(D) The flora and fauna of a geographical area.

62. If a tropical rainforest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because:

[UPSC 2011]

(A) The soil of rain forest is deficient in nutrients.
(B) Propagules of the trees in a rainforest have poor viability.
(C) The rainforest species are slow-growing.
(D) Exotic species invade the fertile soil of rainforest.

63. Which of the following statements is/are correct?

(A) International Day for Biodiversity is sanctioned by United Nations and observed on 22 May.
(B) World Wetland Day occurs on the day of the signing of the convention on wetlands, called Ramsar Convention, on 2 February, every year.
(C) International Day for Preservation of Ozone layer is observed on 16 September, the date of the signing of the Montreal Protocol on Substances that deplete the ozone layer.
(D) All of the above

64. Which of the following statements is/are correct?

(A) The concept of the food chain and food web was given by Charles Elton.
(B) Complex Food Web improves adaptability, competitiveness of organism and increases the stability of ecosystem.
(C) In a food web, species are connected by a line/arrow, which is known as a link, and species are referred to as a node.
(D) All of the above

65. Consider the following statements regarding Earth Hour.

I. It is an initiative by Worldwide Fund for Nature (WWF).

II. It is a movement to raise awareness about climate change and the need to save the Earth.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

66. Consider the following statements regarding biomagnification.

I. It is the phenomenon of increasing concentration of toxic chemicals in the tissues of an organism through the food chain.

II. The concentration of toxic compounds increases with decreasing the tropic level.

III. This process is also known as bio amplification.

Which of the above statements is/are correct?

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III only

67. Which of the following add/adds nitrogen to the soil? **[UPSC 2013]**

I. Excretion of urea by animals
II. Burning of coal by man
III. Death of vegetation

Select the correct answer using the codes given below.

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

68. Which of the following is/are unique characteristic(s) of equatorial forests? **[UPSC 2013]**

I. Presence of tall, closely set trees with crowns forming a continuous canopy.
II. Coexistence of a large number of species.
III. Presence of numerous varieties of epiphytes.

Select the correct answer using the codes given below.

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

69. Eutrophication is a process in which a water body becomes rich in nutrients as well as deprived of oxygen.

Which of the following nutrients primarily cause it?

I. Nitrates
II. Sulphates
III. Phosphates

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

70. Which one of the following regions of India has a combination of mangrove forest, evergreen forest, and deciduous forest? **[UPSC 2015]**

(A) North coastal Andhra Pradesh
(B) Southwest Bengal
(C) Southern Saurashtra
(D) Andaman and Nicobar islands

71. If you travel through the Himalayas, you are likely to see which of the following plants naturally growing there? **[UPSC 2014]**

I. Oak
II. Rhododendron
III. Sandalwood

Select the correct answer using the codes given below.

(A) I and II only
(B) III only
(C) I and III only
(D) I, II, and III

72. Which of the following ecological process occurs when a series of communities replace one another due to large-scale destruction?

(A) Succession
(B) Eutrophication
(C) Homeostasis
(D) Migration

73. Which of the following regions have coral reefs? **[UPSC 2014]**

I. Andaman and Nicobar Islands
II. Gulf of Kutch
III. Gulf of Mannar
IV. Sunderbans

Select the correct answer using the codes given below.

(A) I, II, and III only
(B) II and IV only

(C) I and III only
(D) I, II, III, and IV

74. Consider the following statements regarding ecotone. **[UPSC 2016]**

I. An ecotone is a zone of junction or a transition area between two biomes.
II. A well-developed ecotone contains some organisms, which are entirely different from that of the adjoining communities.

Which of the following statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

75. Consider the following statements.

I. Pelagic biome is part of the open sea or ocean that is not near the coast or seafloor.
II. Littoral biome is found in sea, lake, or river that is close to shore.
III. Maximum biodiversity found in neritic biome or continental shelf.

Which of the following statements is/are correct?
(A) I and II only
(B) III only
(C) II and III only
(D) All of the above

76. Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity? **[UPSC 2013]**
(A) Oceans, lakes, grasslands, mangroves
(B) Mangroves, oceans, grasslands, lakes
(C) Mangroves, grasslands, lakes, oceans
(D) Oceans, mangroves, lakes, grasslands

77. Which of the following can cause coral bleaching?

I. Change in ocean temperature
II. Ran-off polluted water
III. Over-exposure to sunlight
IV. Extreme low tides

Select the correct answer using the codes given below.
(A) I and II only
(B) III and IV only
(C) I, II, and III only
(D) I, II, III, and IV

78. Snowfall is a:
(A) Biotic factor
(B) Abiotic factor
(C) Both of them
(D) None of them

79. Epiphytes are plants that depend on other plants for which of the following purposes?
(A) Shade
(B) Water
(C) Food
(D) Mechanical support

80. Consider the following statements regarding autotrophs.

I. These are primary producers, i.e. green plants including certain bacteria and algae.
II. They synthesise carbohydrates from carbon dioxide and water.

Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

81. Which type of succession occurs when plants recognise an area in which the climax community has been disturbed?
(A) Primary succession
(B) Secondary succession
(C) Allogenic succession
(D) Autotrophic succession

82. Which of the statements is/are correct?
(A) Ecotone describes variation in the forms of species based on genetic differences.
(B) Ecocline describes variation in species of flora and fauna.
(C) In ecocline, based on technical criteria, it is easy to differentiate two species.
(D) All of the above

83. What is the correct order of size of the ecological niche of different consumers?
(A) Herbivores < carnivores < omnivores
(B) Herbivores < omnivores < carnivores
(C) Omnivores < carnivores < herbivores
(D) Omnivores < herbivores < carnivores

84. Which one of the following is not an abiotic component of the environment?
(A) Soil
(B) Fire
(C) Microorganisms in the soil
(D) Water

85. When an ecosystem is considered healthy?
(A) When an ecosystem is sustainable.
(B) All elements of the ecosystem live in balance.
(C) All elements of the ecosystem are capable of reproducing themselves.
(D) All of the above

86. Consider the following statements regarding autotrophs.
I. Autotrophs are primary producers i.e. green plants including certain bacteria and algae.
II. They synthesise carbohydrates from simple inorganic raw materials like carbon dioxide and water.
III. They use solar energy.
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

Answer Key

1. (D)	**2.** (A)	**3.** (C)	**4.** (D)	**5.** (A)	**6.** (C)	**7.** (D)	**8.** (B)	**9.** (B)	**10.** (C)
11. (C)	**12.** (C)	**13.** (A)	**14.** (A)	**15.** (D)	**16.** (C)	**17.** (D)	**18.** (C)	**19.** (B)	**20.** (C)
21. (C)	**22.** (C)	**23.** (A)	**24.** (B)	**25.** (D)	**26.** (B)	**27.** (C)	**28.** (A)	**29.** (D)	**30.** (A)
31. (C)	**32.** (B)	**33.** (B)	**34.** (B)	**35.** (D)	**36.** (B)	**37.** (D)	**38.** (A)	**39.** (B)	**40.** (A)
41. (D)	**42.** (C)	**43.** (D)	**44.** (C)	**45.** (B)	**46.** (D)	**47.** (B)	**48.** (D)	**49.** (C)	**50.** (D)
51. (C)	**52.** (A)	**53.** (D)	**54.** (A)	**55.** (C)	**56.** (A)	**57.** (A)	**58.** (D)	**59.** (B)	**60.** (D)
61. (C)	**62.** (A)	**63.** (D)	**64.** (D)	**65.** (C)	**66.** (C)	**67.** (C)	**68.** (D)	**69.** (C)	**70.** (D)
71. (A)	**72.** (A)	**73.** (A)	**74.** (C)	**75.** (D)	**76.** (C)	**77.** (D)	**78.** (B)	**79.** (D)	**80.** (C)
81. (B)	**82.** (C)	**83.** (A)	**84.** (C)	**85.** (D)	**86.** (D)				

2 Biodiversity and Its Conservation

Biodiversity

- The term 'biodiversity' is the abbreviation of 'biological diversity'. The term was first coined by RF Dasmann in 1968 and popularised by Edward Wilson.
- Biodiversity refers to the variety and abundance of living organisms living in a particular region. It is the variability within species, between species, and between ecosystems.
- The first global agreement to cover all aspects of biological diversity was the result of Earth Summit held in Rio de Janeiro, Brazil in 1992. The agreement is called Convention on Biological Diversity (CBD).
- The 1992 Earth Summit in Rio de Janeiro defined biodiversity as, "The variability among living organisms from all sources, including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part: this includes diversity within species, between species and of ecosystems".
- In short, it refers to the variety and variability of all plants, animals, and microorganisms found on the Earth.
- But, defining biodiversity or measuring it is not so simple. Living organisms range in size from microscopic microbes to large mammals. According to the most widely used classification system, these are divided into five major kingdoms: Monera, Protista, Fungi, Plantae, and Animalia, which are further subdivided into numerous categories.
- Biodiversity is not evenly distributed; it varies greatly across the globe as well as within regions.
- The diversity of all living things (biota) depends on temperature, precipitation, altitude, soils, geography, and the presence of other species.
- Terrestrial biodiversity is greater near the equator and lowers in the polar regions, which seems to be the result of the warm climate and high primary productivity.
- Terrestrial biodiversity declines from the equator to the poles, this characteristic is unverified in aquatic ecosystems, especially in marine ecosystems.

Levels of Biodiversity

Biodiversity is considered to exist on the following three levels.

Species Diversity

- Species diversity is determined not only by the number of species within a biological community i.e., species richness but also by the relative abundance of individuals in that community.
- Species abundance is the number of individuals per species, whereas relative abundance is the distribution of individuals among species in a community homogenously. Two communities may have the same number of species, but their relative abundance may differ.
- Species diversity can be measured in the following terms.
  - **Species richness:** It is a simple count of species. Relative abundance is the measure of the relative proportion of different species occurring in a community.
  - **Species evenness:** species evenness quantifies how equally the species are distributed.
  - **Taxonomic diversity:** refers to the genetic relationships between different groups of species.

Ecosystem Diversity

Ecosystem diversity refers to the biodiversity of an ecosystem. It focuses on the diversity in ecosystems within a geographical area and their overall impact on human existence and the environment, particularly under challenging conditions.

Genetic Diversity

Genetic diversity mentions a wide range of genes present within plant, animal, and microbial species. The level of diversity grows as size and environmental conditions are increased.

Terms in Biodiversity

- **Intraspecific diversity** is the variation within a population from the same species while interspecific diversity occurs in a community made up of different species.
- **Population diversity** may be measured in terms of the variation in genetic and morphological features that define the different populations.
- **Phenotype** refers to all the observable characteristics of an organism that result from the interaction of its genotype (total genetic inheritance) with the environment.
- **Functional diversity** is a task of functionally disparate (different in kind) species within a population. For example, different feeding mechanisms, different motility, predator vs prey, etc.

Measurement of Biodiversity

The two main factors considered when measuring diversity are richness and evenness. Species richness can be divided into three types.

α-Diversity

Alpha diversity is also known as local diversity. It refers to the diversity within a specific region, community, or ecosystem and is evaluated by the number of taxa (generally species) within the ecosystem.

β-Diversity

Beta diversity represents species diversity between ecosystems; it compares the number of taxa unique to each ecosystem.

γ-Diversity

Gamma diversity is a measurement of the overall diversity of different ecosystems within a region. For example, the Biodiversity of a continent with its seas, oceans, lakes, rivers, mountains, plains, forests, etc.

Solved Examples

1. Which of the following can be threats to the biodiversity of a geographical area?

[UPSC 2012]

I. Global warming
II. Fragmentation of habitat
III. Invasion of alien species
IV. Promotion of vegetarianism

Select the correct answer using the codes given below.

(A) I, II, and III only
(B) II and III only
(C) I and IV only
(D) I, II, III and IV

Solution: (A)

Species Diversity and Ecosystem Stability

A stable community shall exhibit minimal variation in productivity from year to year. it must be resistant or resilient to occasional disruptions (natural or anthropogenic), and resistant to invasions by alien species. David Tilman's long-term ecosystem research utilising outdoor plots showed that plots with greater species diversity had less change in total biomass from one year to the next. Additionally, he demonstrated that an increase in biodiversity contributed to a rise in productivity.

Rivet Popper Hypothesis

- The rivet popper hypothesis was given by ecologist Paul Elrich to illustrate the significance of species diversity in preserving an ecosystem's health.
- Each component of an aeroplane (ecosystem) is secured with hundreds of rivets (species). If every passenger in a plane removes a rivet to take home (causing the extinction of a species), the initial impact on flight safety (ecological health) may be minimal, but as more and more rivets are removed over time, the plane becomes dangerously fragile.

- The loss of rivets on the wings (important species that drive significant ecosystem functions) poses a far greater risk to flight safety than losing a few rivets on the seats or windows inside the aircraft.

Simpson's Index

The Simpson's reciprocal index can be used to quantify the relative biodiversity of a community. It considers the number of species (diversity) and the number of individuals per species (evenness). A bigger index value denotes a greater level of biodiversity within the community.

Shanon-Wiener Index

- The Shanon–Wiener Index is utilised to characterise the disorder and unpredictability of individual species. The greater the degree of unpredictability, the greater the diversity.
- The Shannon-Weiner diversity index number of species includes two factors: species richness and the average or evenness of individual distribution.
- A high species count can boost biodiversity. Increasing the regularity of the distribution of individuals among species will similarly boost biodiversity.
- The diversity index is greatest when each individual belongs to a different species. If every individual in a population belongs to the same species, its diversity index is the lowest.

Solved Examples

1. Biodiversity forms the basis for human existence in the following ways.

[UPSC 2011]

I. Soil formation
II. Prevention of soil erosion
III. Recycling of waste
IV. Pollination of crops

Select the correct answer using the codes given below.

(A) I, II, and III only
(B) II, III, and IV only
(C) I and IV only
(D) I, II, III, and IV

Solution: (D)

Importance of Biodiversity

- Humans depend for their sustenance, health, well-being, and cultural growth on nature.
- Biodiversity also has enormous social and cultural importance.
- The various benefits of biological diversity can be grouped under three categories:
  - Ecosystem services
  - Biological resources
  - Social benefits
- **Ecosystem services:** living organisms provide many ecological services free of cost that are responsible for maintaining ecosystem health such as soil protection, protection of water resources, pollution reduction, nutrient storage and cycling, and climate stability.
- **Biological services:** various biological services of economic importance are provided such as food, fibre, medicines, fuelwood, and ornamental plants.
- **Social benefits:** the natural environment serves the inspirational, aesthetic, spiritual, and educational needs of the people and all cultures. In a majority of Indian places, plants such as tulsi, peepal, etc., are planted and considered sacred and worshipped. Biosphere reserves, national parks, wildlife sanctuaries, gardens, etc., have high entertainment and recreation value. Activities such as ecotourism, film-making, and literary activities are closely related.

Biogeographic Zones of India and Their Biodiversity

Biogeographic zones depict the principal species classifications. Each zone represents a unique combination of physical, climatic, and historical characteristics. The Himalayas and the Gangetic Plains are two adjacent yet radically distinct regions. There are ten separate biogeographic zones in India.

Trans Himalayan Zone

This zone encompasses Ladakh and Lahul–Spiti, with an altitude between 4500–6000 m at mean sea level, and has a total area of approximately 186200 km^2. Due to its extreme weather conditions and unfriendly topography, the environment in this region is extremely fragile.

Flora	Fauna
• Ladakh and Lahul–Spiti are primarily covered by a patchy alpine steppe and with many endemic species.	• Wild yaks, Tibetan asses, gazelles, antelopes, and ibexes form a separate grazing population on the plateau's flatter terrain. • Snow leopard, Indian wolf, fox, and smaller animals, such as marbled pole cat and marmot, are the carnivores. • The lakes and marshes are home to unique birds such as Black-necked Crane, a migratory bird.

Table 2.1

The Himalayas

The Indian Himalayan mountain ranges span more than 2000 km from east to west. It is one of India's most diverse biogeographic zones in terms of habitat and species diversity.

Flora	Fauna
• Mixed deciduous community merging into chir pine and ban oak in the lower sub-tropical foothills. • Forests of maple, walnut, oak, and a variety of conifers, such as blue pine, fir, and spruce, in temperate areas. • Scrub vegetation of birch and rhododendrons interspersed with grasslands in the sub-alpine region. • Blue pine and deodars in the drier western region.	• Ibex, markhor, and hangul are found in the central part of the Himalayas. • Mishmi takin, snow leopard, Indian rhinoceros, and clouded leopards are found in the eastern part. The Kashmir stage or hangul is found in the western part.

Table 2.2

The Desert

This region is also known as the Thar desert. It encompasses western Gujarat and western Rajasthan. Parts of Punjab and Haryana were formerly a part of this desert, but its irrigation infrastructure has transformed. Desert flora and fauna are unique not because of their abundance or diversity, but because of their amazing ecological adaptations to desert environments.

Flora	Fauna
• Tropical thorn woods and tropical dry deciduous forests, sand deserts with seasonal salt marshes, and mangroves in the principal estuaries are the natural vegetation. • Phog is a typical shrub found on dunes. Sewan grass covers vast expanses known as pali.	• The Thar desert is home to the vast majority of insect species. There are 43 reptile species and moderate endemism among birds in this area. • Once the desert's main mammal, the blackbuck is now restricted to a few isolated enclaves. The Gazelle is the only species of Indian antelope with horned females. The wild ass, a unique subspecies, is currently restricted to the Rann of Kutch, which is also the only nesting habitat for flamingos on the Indian subcontinent. There are also other species, such as the desert fox, great Indian bustard, chinkara, and desert cat. • Only the Thar region is home to the desert fox, desert cat, great Indian bustard, and a few species of sand grouse.

Table 2.3

The Semi-Arid

- This region comprises 15 percent of India's entire area, about 508000 km^2.
- This zone is the favourite place of a large number of wildlife due to the presence of various types of grass and palatable shrubs.

Flora	Fauna
• Tropical thorn woods and tropical dry deciduous forests, and mangroves are the natural vegetation. • The plant species present in this region are of African origin. Acacia and prosopis trees grow sparsely on the sandy plains. On the gravelly plains, calotropis, gymnosporia, and other plants can be found. • Euphorbia bushes cover the rocky areas, while salvadora and tamarix species are primarily found around salty depressions.	• Large herbivores, such as blackbuck, chowsingha, gazelle, and nilgai, make up the fauna. This region's endemic species is the lion of Gir.

Table 2.4

The Western Ghats (Sahyadri)

- The Western Ghats, also known as Sahyadri, is a mountain range that runs parallel to the western coast of the Indian Peninsula.
- It is a UNESCO World Heritage Site and is one of the eight 'hottest hot spots of biological diversity in the world. It is sometimes called 'the Great Escarpment of India.'
- Its range starts from south of the Tapti River and runs approximately 1600 km through the states of Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu.
- At least 325 globally threatened species are found in the Western Ghats. Since the area is ecologically sensitive to development that is the reason the Government of India established many protected areas such as biosphere reserves, national parks, wildlife sanctuaries, etc.
- Union Government had set up two ecology expert panels headed by Madhav Gadgil and Kasturirangan, respectively.
- Gadgil Committee was set up in March 2010 and it submitted its report to Government in August 2011. The committee suggested declaring the entire landscape as an ecological sensitive zone (E.S.Z.), i.e.,
  - E.S.Z.-I (60%) Comprises existing sanctuaries, such as national parks, and biosphere reserves.
  - E.S.Z.-II (15%) Only existing mines. No new mine is allowed.
  - E.S.Z.-III (25%) Developmental activities can be allowed with precautions. The new mining project can take place in it.
- Gadgil's report was criticised as too environmentalist. Therefore, a high-level working group (HLWG) headed by Kasturirangan was set up and gave its report in April 2013. He suggested 37% area, 60000 km^2, as ecologically sensitive.

Flora	Fauna
• Tropical evergreen forests, moist deciduous forests, dry deciduous forests, scrub jungles, sholas, savannas, and swamps are the various vegetation.	• The Ghats of Maharashtra, Mysuru, Malabar, and Tamil Nadu are home to distinct subspecies of the Ratufa indica squirrel. The grizzled squirrel is only found in two areas in Tamil Nadu's dry forests. • Carnivores comprise the Malabar civet in the southern evergreen forests and the rusty-spotted cat in the northern deciduous forests. • Malabar grey hornbill, nilgiri tahr in Nilgiris to Agastyamalai montane grassland.

Table 2.5

- However, in March 2014, the Ministry of Environment, Forest, and Climate Change issued a draft notification of Western Ghats and declared 56825 km^2 as an ecologically sensitive area (ESA).

The Deccan Peninsula

- This zone comprises approximately 1421000 km^2 around 43% of India's total land.
- Although a huge portion of this zone has been drastically affected by people, there are still forest regions in Madhya Pradesh, Maharashtra, and Odisha, in particular.

Flora	Fauna
Deciduous forests, thorn forests, and degraded shrublands. On the coastal side of the plains of Andhra Pradesh and Tamil Nadu, there are dry evergreen woods or thorn scrubs.	Chital, sambar, nilgai, barking deer, and gaur can be found across the zone. Some animals, such as the blackbuck, are restricted to dry, open regions. Elephant (Bihar-Odisha and Karnataka-Tamil Nadu) and the wild buffalo (in a small area at the junction of Odisha, MP, and Maharashtra).

Table 2.6

The Gangetic Plain

- This region has one of the world's most productive regions and supports a dense and expanding human population. It encompasses around 359400 km^2.
- Most of the area's native vegetation has been eradicated as a large amount of it has been converted to agricultural use.

Flora	Fauna
• Moist deciduous forests cover the alluvial plain formed by the lower Ganges and Brahmaputra rivers, the biggest river delta in the world. The ecoregion is today one of the most heavily populated places on Earth, and intensive agriculture has mostly replaced its forests.	• Nilgai, blackbuck, and chinkara currently exist in the western areas, interspersed with dense cultivation in small numbers. • The wetlands and rivers contain crocodile mugger, gharial, Gangetic dolphin, freshwater turtle.

Table 2.7

North-East India

North-east India is the intersection of the Indian, Indo-Malaysian, and Indo-Chinese areas as well as the Himalayan and peninsular Indian regions. Due to the vast ecological diversity and quantity of endemic species, it is a crucial zone on the Indian subcontinent.

Flora	Fauna
A unique natural vegetation, swamps, grasslands, and fringing woodlands, is observed in the Brahmaputra Valley region.	Wildlife such as the rhinoceros, buffalo, swamp deer, hog deer, pygmy hog, and hispid hare is found in the region. Moreover, this region is home to the world's greatest elephant population.

Table 2.8

The Islands

- The Andaman and Nicobar Islands in the Bay of Bengal consists of 348 islands that are aligned north to south, and the Lakshadweep Islands in the Arabian Sea.
- A high degree of endemism characterises the region's distinct plant and animal species. These islands have a deficient animal population. This could be largely attributable to the Andaman and Nicobar Islands' remoteness and tiny size.

Flora	Fauna
Andaman • Tropical rainforest canopy, moist deciduous forests, and wet evergreen forests type are common vegetation. • Grasslands are exclusive to the Nicobars, whereas deciduous forests are prevalent in the Andamans but nearly nonexistent in the Nicobars.	**Andaman** • Andaman wood pigeon, the state bird of Andaman, Narcondam hornbill, Andaman scop's owl, the blue-eared kingfisher, fulvous breasted woodpecker, wild boars, elephants, and saltwater crocodiles.
Lakshadweep • Banana, vazha, colocasia, chambu, breadfruit, chakka, and wild almond are widely found • The shrub jungle plants kanni, punna, chavok, and cheerani are irregularly distributed across the island.	**Lakshadweep** • Tharathasi and karifutu are two common oceanic birds found on one of the PITTI uninhabited islands of Lakshadweep. Molluscan forms and the hermit crab is the most common species. Colourful coral fish, such as parrotfish, butterflyfish, and surgeonfish, are abundantly found.

Table 2.9

The Coasts

The coastline of India is vast. In the west, the Arabian Sea washes Gujarat, Maharashtra, Goa, Karnataka, and Kerala coasts. On the east coast, the Bay of Bengal washes West Bengal Sunderbans, Odisha, Andhra Pradesh, and Tamil Nadu's coastlines. Along the southern beaches of Tamil Nadu, the Gulf of Manaar, and the Indian Ocean bathe the southern point of the Indian Peninsula.

Flora	Fauna
• Mangrove communities range from estuaries to lagoons, and deltas. • Sandy beaches, including elevated beaches and unique plant communities such as casuarina, calophyllum, and pandanus. • Mudflats with a spectrum of successional stages to entirely terrestrial vegetation. • Coral formations and rugged coasts.	• The dugong, the humpback dolphin, the Olive Ridley, green, hawksbill, leather, and loggerhead sea turtles; the Sunderbans' estuarine turtle Batagur Basker, and the softshell estuarine turtle. Mudskippers, semi-terrestrial spotted deer, and Sunderban tiger all live in the mangroves' higher regions.

Table 2.10

Biodiversity Hotspots

- Conservation International (CI), based in Crystal City, Arlington, Virginia, is an American non-profit environmental organisation which pioneered the definition of hotspots. It adopted the idea of protection of hotspots

as the guiding investment principle from Norman Myers's study of Hotspots.

- Hotspot regions have high species diversity, endemism rate, and threat. Myers found 18 hot spots or locations that include approximately 50000 indigenous plant species in only 746000 km^2 or 0.5% of the Earth's land surface.
- To qualify as a biodiversity hotspot, a region must meet two strict criteria.
  - It must have at least 1500 vascular plants as endemics, which is to say, it must have a high percentage of plant life found nowhere else on the planet. A hotspot, in other words, is irreplaceable.
  - It must have 30% or less of its original natural vegetation. In other words, it must be threatened.

Vascular vs Non-vascular Plants

- Vascular plants transport water through the xylem, and nutrients through the phloem to all the parts of the plant, while non-vascular plants lack a vascular system (xylem and phloem).
- Non-vascular plants are held in place by rhizoids, which are tiny hairs found in moist areas, that allow the plants to obtain sufficient water without roots, while vascular plants have an intricate root system.
- Non-vascular plants reproduce more easily than vascular plants. Non-vascular plants reproduce asexually by the production of single-celled spores or vegetative propagation, in which a new plant grows from a piece of the original plant.

Examples of vascular and non-vascular plants are as follows.

- Vascular plants: Clubmosses, horsetails, ferns, gymnosperms, and angiosperms (flowering plants).
- Non-vascular plants: Mosses, liverworts, and hornworts.

Biodiversity Hotspots in India

As per Conservation International, India has four biodiversity hotspots. Three are described here. Refer to page 45 for the fourth, Western Ghats.

Himalayas

- Indian Himalayan region, which includes Tibet, Nepal, Bhutan, China, and Myanmar, is severely degraded due to overgrazing of lowland and alpine habitats by domestic livestock (cattle, yak, etc). Tigers and rhinoceroses are poached for their body parts used in traditional Chinese medicine while snow leopards and red pandas for their exquisite furs.
- Tigers and rhinoceroses are poached for their body parts, which are used in traditional Chinese medicine while Snow leopards and red pandas are hunted for their exquisite furs.
- In total, 25% of the Himalayan region is still intact with original vegetation, which houses a variety of plant species, mammals, birds, amphibians, reptiles, and freshwater fishes.

Indo-Burma

- Indo-Burma region encompasses all of North-Eastern India, excluding Assam and the Andaman Islands.
- Due to excessive agriculture and deforestation, resources were exploited in this region, which became the most vulnerable biodiversity hotspot. Only 5% of natural habitats exist in a relatively pure state.
- This hotspot is home to freshwater turtle species, the majority of which are threatened with extinction due to over-exploitation and substantial habitat loss. The birdlife in this region is extraordinarily rich, which includes the threatened white-eared night-heron and the orange-necked partridge.

Sundalands

- Sundaland includes the Nicobar Islands and Indonesia, Malaysia, Singapore, Brunei, and the Philippines.
- The greatest danger to the biodiversity of Sundaland is rubber, oil palm plantations, and pulp production.
- The orangutan population, which is exclusive to this hotspot, is severely declining. The islands of Java and Sumatra feature some of the final refuges for two rhinoceros species in Southeast Asia.

Threat to Biodiversity

- In the name of growth and development, man has recklessly destroyed everything in his way, including the precious biota of nature.
- Almost every corner of the Earth is facing an ecological crisis brought forth by the excesses of humans. The rate of extinction of species is proof of this.

Major Causes of Threat

- Habitat loss and fragmentation
- Over-exploitation
- Alien species invasions
- Co-extinctions
- Poaching
- Human-wildlife conflicts
- Shifting cultivation, deforestation, grazing, forest fire, urbanisation, etc.

Species Extinction

- Extinction is the end of an organism or of a group of organisms meaning that the last individual of the species died though its capacity to breed and recover might have been lost earlier.
- There are three types of extinction of species
  - Natural extinction
  - Mass extinction
  - Anthropogenic extinction

Reasons for the Extinction

- Endemism and rarity go hand in hand as the rare species are concentrated in one geographic location alone.
- A small population of the species leads to inbreeding, which further reduces the genetic vibrancy of the species.
- When the habitat in which a species thrives is destroyed, the species also is threatened with extinction if it does not adapt to the changing conditions.
  - Restricted food source
  - Low reproductive potential

International Union for Conservation of Nature and Natural Resources (IUCN)

- This 'umbrella' organisation addresses the most critical environmental and development concerns confronting biodiversity and species.
- The IUCN is an international non-governmental organisation (NGO) that focuses on environmental conservation and the effective use of natural resources.
- IUCN, founded in 1948 in Gland, Switzerland, works on biodiversity, climate change, energy, human livelihoods, and promoting a sustainable global economy by promoting scientific research, managing field projects, and bringing together governments, NGOs, the United Nations, and corporations to develop policy, laws, and best practices.
- Red Data list is published by IUCN, which ranks the most endangered species.

IUCN Red List of Endangered Species

- The International Union for Conservation of Nature (IUCN) Red List of Threatened Species, often known as the IUCN Red List, is one of the most well-known objective systems started in 1994 for classifying the extinction risk of plants, animals, and other things. It includes explicit criteria and categories for classifying the conservation status of individual species based on their extinction risk.
- The IUCN system uses a set of five quantitative criteria to assess the extinction risk of a given species. In general, these criteria consider
  - The rate of population decline
  - The geographic range
  - Whether the species already possesses a small population size
  - Whether the species is very small or lives in a restricted area
  - Whether the results of a quantitative analysis indicate a high probability of extinction in the wild
- A species is categorised into one of many categories after a comprehensive investigation. (The specifics of each category have been reduced to highlight two or three key elements). Furthermore, three of the categories (CR, EN, and VU) fall under the umbrella word 'threatened'. The IUCN Red List of Threatened Species categorises species into several groups.
- **Extinct (EX)** is a designation applied to species in which the last of the species has died or

where systematic and time-appropriate surveys have been unable to log even a single member of the species.

- **Extinct in the wild (EW)** is a category containing those species whose members survive only in captivity or as artificially supported populations far outside their historical geographic range.
- **Critically endangered (CR)** is a category of those species that are at an extremely high risk of extinction as a result of rapid population decline of 80 to more than 90 per cent over the previous 10 years or three generations, a current population size of fewer than 50, or other factors.
- **Endangered (EN)** is a designation applied to species that possess a very high risk of extinction as a result of rapid population decline of 50 to more than 70 per cent over the previous 10 years or three generations, a current population size of fewer than 250, or other factors.
- **Vulnerable (VU)** is a category of those species that possess a very high risk of extinction as a result of rapid population decline of 30 to more than 50 per cent over the previous 10 years or three generations, a current population size of fewer than 1000, or other factors.
- **Near threatened (NT)**, is the designation given to species that are close to becoming threatened or may meet the criteria for threatened status in the near future.
- **Least concern (LC)**, is a category that has species that are pervasive and abundant after careful assessment.
- **Data deficient (DD)** is a category applied to those species of which the amount of available data related to their risk of extinction is lacking in some way. Consequently, a complete assessment cannot be performed. Thus, unlike the other categories in this list, this category does not describe the conservation status of a species.
- **Not evaluated (NE)**, includes any of the nearly 1.9 million species described by science but not assessed by the IUCN.
- All else being equal, a species experiencing a 90 per cent decline over 10 years (or three generations), for example, would be classified as critically endangered. Similarly, another species experiencing a 50% decline over the same time would be classified as endangered, while one experiencing a 30% decline would be classified as vulnerable. It is critical to understand, however, that a species cannot be classified solely on the basis of one criterion; the scientist assessing the species must consider all five criteria when determining the status of the species.

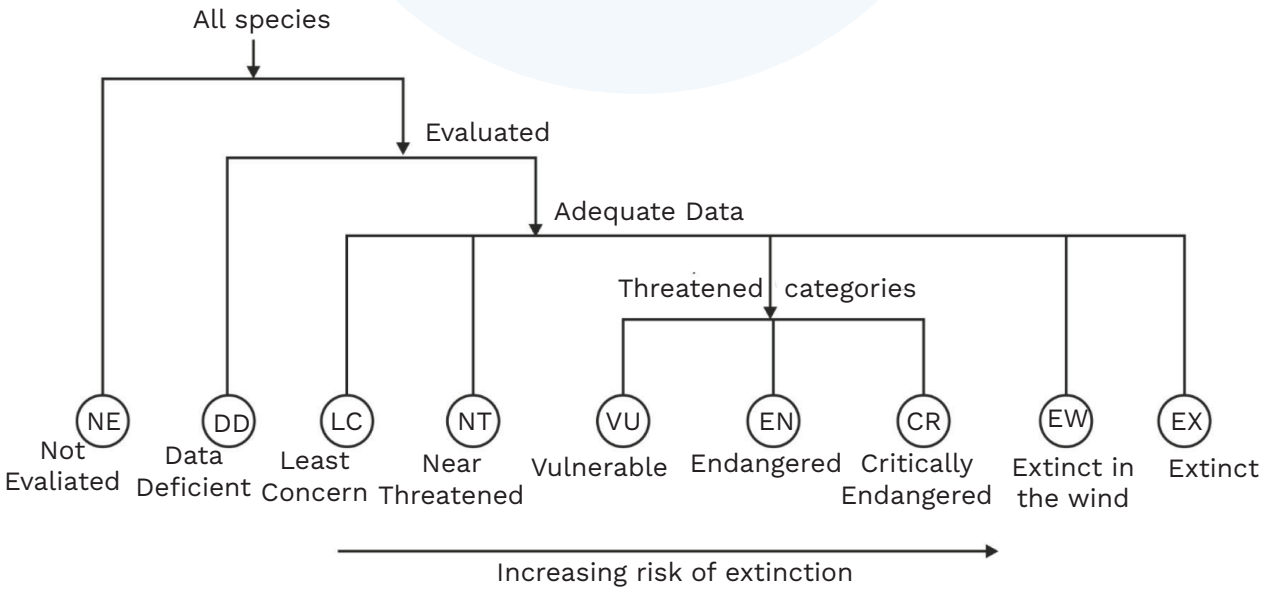


Fig. 2.1

Solved Examples

1. The Red Data Book published by the International Union for Conservation of Nature (IUCN) contains lists of: **[UPSC 2011]**

I. Endemic plant and animal species present in the biodiversity hotspots
II. Threatened plant and animal species
III. Protected sites for conservation of nature and natural resources in various countries

Select the correct answer using the codes given below.

(A) I and III only
(B) II only
(C) II and III only
(D) III only

Solution: (B)

Illegal Wildlife Trade

- According to the Worldwide Fund for Nature, 'illegal wildlife trade is projected to approach $20 billion annually, making it the fourth-largest unlawful activity in the world, after narcotics, human trafficking, and trade in counterfeit goods. Illegal wildlife trafficking not only poses a threat to the conservation of species but also to human health and a country's natural resources and local populations.'
- The trade and poaching demand a variety of products, including exotic pets and luxury items, traditional medicines, clothing, and jewellery made from mongoose hair, snake skins, rhinoceros's horns, tiger, leopard claws, etc.
- There are three main types of hunting.
  - Commercial hunting: in which the animals are killed for profit from the sale of their furs, bones, or other parts.
  - Subsistence hunting: the killing of animals to provide food for survival.
  - Sport hunting: the killing of animals for recreation.
- The porous international land borders of India are the primary cause of the continuous wildlife trafficking in India. The Gulf, China, and Southeast Asia are the biggest consumers.
- Constitutional guarantee, Article 51 (A) (g) imposes a constitutional obligation on every Indian to maintain and enhance the natural environment, including forests, lakes, rivers, and wildlife, and to have compassion for all living creatures.

International Organisations for Control of Wildlife Trafficking

- The Trade Records Analysis of Flora and Fauna in Commerce (TRAFFIC), founded in 1976, is a network for monitoring the wildlife trade and a collaborative effort of the WWF and the IUCN. It collaborates closely with the Central and State Governments and agencies which aid in research, monitoring and influencing action to prevent illicit wildlife trafficking and bring long-term sustainability to wildlife commerce.
- The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), the Washington Convention, provides the terms to regulate international wildlife trade. Under its leadership, countries throughout the world have taken steps to prohibit and control this illegal trade.
- The Monitoring the Illegal Killing of Elephants (MIKE) programme is an international collaboration that measures the levels, trends, and causes of elephant mortality, thereby providing an information base to support international decision-making related to the conservation of elephants in Asia and Africa. The MIKE programme was established by CITES by Resolution 10.10 adopted at the 10th Conference of the Parties in 1997.
- The International Fund for Animal Welfare (IFAW) is one of the world's major animal welfare and conservation organisation which rescues individual animals preserves habitats and advocates for increased protection.
- In 2015, the United Nations General Assembly voted overwhelmingly on a resolution

to combat illegal wildlife trafficking. The Sustainable Development Goals include explicit objectives for combating poaching and the trafficking of endangered species.

Convention on Biological Diversity (CBD)

- The convention on Biological Diversity is an international legally binding treaty. It was started in 1992 at United Nations Conference on Environment and Development (UNCED). Its main goals are Conservation of biodiversity, sustainable use of biodiversity, and Fair and equitable sharing of benefits arising from the use of genetic resources.
- Under this, two protocols were laid down.
  - Cartagena Protocol on biosafety
  - Nagoya Protocol on access to genetic resources

Cartagena Protocol

It is the first international regulatory framework for the safe transfer, handling, and use of living modified organisms (LMOs), or hybrid organisms, that was negotiated under the aegis of the Convention on Biological Diversity. It came into force in September 2003.

Nagoya Protocol

It is an international agreement which aims at sharing the benefits arising from the utilisation of genetic resources fairly and equitably. It ensures appropriate access to genetic resources by appropriate transfer of relevant technologies, by taking into account all rights over those resources and to the technology as well. It also ensures appropriate funding, thereby contributing to the conservation of biological diversity and the sustainable use of its components. Nagoya Protocol was adopted in CBD 10 at Nagoya in Japan in October 2010.

Conservation Strategies

Conservation needs different strategies; they can be species-based or habitat-based, or ecosystem-based. Some species are given importance at a national level while some need treatment at international levels. Conservation efforts can be largely grouped into the following two categories

- The term 'in-situ conservation' refers to conservation that takes place on-site. It is the process of safeguarding an endangered plant or animal species in its natural habitat, either by preserving or cleaning up the habitat itself or by defending the species against predators. The advantage of in-situ conservation is t hat it helps to maintain recovering populations in their natural habitats where they have developed their distinctive properties.
- The term 'ex-situ conservation' refers to 'off-site conservation'. It is the process of relocating an endangered plant or animal population from a hazardous or threatened habitat and placing it, or a portion of it, in the care of humans. While ex-situ conservation utilises some of the oldest and most well-known conservation techniques available to humans, it also incorporates newer, sometimes contentious laboratory techniques.

In-Situ Conservation Methods

- Habitat protection: This strategy focuses on preserving species in their natural habitats, which is widely believed to be the most effective way to preserve the Earth's biological diversity. Thus, protected areas contribute significantly to the in-situ conservation of species, particularly threatened species, by ensuring habitat conservation.
- Species-specific programme: Certain species are identified as they require a coordinated and targeted protection effort. Project Tiger, Project Elephant, and Project Crocodile are examples of species-specific habitat conservation initiatives.

Solved Examples

1. Which one of the following is not a site for in situ method of conservation of flora?

[UPSC 2011]

(A) Biosphere reserves
(B) Botanical gardens
(C) National parks
(D) Wildlife sanctuaries

Solution: (B)

Ex-Situ Conservation Strategies

Ex-situ conservation has some limitations in terms of animal conservation. These include

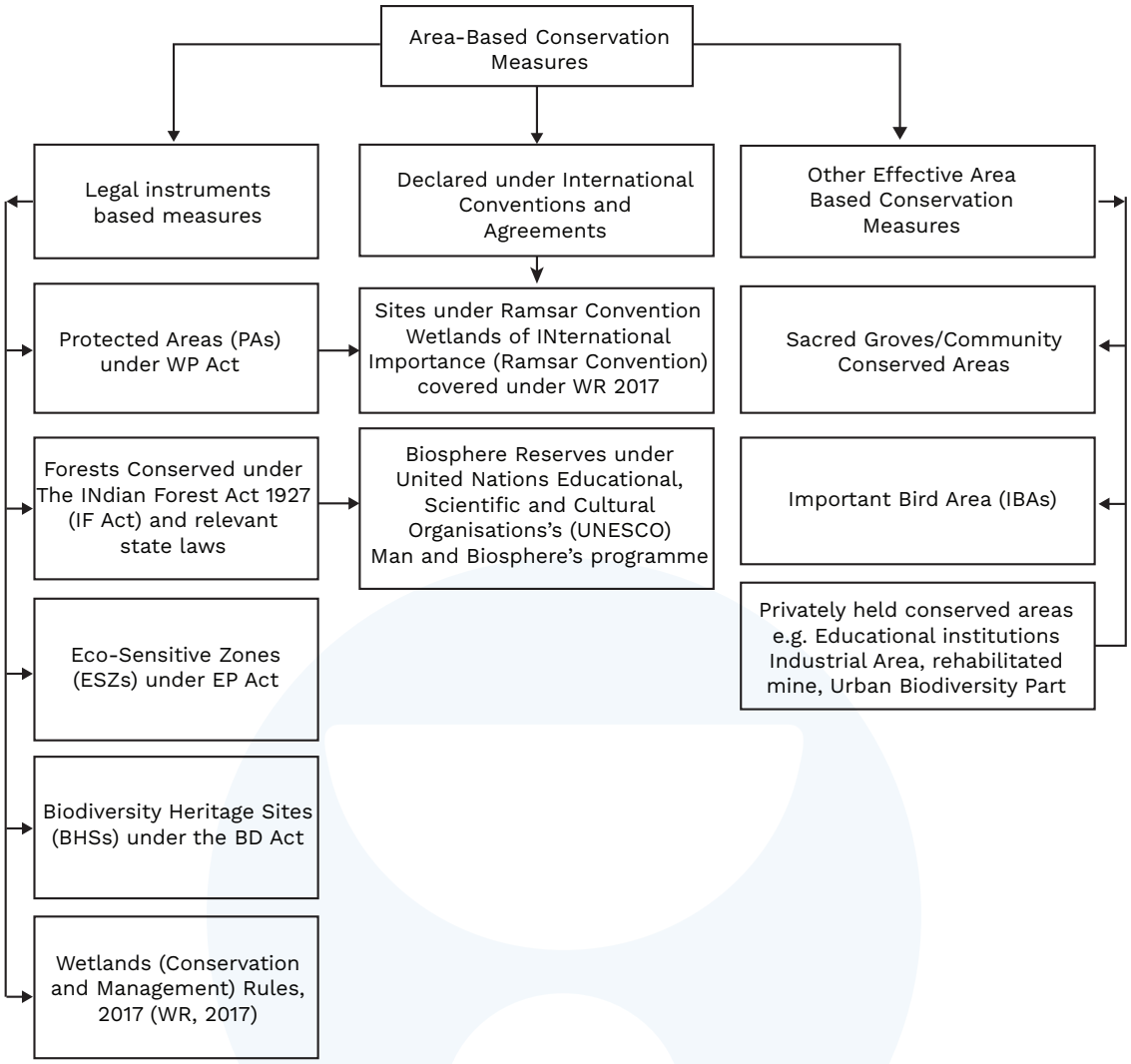


Fig. 2.2

adaptation issues, genetic variability loss as a result of inbreeding, concentration in small areas, surplus animals, and funding continuity. Captive population research can shed light on the species' basic biology and suggest new conservation strategies. Several significant ex-situ conservation strategies include the following.

- Botanical gardens, zoos, and similar establishments: to complement in-situ conservation efforts, various agencies are establishing botanic gardens, zoos, and medicinal plant parks. The Indian Botanical Garden in Howrah (West Bengal) dates all the way back over 200 years.
- Gene banks and seed banks: gene banks and seed banks are used to collect and preserve genetic resources ex-situ. The National Bureau of Plant Genetic Resources (NBPGR) in New Delhi conserves seeds of wild relatives of crop plants as well as cultivated varieties; the National Bureau of Animal Genetic Resources in Karnal, Haryana, safeguards domesticated animals' genetic material; and the National Bureau of Fish Genetic Resources in Lucknow safeguards fish genetic material.
- Cryopreservation or freeze preservation is especially advantageous for preserving vegetatively propagated crops. Cryopreservation is the process of storing material at the extremely low temperature of liquid nitrogen (–196°C) and essentially suspends all metabolic processes and activities. Cryopreservation has been successfully used to preserve meristems, zygotic and somatic embryos, pollen, protoplasts, and suspension cultures of a variety of plant species.

- Conservation at the molecular level (DNA level): in addition to the preceding, germplasm conservation at the molecular level is now feasible and gaining attention. All types of DNA can be used for genetic conservation, including cloned DNA and material containing DNA in its native state. Additionally, non-viable material representing valuable genotypes stored in gene banks can all be used to generate DNA libraries from which a relevant gene or gene combination can be recovered.

Cultural Model of Conservation and Its Application

- This is based on respect for the rights of indigenous peoples and other bearers of 'traditional knowledge' and prevents social conflicts.
- It involves forest dwellers in forest management and governance and acknowledges the traditional rights of tribal over minor forest produce and provisions for making conservation more effective and more transparent.
- It is different from the colonial Model of Conservation where human presence is taken as a threat to nature, and which denied indigenous peoples' rights and provoked long-term social conflict.
- **The Kinshasa Resolution of 1975** (under the auspices of the IUCN) establishes international recognition for cultural conservation models. It recognises the importance of traditional ways of life and land ownership and calls on governments to preserve and promote them.
- **Bishnoi Tribe of Rajasthan** regards trees as sacred and protect the entire ecosystem within their villages, including the animals and birds. The tribe has formed its own Tiger Force, a brigade of youth dedicated to wildlife protection.
- **Andhra Pradesh's Chenchu tribe** is actively involved in tiger conservation at the Nagarjuna Sagar Srisailam Tiger Reserve (NSTR). For centuries, the tribe has coexisted peacefully with tigers and other wild animals, without disrupting the ecological balance that ensures sufficient water and fodder for herbivores.
- **Maldhari Tribe in Junagadh (Gujarat):** the success of lion conservation in the Gir forest area is attributable to the tribe's peaceful coexistence with lions.
- **Arunachal Pradesh's Bugun tribe** uses community-led conservation initiatives and traditional knowledge, the tribe assisted in the protection of the critically endangered Bugun Liocichla bird. Singchung Bugun Community Reserve was awarded the 2018 India Biodiversity Award for its efforts.
- **Arunachal Pradesh's Nyishi tribe** is conserving hornbills in the Pakke/Pakhui Tiger Reserve. Recently, the Arunachal Pradesh government designated the Pakke Paga Hornbill Festival (PPHF) as a state festival'. The PPHF is the state's only conservation festival.
- **Chipko Movement** (1973) was started in the Chamoli district of Uttarakhand by Gaura Devi, Chandi Prasad Bhatt, and Sunderlal Bahuguna.
- **Appiko Movement** was started in September 1983, in Uttarakhand and Shimoga district of Karnataka by Panduranga Hegde.
- **Narmada Bachao Andolan** was started by Medha Patkar, Baba Amte in 1985.

Coral Reefs

- Coral reefs are diverse underwater ecosystems held together by calcium carbonate structures secreted by polyps. These are some of the most diverse ecosystems in the world, housing tens of thousands of marine species.
- Most coral reefs are built from stony corals, which in turn consist of polyps that cluster in groups. About one-third of all marine fish species live in coral reefs.
- An individual coral is known as a polyp, a very small and simple organism consisting mostly of a stomach topped by a tentacle-bearing mouth. The polyps extend their tentacles at night to sting and ingest tiny organisms called plankton and other small creatures.
- Coral reefs are found all around the world in tropical and subtropical oceans. They are usually found in shallow areas at a depth of fewer than 150 feet. However, some coral reefs extend up to about 450 feet deep.

- Survival conditions of corals are temperature of 20°C–21°C, depth of 0–60 m, and salinity of 27%–30%.
- Loss of zooxanthellae (algae) from coral is referred to as Coral Bleaching.
- Causes of depletion of coral reefs are global warming, ocean acidification pollution from land, el-nino, and overfishing.
- The benefits of corals are they work as a sink for CO_2. Corals have a huge reserve of $CaCO_3$. Corals are known as heath indicators for the ecosystem.

Wetlands

- Wetlands are complex ecosystems which encompass a wide range of inland, coastal, and marine habitats.
- A wetland is a land area which is saturated with water either permanently or seasonally. It is distinct from other water bodies based on Waterland and on the type of plants that are found within wetlands including marshy areas, swampy areas, bogs, and fens.
- According to Ramsar Convention (held at Ramsar, Iran in 1971), a wetland can be defined as an area of marsh, fen, peatland, or water, whether natural or artificial, permanent or temporary, with water static or flowing, fresh, brackish or salt, including areas of marine water and the depth of which should not be more than 6 metre at the time of low tide.

Ramsar Convention

- Ramsar Convention is an international treaty for 'the conservation and sustainable use of wetlands'. It is also known as the Convention on Wetlands. The 2nd of February each year is World Wetlands Day.
- Under the 'three pillars' of the Convention, the Contracting Parties commit to
  - Towards the wise use of all their wetlands.
  - Designate suitable wetlands for the list of Wetlands of International Importance (the 'Ramsar List') and ensure their effective management.
  - Work cooperates internationally on transboundary wetlands, shared wetland systems, and shared species.
- During the joining, each contracting party agrees to include at least one wetland site on the List of Wetlands of International Importance. The inclusion of a 'Ramsar Site' on the List represents the government's willingness to take the required measures to preserve its natural integrity.
- There are more than 2300 Ramsar Sites on the territories of 171 Ramsar Contracting Parties around the globe. With 175 and 142, respectively, the United Kingdom and Mexico are the countries with the most Sites. The greatest area under Ramsar protection is located in Bolivia.
- The Montreux Record is a registry of wetland areas on the List of Wetlands of International Importance where ecological changes have occurred, are occurring, or are anticipated to occur as a result of technological advancements, pollution, or other human involvement. It is kept as part of the Ramsar List.
- If a wetland of international importance is added to the 'Montreux Record,' it indicates that ecological changes have occurred, are occurring, or are expected to occur due to human activity.
- According to Ramsar Convention, there are three types of wetlands.
  - Marine and coastal wetland
  - Inland wetland
  - Man-made wetland

Mangroves

- The word 'mangrove' covers a number of forest ecosystems characterised by hydrophytic vegetation, which can tolerate excessive salt content and regular water logging condition. These are a group of trees and shrubs that live in the coastal intertidal zone.
- There are about 80 different species of mangrove trees. All of these trees grow in areas with low oxygen soil, where slow-moving waters allow fine sediments to accumulate.
- Mangrove forests only grow at tropical and subtropical latitudes near the equator because they Cannot withstand freezing temperatures.
- Sundari tree, Grey Mangrove, and Rhu tree are some examples of Mangroves.

- Mangrove forests stabilise the coastline, reducing erosion from storm.

National Park

- A national park is the ecosystem where plant and animal species, geomorphologic sites, and habitats are of special scientific, educational, and natural landscape found.
- It is in use for conservation purposes, which is a reserve of natural, semi-natural, or developed land that a sovereign state declares or owns.
- National parks in India are the International Union for Conservation of Nature (IUCN) Category II Protected Areas.

Wildlife Sanctuary

- A wildlife sanctuary is a place of refuge where abused, injured, and abandoned captive wildlife may live in peace and dignity for the remainder of their lives. They were established by IUCN Category II Protected Areas.
- India has 515 Wildlife Sanctuaries. Among these 48 Tiger Reserves are governed by Project Tiger.
- They cover an area of 118,150 km^2, which is 3.59% of the geographical area of the country.

Biosphere Reserve

- A biosphere reserve is the area of terrestrial and coastal ecosystems that promotes solutions to reconcile the conservation of biodiversity with its sustainable use.
- It is internationally recognised, nominated by national governments, and remains under the sovereign jurisdiction of the states where they are located.

A biosphere reserve comprises the following zones.

- **Core zone:** this zone is managed for minimum human interference to serve as a baseline for the biological region. Research education and training activities are carefully controlled.
- **Buffer zone:** this zone is managed for research, education, and training activities. Manipulative methods and techniques are permitted. Traditional activities including timber extraction, hunting, fishing, and grazing are permitted in a controlled manner.
- **Restoration zone:** this zone has studied and reclaimed lands and natural resources where a significant anthropogenic (human) modification has exceeded an ecological threshold.
- **Cultural zone:** this zone is managed to protect and study in harmony with the environment's ongoing cultural and land-use practices. Local residents and their associated activities are subject to stringent regulations.

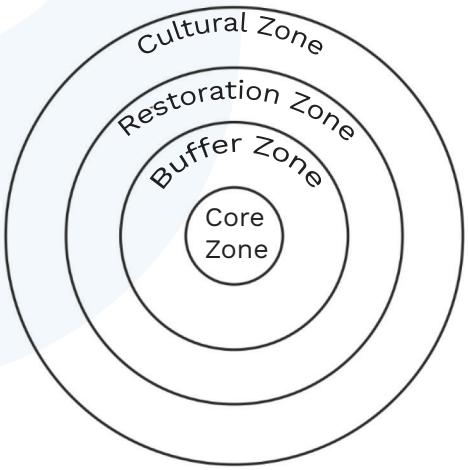


Fig. 2.3

Practice Exercise

1. Consider the following statements regarding the structure and design of biosphere reserves.

I. Biosphere reserve is divided into two interrelated zones namely core and buffer.

II. Core area is kept undisturbed and also contains exceptional scientific interest as it may contain endemic species.

III. Buffer area is an outermost zone, usually not delimited and all kinds of activities which are not harmful to the environment are allowed.

Select the correct statement(s) using the codes given below.

(A) I and II only
(B) I only
(C) II and III only
(D) II only

2. Research, education, and training activities are primarily managed in which zone of a biosphere reserve?

(A) Transition zone
(B) Buffer zone
(C) Cultural zone
(D) Core zone

3. Consider the following statements regarding TRAFFIC—a wildlife trade monitoring network.

I. It has the objective to ensure that trade in wild plants and animals is not a threat to the conservation of nature.

II. It is a joint conservation programme of the World Wildlife Fund (WWF) and the International Union for Conservation of Nature (IUCN).

Which of the above statement(s) is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

4. Which of the following is a cause of loss of biodiversity?

(A) Habitat degradation
(B) Pollution
(C) Invasion of non-native species
(D) All of the above

5. Symbol of WWF is:

(A) Red panda
(B) Giant panda
(C) Black panda
(D) Teddy panda

6. Ex-situ area of biodiversity conservation is:

(A) Biosphere reserve
(B) Botanical gardens
(C) Wetlands
(D) Ramsar wetlands

7. Consider the following statements.

I. Buffer zone of the biosphere reserve is an area of active cooperation between the management and the local people.

II. Core zone of the biosphere consists of an undisturbed and legally protected ecosystem.

Which of the statement(s) given above is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

8. What is not true for global biodiversity?

(A) Biodiversity is affected by climate change.
(B) Biodiversity supports and provides many ecosystem services.
(C) Biodiversity contributes to both climate change mitigation and adaptation.
(D) Biodiversity is fully documented.

9. The Himalayan range is very rich in species diversity. Which one among the following is the most appropriate reason for this phenomenon? **[UPSC 2011]**

(A) It has a high rainfall that supports luxuriant vegetative growth.
(B) It is a confluence of different bio-geographical zones.
(C) Exotic and invasive species have not been introduced in this region.
(D) It has less human interference.

10. Which one of the following national parks has a climate that varies from tropical, temperate, and Arctic? **[UPSC 2015]**
(A) Khangchendzonga National Park
(B) Nandadevi National Park
(C) Neora Valley National Park
(D) Namdapha National Park

11. Consider the following statements.
I. Conserving the animals and plants in their natural habitat is known as ex-situ conservation.
II. Seed banks, botanical gardens, and zoological parks are the examples of ex-situ conservation.
Which of the above statement(s) is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

12. Regarding an organisation known as 'Birdlife International,' which of the following statements is/are correct? **[UPSC 2015]**
I. It is a global partnership of conservation organisations.
II. The concept of biodiversity hotspots originated from this organisation.
III. It identifies the sites known/referred to as important bird and biodiversity areas.
Select the correct answer using the codes given below.
(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

13. Which one of the following is not a site for in-situ method for the conservation of flora?
(A) Biosphere reserves
(B) Botanical garden
(C) National park
(D) Wildlife sanctuary

14. Vikramshila Gangetic Dolphin Sanctuary, India's only dolphin sanctuary, is located in which among the following states of India? **[SSC 2019]**
(A) Uttarakhand
(B) Bihar
(C) West Bengal
(D) Uttar Pradesh

15. Which of the following is/are incorrect pairs regarding the classification of biodiversity?
I. Alpha diversity: Between community diversity.
II. Beta diversity: Within community diversity.
III. Gamma diversity: overall biodiversity.
Select the correct option from the codes given below.
(A) I and II only
(B) II only
(C) I and III only
(D) I, II, and III

16. Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins? **[UPSC 2014]**
I. Construction of dams and barrages on rivers.
II. Increase in the population of crocodiles in rivers.
III. Getting trapped during fishing in finishing nets accidentally.
IV. Use of synthetic fertilisers and other agricultural chemicals in crop fields in the vicinity of rivers.
Select the correct answer using the below.
(A) I and II only
(B) II and III only
(C) I, III, and IV only
(D) I, II, III, and IV

17. Which one of the following is the national aquatic animal of India? **[UPSC 2015]**
(A) Saltwater crocodile
(B) Olive ridley turtle
(C) Gangetic dolphin
(D) Gharial

18. Consider the following fauna of India. **[UPSC 2013]**

I. Gharial
II. Leatherback turtle
III. Swamp deer

Which of the above is/are endangered?

(A) I and II only
(B) III only
(C) I, II, and III
(D) None of these

19. Which of the following statements about the convention on biological diversity (CBD) is incorrect?

(A) It is an international and legally binding treaty, adopted in 1992 at the United Nations Conference on Environment and Development (UNCED).
Cartagena and Nagoya Protocols are laid down under CBD.
(B) India has not yet ratified this convention.
(C) Aichi targets were adopted by the CBD at its Nagoya conference in 2010.
(D) None of the above

20. Which of the following protocols is based on the access to genetic resources and the fair and equitable sharing of benefits arising from their utilisation?

(A) Cartagena Protocol
(B) Nagoya Protocol
(C) Geneva Protocol
(D) Sofia Protocol

21. Among the following organisms, which one does not belong to the class of the other three?

(A) Crab
(B) Mite
(C) Scorpion
(D) Spider

22. Consider the following statements. **[UPSC 2010]**

I. Biodiversity hotspots are located only in tropical regions.
II. India has four biodiversity hotspots i.e., Eastern Himalayas, Western Himalayas, Western Ghats, and Andaman and Nicobar Islands.

Which of the statement(s) given above is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

23. Which one of the following statements is incorrect regarding the Gadgil Committee on biodiversity?

(A) Committee was regarding the biodiversity hotspot of Eastern Himalaya.
(B) Committee suggested declaring the entire landscape as an ecological sensitive zone (ESZ).
(C) ESZ-I, as per the Gadgil Committee, comprised of existing sanctuaries like national parks and biosphere reserves.
(D) In ESZ-II, only existing mines are allowed but not new mines.

24. Which one among the following is not an example of ex-situ biodiversity conservation?

(A) Gene bank
(B) Zoo
(C) National park
(D) Botanical garden

25. Which of the following mammals is critically endangered in India?

(A) Red panda
(B) Lion-tailed macaque
(C) Eid's deer
(D) Malabar civet

26. Who among the following heads the National Board for Wildlife (NBWL)?

(A) Union Minister of Environment, Forests, and Climate Change
(B) Prime Minister of India
(C) Chairperson of National Green Tribunal (NGT)
(D) Union Minister of State for Environment and Wildlife

27. Which of the following zones of a biosphere reserve is managed for the minimum human interference?

(A) Core zone
(B) Buffer zone
(C) Restoration zone
(D) Cultural zone

28. Which of the following is not a method of in-situ conservation?
(A) Biodiversity hotspots
(B) Sacred groves
(C) National parks
(D) Zoological garden

29. MIDORI prize is related to which of the following fields?
(A) Genetics resources
(B) Pollution control measures
(C) Biodiversity
(D) Biosafety

30. Which of the following is not a factor in the extinction of wildlife?
(A) Introduction to foreign species
(B) Control of pests and predators
(C) Captive breeding
(D) Alteration of habitat

31. The conservation within the nature is:
(A) In-situ conservation
(B) Ex-situ conservation
(C) In-vivo conservation
(D) Ex-vivo conservation

32. Consider the following statements regarding genetic diversity.
I. Genetic diversity is the total number of genetic characteristics in the genetic makeup of a species.
II. Extinction is not only the loss of whole species but also preceded by a loss of genetic diversity within the species.
Which of the above statement(s) is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

33. Regarding National Green Tribunal, consider the following statements.
I. The tribunal has been established for the effective and expeditious disposal of cases relating to environmental protection and conservation of forests.
II. No civil court shall have jurisdiction to entertain any appeal in respect of any matter, which the tribunal is empowered to determine under its appellate jurisdiction.
III. At present only Delhi and Bhopal are the places sitting for the tribunal.
Which of the statements given above are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

34. The *Red Data Book* published by the International Union for Conservation of Nature (IUCN) contains the lists of:
I. Endemic plant and animal species present in the biodiversity hotspots.
II. Threatened plant and animal species.
III. Protected sites for the conservation of nature and natural resources in various countries.
Select the correct answer using the codes given below.
(A) I and III only
(B) II only
(C) II and III only
(D) III only

35. Consider the following statements.
I. IUCN classify protected areas according to their management objectives.
II. National parks of India are in the IUCN category II protected areas.
Which of the statement(s) given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

36. Consider the following statements regarding the International Union for conservation of nature (IUCN):
I. It is the world's largest and most diverse environmental network, which maintains the lists of threatened plants and animal species.
II. It is headquartered in Gland, Switzerland.

Which of the above statement(s) is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

37. *Red Data Book* refers to:
(A) A book with a red cover
(B) A book with a list of extinct species
(C) A book with a list of rare species
(D) A book written by Karl Marx

38. CITES means:
(A) Conservation for Indian Tiger and Ecosystems
(B) Conservation of International Trade in Endangered Species
(C) Convention on International Trade in Endangered Species
(D) None of the above

39. Regarding the Convention on Biological Diversity (CBD), consider the following statements.
I. The governing body of the CBD is the Conference of the Parties (COP)
II. The secretariat of the CBD is based in Montreal, Canada
Which of the statement(s) given is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

40. Which of the following can be the Cartagena Protocol of biosafety?
I. Banning of imports of genetically modified (GM) crop seeds.
II. Labelling of GM-related products.
III. Scrapping of intellectual property rights of GM crops and opening up for more transparent research until safety is proved.
Select the correct answer using the codes given below.
(A) I and II only
(B) II only
(C) I and III only
(D) All of the above

41. Consider the following statements.
I. IUCN has defined the different protected areas under six categories.
II. Category l has two sub-part l(a) and l(b); in which l(a) is for strict nature reserve managed for science and l(b) for wilderness area managed for wilderness protection.
Which of the above statement(s) is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

42. Which one of the following protocols is related to the movement of Living Modified Organisms (LMOs)?
(A) Nagoya protocol
(B) Cartagena protocol
(C) Kyoto protocol
(D) Montreal protocol

43. Consider the following statements regarding National Green Tribunal (NGT).
I. It is a statutory body set up under the provision of the National Green Tribunal Act, 2010.
II. It is headquartered in New Delhi and has regional offices in Bhopal, Pune, Chennai, and Kolkata.
III. After Australia and New Zealand, India is the third country to set up NGT.
Which of the above statements are correct?
(A) I and II only
(B) I and III only
(C) II and II only
(D) All of the above

44. Consider the following statements regarding the convention on biological diversity.
I. Convention on Biological Diversity (CBD) is an international and legally binding treaty, started in 1992 at United Nations Conference on Environment and Development (UNCED).

II. Cartagena and Nagoya protocols are laid down under CBD.
III. CBD has 168 signatory members and 196 parties have ratified it.

Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

Answer Key

1. (D)	**2.** (B)	**3.** (C)	**4.** (D)	**5.** (B)	**6.** (B)	**7.** (B)	**8.** (D)	**9.** (B)	**10.** (D)
11. (B)	**12.** (C)	**13.** (B)	**14.** (B)	**15.** (A)	**16.** (C)	**17.** (C)	**18.** (A)	**19.** (C)	**20.** (B)
21. (A)	**22.** (D)	**23.** (A)	**24.** (C)	**25.** (D)	**26.** (B)	**27.** (A)	**28.** (D)	**29.** (C)	**30.** (C)
31. (A)	**32.** (C)	**33. (**A)	**34.** (B)	**35.** (C)	**36.** (C)	**37.** (C)	**38.** (C)	**39.** (C)	**40.** (A)
41. (C)	**42.** (B)	**43.** (D)	**44.** (D)						

3 Environmental Degradation and Pollution

Pollution

Pollution can be described as the introduction of harmful substances whose nature and quantity modify the properties of the surrounding environment. Pollutants are substances that cause environmental damage. Human actions trigger the emission of pollutants into the environment.

Classification of Pollutants

Pollutants are generally classified into two broad groups.

Biodegradable Pollutants

Biogeochemical cycle activities break down these pollutants/contaminants. If not degraded at the appropriate time, they act as a pollutant. Household garbage and agricultural waste are some examples of biodegradable pollutants.

Non-biodegradable Pollutants

Non-biodegradable pollutants/contaminants do not decompose into non-toxic substances. They are pollutants regardless of their concentration. They penetrate the food chain and cause biomagnification. These contaminants may exist in solid, liquid, or gaseous forms. Examples include pesticides, plastic, radioactive wastes, etc.

Types of Pollution

A broad classification of pollution based on the environment it occurs most in is air pollution, water pollution, and soil pollution.

Air Pollution

- Air pollution is defined as an imbalance in the quantity of air which adversely affects the biotic community. In other words, the contamination of undesirable air pollutants adversely affects the environment because of the lowering quality of air.
- Air pollutants are categorised into two groups.
  - Gaseous/primary air pollutants
  - Particulate/secondary air pollutants

Gaseous/Primary Air Pollutants

These are derived from the direct burning of sources such as coal, wood, etc., which cause the release of pollutants. For example, oxides of nitrogen, sulphur, carbon, hydrogen sulphide, ozone, etc.

Pollutant	Source	Harmful Effects
Oxides of Carbon (CO and CO_2)	Burning of wood and coal and other fossil fuels such as petroleum	Global warming, respiratory issues
Oxides of Sulphur (SO_2 and H_2S)	Volcanic eruptions, and when Sulphur containing fuel is burnt, etc.	Acid rain, respiratory issues, and loss of chlorophyll in plants (chlorosis)
Oxides of Nitrogen (NO and N_2O)	Naturally (lightning) and in auto-mobile exhaust (at high temperature)	Eyes and lungs irritation, reduced plant productivity, and acid rain
Hydrocarbons (Benzene and Ethylene)	Automobiles and petroleum industries (incomplete combustion of fuels)	Respiratory issues and carcinogenic

Table 3.1

Secondary/Particulate Pollutants

- The diameter of their particles ranges from 0.001 to 500 µm. Particles less than 10 µm float and move freely with air circulation. Particles with diameters greater than 10 µm settle down. Aerosols are composed of particles less than 0.02 µm in size.
- The impact of particulate matter (PM) is mostly determined by particle size. Particulate matter larger than 5 microns are likely to lodge in the nasal canal, whereas particles of approximately 10 microns can easily enter the lungs. Dust, smoke, smog, etc. are particle pollutants.

Pollutant	Source	Harmful Effects
Suspended Particulate Matter Fly Ash, Lead, and other metals and Dust	Thermal power plants, Construction activities, metallurgical processes, and Automobile exhaust $N_2(g) + O_2(g) \rightarrow 2NO(g)$ $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$	Smog (Smoke + Fog) leads to Breathing and visibility issues Lead interferes with the development of red blood cells Carcinogenic
Fibres (Cotton, wool)	Textiles and weaving enterprises	Lung disorders

Table 3.2

- **Dust** consists of fine solid particles (over 1 µm in diameter) formed by the crushing and grinding of solid materials. Typical examples of this sort of particulate pollution are sawdust, pulverised coal, dust storms, etc.
- **Mists** are formed by the spray liquid particles and condensation of vapours in the air.
- **Fumes** are formed by the condensation of vapours during sublimation, distillation, and many other chemical reactions. For example, metals and oxides of metals produce fume particles.
- **Smog,** the word is derived from smoke and fog. There are two types of smog.
  - **Classical Smog**
    It develops in a cool and humid climate. It is a mixture of smoke, fog, and sulphur dioxide. It is also known as reducing smog because it is a reducing chemical mixture.
  - **Photochemical Smog**
    - It is formed in an environment that is warm, dry, and sunny. The primary components of photochemical smog are created by the interaction of sunlight with unsaturated hydrocarbons and nitrogen oxides emitted by automobiles and industries. As photochemical smog contains a high concentration of oxidising chemicals, it is known as oxidising smog.
    - Ozone, nitric oxide, acrolein, formaldehyde, and peroxyacetyl nitrate are typical photochemical smog constituents (PAN).
    - Ozone and PAN are both potent eye irritants. Ozone and nitric oxide irritate the nose and throat, and their high concentration results in headaches, chest pain, and breathing difficulties. Photochemical smog causes the rubber to crack and causes considerable

Solved Examples

1. According to UNEP, which of the following is/are the major components of air pollution? **[UPSC ESE 2021]**

I. Sulphur dioxide
II. Ozone
III. Carbon monoxide
IV. Nitrogen dioxide

Select the correct answer using the codes given below.

(A) II and III only
(B) II only
(C) II, III, and IV only
(D) I, II, III, and IV

Solution: (D)

damage to the life of plants. It corrodes metals, stones, and painted surfaces.

- Different types of contaminants are emitted when fossil fuels are burned. Hydrocarbons (unburned fuels) and nitric oxide are some of the contaminants emitted. Due to the interaction of these contaminants with sunlight, a chain reaction starts when their concentrations reach a certain threshold.
  - In automobile exhausts
    $N_2(g) + O_2(g) \rightarrow 2NO(g)$
  - In atmosphere
    $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
  - After some time, this NO_2 absorbs energy from sunlight and breaks up into nitric oxide and a free oxygen atom.
    $NO_2(g) \rightarrow 2NO(g) + O$
  - Free oxygen atoms are very reactive and combine with the O_2 in the air to produce ozone.
    $O + O_2(g) \rightarrow O_3(g)$
  - The ozone formed in this reaction is tropospheric ozone or bad ozone, in contrast with stratospheric ozone, which is good and protects the Earth from harmful UV rays.
  - The ozone formed in the above reaction reacts rapidly with the NO(g) formed in the reaction (automobile exhaust) to regenerate NO_2.
    $NO(g) + O_3 \rightarrow NO_2(g) + O_2$
  - The NO_2 is a brown gas and at sufficiently high levels can contribute to haze.
  - Ozone is a dangerous gas, as both NO_2 and O_3 are powerful oxidising agents that can combine with unburned hydrocarbons in polluted air to produce chemicals such as formaldehyde, acrolein, and peroxyacetyl nitrate (PAN).

Diseases Caused by Air Pollutants

- **Silicosis** or silico-tuberculosis is caused by inhaling free silica or silicon dioxide while working in sectors involving pottery, ceramics, etc. The people suffer from persistent cough and chest pain.
- **Asbestosis** is caused by asbestos a carcinogen used to construct ceilings.
- **Byssinosis** is an impairing lung illness characterised by persistent cough and bronchitis due to long-term inhalation of cotton fibres.
- **Pneumoconiosis** is caused by inhaling coal dust, and it leads to people having

Bharat Stage (BS) Emission Standards

- Bharat Stage Emission Standards are the standards established by the Indian government that regulate the number of air pollutants that can be emitted from internal combustion (IC) engines and vehicles. These cannot be sold on the open market if they release more pollutants than the authorised level. The Central Pollution Control Board (CPCB) of the Ministry of Environment, Forests, and Climate Change has enacted these regulations.
- In 1991, India adopted pollution standards for petrol vehicles, and in 1992, diesel vehicles. Since the year 2000, four-wheeled cars in India have adhered to Bharat Stage Emission Standards, also known as Euro Standards.
- Bharat stage III norms have been enforced across India since October 2010. The recommendation is based on the Mashelkar committee report for the auto-fuel policy launched in the country for improvement of air quality, In a few cities, Bharat stage IV norms placed since April 2010. Bharat stage IV is proposed to be enforced throughout India by April 2017. It has already been put into use in 13 major cities.
- Bharat stage VI norms will be implemented from 1 April 2020. Specifically, gasses such as carbon dioxide (CO_2), carbon monoxide (CO), Hydrocarbons, and oxides of Nitrogen, are the by-products of an IC engine running. It is said that Nitrogen Oxide compounds emissions will be brought down by 68% in diesel and 25% in petrol cars.

National Ambient Air Quality Monitoring Program (NAMP)

- This initiative is carried out by CPCB. Under NAMP, four air pollutants, namely sulphur

dioxide (SO_2), oxides of nitrogen (NO_x), respirable suspended particulate matter (RSPM/PM_{10}), and fine particulate matter ($PM_{2.5}$), have been designated for continuous monitoring at all locations.

- The objectives of the NAMP are to:
  - Assess the current state and future projections of ambient air quality.
  - Determine if the stipulated ambient air quality requirements have been breached and identify non-compliant cities.
  - Attaining the information and understanding required to develop preventive and corrective measures and to comprehend the natural cleansing process occurring in the environment through pollution dilution, dispersion, dry deposition, wind-based movement, precipitation, and chemical transformation of generated pollutants.
  - The monitoring of meteorological data such as wind speed and direction, relative humidity (RH), and temperature is also incorporated into the monitoring of air quality.

National Air Quality Index (NAQI)

- It was launched by MoEFCC in 2014 under Swachha Bharat Abhiyan as 'One Number-One Colour One Description' for the people to assess the air quality in their surroundings.
- There are six AQI categories, namely Good, Satisfactory, moderately polluted, Poor, Very Poor, and Severe.
- The intended AQI will take into account eight pollutants(PM_{10}, $PM_{2.5}$, SO_2, NO_2, CO, O_3, NH_3, Pb) for which short-term (up to 24-hour average) National Ambient Air Quality Standards are specified. Based on the measured ambient concentrations, respective standards and expected health impacts are compiled

AQI	Associated Health Impacts
Good (0.50)	Minimal impact
Satisfactory (51–100)	May cause minor breathing discomfort to sensitive people.
Moderately Polluted (101–200)	May cause breathing discomfort for people (children and older adults) with lung diseases such as asthma and discomfort for people with heart disease.
Poor (201–300)	May cause breathing discomfort to people on prolonged exposure and discomfort to people with heart disease.
Very poor (301–400)	May cause respiratory illness to people on prolonged exposure. The effect may be more pronounced in people with lung and heart disease.
Severe (401–500)	May cause respiratory impact even on healthy people and serious health impacts. On people with lung/heart disease. The health impacts may be experienced even during light physical activity.

Table 3.2

Eight Pollutants That Are Recognised under NAQI

- **PM_{10}:** coarse pollutants with a diameter of 10 microns or less. Mostly dust and smoke create breathing difficulty, particularly among asthma patients and the elderly.
- **$PM_{2.5}$:** respirable particles with a diameter of 2.5 microns or less. Due to their small size, they cause the formation of smog and Haze. They aggravate asthmatic and heart disease.

- **NO_2:** it is a lung irritant. Red Haze formed in traffic is formed due to NO_2. It damages plants, and leaves and slows down photosynthesis.
- **SO_2:** it causes respiratory diseases such as asthma bronchitis and emphysema. It also causes eye irritation and tears. In plants, it increases stiffness in flower buds, and once stiffness in flower buds increases, they fall from plants.
- **CO:** CO has more affinity to bind with haemoglobin than oxygen. As a result, the O_2 carrying capacity of blood decreases, which results in headache, nausea, weak eyesight, premature birth, abortion, etc.
- **O_3:** it is a toxic gas. It causes dryness of the eyes, chest pain, and cough. Collectively, ozone, PAN, nitric oxide, and formaldehyde result in photochemical smog. Photochemical smog occurs when sunlight acts on vehicular pollutants.
- **NH_3:** ammonia in the air, along with NO_2 and SO_2, causes smog and haze problems. NH_3 is responsible for eutrophication and algae blooms in the water bodies.
- **Pb (lead):** lead interferes with red blood cells.

Drawbacks of NAQI

- Industrial units have to install instruments to measure air quality and send data to the state pollution board. If too much pollution occurs, then units will be shut down. With this, the economic impact on the industrial sector of the country will also be suffered.
- Industrial units deliberately use indigenously made faulty machines to measure air pollutants, so the reported number of particulate matter is less than the actual pollution.
- One index is used for entire India. But there is a need for a separate index for residential and industrial areas. Similarly, there is also a need for a separate index for cities, towns, and villages.
- It does not monitor benzene in the air, which is a carcinogenic chemical and can cause cancer.

Solved Examples

1. Consider the following. **[UPSC 2019]**

I. Carbon monoxide
II. Methane
III. Ozone
IV. Sulphur dioxide

Which of the above are released into atmosphere due to the burning of crop/ biomass residue?

(A) I and II only
(B) II, III, and IV only
(C) I and IV only
(D) I, II, III, and IV

Solution: (D)

2. Which of the following are some important pollutants released by steel industry in India? **[UPSC 2014]**

I. Oxides of sulphur
II. Oxide of nitrogen
III. Carbon monoxide
IV. Carbon dioxide

Select the correct answer using the codes given below.

(A) I, III, and IV only
(B) II and III only
(C) I and IV only
(D) I, II, III, and IV

Solution: (D)

System of Air Quality Forecasting and Research (SAFAR)

- It is a computerised system jointly developed by the Indian Institute of Tropical Meteorology (IITM), Pune and Indian Meteorological Department (IMD), Delhi.
- It can forecast the quality of air 24–72 hours in advance. It provides data on air quality index and health advisory. It also provides data on UV rays, known as UV-Index.
- In India, the first time SAFAR was installed in Delhi was during Commonwealth Games in 2010.

Control of Air Pollution

- The need and awareness for the usage of eco-friendly, cleaner, cost-effective fuels, such as liquefied natural gas (LNG) are increasing. Using environmentally-friendly industrial techniques to minimise the discharge of pollutants and hazardous waste is followed. For example, installing systems that decrease the emission of pollutants.
- **Filters:** these devices remove particles from the gas stream. A filter's medium may consist of fibrous materials such as cloth, granular materials such as sand, and stiff materials such as a screen.
- **Electrostatic precipitators (ESP):** the emitted dust is charged with ions, and the ionised particulate matter is collected on a surface with the opposite charge. The particles are removed from the collection surface through periodic shaking or tapping of the surface. They are used in boilers, furnaces, thermal power, cement, steel industries, etc.
- **Inertial collectors:** it operates on the idea that the inertia of suspended particulate matter (SPM) in gas is greater than its solvent, and as inertia is a function of the particle's mass, this device collects heavier particles more effectively.
- Scrubbers are wet collection devices. They remove aerosols from a gas stream by capturing moist particles on a surface and then removing them, or by wetting the particles with a scrubbing liquid. As the particles migrate from the supporting gaseous medium across the interface to the liquid scrubbing medium, the particles get trapped.
- Conventional biomass fuel usage generates air pollution as a result of the incomplete combustion of biomass, which produces many harmful products. To decrease reliance on conventional biomass cooking, the Ministry of New and Renewable Energy (MNRE) is implementing the following programmes.
- **The Unnat Chulha Abhiyan:** it was launched in 2014 to promote enhanced biomass cookstoves, which is a clean cooking energy option that reduces firewood usage with greater efficiency and lower emissions.
- **National Biogas and Manure Management Programme (NBMMP)** for establishing domestic biogas plants to meet the cooking energy needs of rural and semi-urban areas and reduce the usage of firewood. Traditional cooking practices cause indoor air pollution, so encouraging the use of solar cookers reduces indoor air pollution.
- **The Pradhan Mantri Ujjwala Yojana (PMUY)** was launched in 2016 by the Ministry of Petroleum and Natural Gas with the slogan 'Swachha Indian, behtar Jeevan' (Clean fuel, better life) to protect the health of women and children by providing them with clean cooking fuel using LPG. The Scheme provides BPL households with a Rs 1,600 subsidy for each LPG connection and an interest-free loan to purchase stoves and refills from Oil Marketing Companies.

NEERDHUR

- It is a multi-fuel improved cookstove developed by CSIRNEERI (National Environmental Engineering Research Institute).
- Apart from wood, it can consume several fuels such as cow dung, charcoal, biomass, and other agricultural residues. It is developed as a hybrid with technology innovations and uses low fuel with low emissions.

Alternative Fuels to Reduce Air Pollution

Propane, ethanol, methanol, P-series fuels, and solar fuels are used as alternative fuels and they can replace petroleum in the future.

Biodiesel

- Biodiesel (Mono alkyl esters) is a clean-burning diesel fuel made from natural, renewable sources such as vegetable oils. Similar to diesel, biodiesel operates in combustion-ignition engines.
- It is made from an oil-producing plant Jatropha through a process called transesterification. In this process, fat is mixed with alcohol and a catalyst, which separates the vegetable oil into two components, that is, glycerine and methyl esters. Glycerine is used in making soaps and methyl esters are used in biodiesel.
- It is biodegradable, non-polluting, and non-toxic, and hence can be safely used in the marine industry.

P-Series Fuel

P-Series fuel is the new and latest alternate fuel to be added. The blend of methyl tetrahydrofuran (MTHF), ethanol, and hydrocarbons constitutes the P-Series fuel.

Solar Fuels

Solar energy systems are used to heat water, generate power, etc. Photovoltaic cells are used to collect solar energy and used it as a fuel for vehicles.

Water Pollution

Water pollution is the presence or addition of harmful substances to water. Natural (erosion, leaching, and decomposition of organic matter) and anthropogenic (industrial, agricultural, and domestic) activities contribute to water pollution.

Sources of Water Pollution

Pesticides

Water bodies may be contaminated by agricultural pesticides such as DDT, Organophosphates, etc. Aquatic species absorb pesticides from water, which then enter the food chain and migrate up the food chain. At higher trophic levels, they become concentrated and may rise to the apex of the food chain.

Solved Examples

1. What can be the impact of excessive/ inappropriate use of nitrogenous fertilizers in agriculture? **[UPSC 2015]**

I. Proliferation of nitrogen fixing microorganisms in soil can occur.
II. Increase in the acidity of soil can take place.
III. Leaching of nitrate to the ground water can occur.

Select the correct answer using the codes given below.

(A) I and III only
(B) II only
(C) II and III only
(D) I, II, and III

Solution: (C)

Industrial Effluents

- Humans and other animals are adversely affected by metals such as lead, zinc, copper, arsenic, cadmium, and mercury in industrial wastewater.
- Mercury is responsible for minamata illness in humans and dropsy in fish.
- Lead poisoning results in dyslexia, while cadmium contamination results in itai-itai disease, etc.

Domestic Sewage

Sewage, especially from towns and cities, causes organic pollution of water bodies leading to eutrophication and the spread of diseases.

Oil Spills

- Oil leaks from ships, rigs, pipelines, etc lead to pollution of the sea. The oil film also known as an oil slick inhibits the transport of oxygen into the water by forming a barrier.
- This oil film imparts a bad flavour to fish and seafood, hence decreasing their market value. It causes the death of birds due to its effect on feathers. Oil in the feathers disrupts the feather alignment which leads to loss of critical body heat and buoyancy.
- Once oils are ingested by a marine organism, they move unimpeded through numerous marine food chains, affecting humans through biomagnification.

Measures to Curb Oil Pollution

- Mechanical barriers or booms must be constructed around the oil spill to prevent it from spreading to other places.
- Skimmers are vessels which can scoop oil from the water's surface.
- A dispersant that breaks oil slick can be used to control the spread of oil. Dispersants work similarly to soap on an oil slick. It reduces the surface tension between oil and water and disperses the oil into smaller droplets.
- Absorbents, such as polystyrene, polypropylene, and polyurethane, are frequently utilised to absorb oil and prevent its spread.

- Microbiological surfactants can also be utilised to prevent its spread. Microbiological surfactants combine with oil to emulsify and disseminate an oil slick. These accelerate the evaporation and decomposition of oil by other natural mechanisms.
- Oil zapper refers mixture of five micro bacterial strains capable to degrade crude oil and oil sludge very fast.

Sources of River Water Pollution

Point Source Pollution

It refers to the pollution which enters a waterway through discrete conveyances such as pipes, canals, etc. such as industrial effluents.

Non-point Source Pollution

- It refers to the pollution that does not enter a waterway from a singular source, but rather accumulates over time. These substances include
- It refers to the pollution which does not enter a waterway from a singular source, but rather accumulates over time. These substances include
- Natural pollutants such as dry leaves, insect and animal carcasses, bird droppings, etc.
- Contaminants from agriculture, such as agricultural runoff, pesticides, fertilisers, etc.
- Contaminants from the industrial sector, such as industrial runoff, including industrial wastes.
- Microbial pollutants including Faecal and Total Coliform, particularly during cultural gatherings.
- Contaminants, such as organic debris, discharged by humans via residential discharges.

Central Pollution Control Board (CPCB)

- The Water (Prevention and Control of Pollution) Act of 1974 established it as a statutory body. In addition, the Air (Prevention and Control of Pollution) Act of 1981 vested it with the corresponding powers and duties. It also acts as a field unit and provides technical assistance to the Ministry of Environment and Forests in accordance with the Environment (Protection) Act of 1986.

Functions

Water Quality Monitoring

To enhance the cleanliness of different water sources in various regions by preventing, controlling, and eliminating water pollution.

Air Quality Monitoring

To improve air quality and prevent, control, or reduce air pollution.

Effects of Water Pollution

- There are many impacts of water pollution and it is dependent on the chemicals that are discharged into it. Organic and inorganic wastes reduce water bodies' dissolved oxygen (DO) levels. It is appropriate to consider water polluted if its DO level is below 8 mg/L.
- Extremely contaminated water has a DO level of less than 4 mg/L. Higher concentrations of organic waste accelerate the pace of decomposition and oxygen consumption, resulting in a decrease in the DO content of water.
- The oxygen demand is proportional to the growing intake of organic residues and is represented as the biochemical oxygen demand (BOD) of water. The DO decreases with an increase in BOD.
- Chemical oxygen demand (COD) is a measure to know the level of pollution in water. It is a measure of the oxygen equivalent required to oxidise the entire organic matter (biodegradable and non-biodegradable) in water.
- Tubifex and some insect larvae may survive in highly polluted water and are recognised as indicator species for polluted water.
- The rate of O_2 dissolution in water decreases as water temperature increases. When added to aquatic bodies, heated wastewater released by factories also reduces the DO concentration.

  Hot water ↑ → ↓ DO

- Biomagnification refers to the process by which some pollutants accumulate in tissues at increasing concentrations across the food chain. These non-biodegradable contaminants tend to accumulate in organisms' fat tissues.

Solved Examples

1. There is a concern over the increase in harmful algal blooms in the seawaters of India. What could be the causative factors for this phenomenon? **[UPSC 2011]**

I. Discharge of nutrients from the estuaries
II. Run-off from the land during the monsoon
III. Upwelling in the seas

Select the correct answer from the codes given below.

(A) I only
(B) I and II only
(C) II and III only
(D) I, II, and III

Solution: (D)

Groundwater Pollutants

Fluoride

A high concentration of fluoride in groundwater is beyond the permissible limit of 1.5 mg/L. Excess fluoride causes neuro-muscular disorders, gastrointestinal problems, teeth deformity, hardening of bones, and skeletal fluorosis.

Arsenic

Chronic exposure to arsenic causes black foot disease. It may also cause diarrhoea and lung and skin cancer.

Nitrates

High nitrate concentration in groundwater in India has been found in almost all hydrogeological formations. Excess nitrate in drinking water causes methemoglobinemia or blue baby syndrome.

Uranium

Aquifers in as many as 16 states of India are contaminated by uranium, whose presence in drinking water has been linked to chronic kidney disease.

Fluoride

Fluoride levels in groundwater are above the allowable limit of 1.5 mg/L. Fluoride, in excess, causes neuromuscular diseases, gastrointestinal issues, malformed teeth, bone hardening, etc.

Arsenic

Black foot disease is caused by chronic exposure to arsenic. It can also lead to diarrhoea and lung cancer.

Nitrates

Nearly all hydrogeological formations in India contain an abundance of nitrate-rich groundwater. Methemoglobinemia or blue baby syndrome is caused by excessive nitrate in drinking water.

Uranium

The presence of uranium in India's aquifers has been related to chronic kidney disease in most parts of the country.

Improving Water Quality

Wastewaters and effluents released from industries and municipalities are treated in effluent treatment plants (ETPs) before being discharged into water bodies. Generally, the following treatments are given in ETPs.

Primary Treatment

The primary treatment process involves the separation of large debris, followed by sedimentation in tanks or clarifiers.

Secondary Treatment

In secondary treatment, the wastewater is drawn into shallow stabilisation or oxidation ponds where microorganisms oxidise their organic matter. This results in the release of CO_2 and the formation of sludge and biosolid. Algae grown in the upper lighted zone of the wastewater provide CO_2, which further oxidises sludge.

Tertiary Treatment

The tertiary treatment stage involves the chemical oxidation of wastewater with oxidising agents such as chlorine gas, perchlorate salts, ozone

gas, and ultraviolet light. Some tertiary treatment methods include the ion-exchange method, reverse osmosis, chemical oxidation processes, etc. After tertiary treatment, the wastewater can be released into natural waters or utilised for irrigation.

Ocean Acidification

- The excess addition of CO_2 to the atmosphere as a result of the combustion of fossil fuels over centuries has increased the acidity of the oceans, which threatens the environment by disrupting the food chain and food web. Since the start of the industrial revolution, the surface ocean's pH value has decreased by 0.1 units.
- When the atmosphere CO_2 is added to the seawater, a chemical reaction occurs that reduces the pH value of ocean water by the formation of carbonic acid. The addition of carbonic acid to the ocean is known as acidification of the ocean.

  $CO_2 + H_2O \rightarrow H_2CO_3$ (Carbonic Acid)
- Due to $CaCO_3$ mineral content increases in the ocean. $CaCO_3$ minerals are the basic building blocks of many marine organisms' skeletons and shells.
- Due to high CO_2 levels in the ocean, polysynthetic algae and seagrasses are benefited as they require CO_2 to live such as plants on land.
- A highly acidic environment has a drastic effect on some calcifying species such as oysters, corals, and calcareous plantations.

Conventions on Pollution

Basel Convention

- On 22 March 1989, the Conference of Plenipotentiaries in Basel, Switzerland, adopted the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal.
- It was convened in reaction to a public uproar following the revelation (in the 1980s in Africa and other developing nations) of imports of toxic garbage.
- The primary purpose of the Basel Convention is to safeguard human health and the environment from the negative impacts of hazardous waste.

Solved Examples

1. Basel Convention provides: **[UPSC ESE 2020]**

(A) Indian standards for pollution measurement and prevention
(B) international guidelines to control the transboundary movements of hazardous wastes between different countries
(C) Indian standards for the disposal of municipal and industrial wastes
(D) international standards to categorise pollution in air wastewater

Solution: (B)

Minamata Convention

- The Minamata Convention on Mercury is a global treaty to safeguard human health and the environment against mercury's harmful effects. It was agreed upon and adopted in 2013.
- The Convention emphasises the importance of regulating anthropogenic mercury emissions and preventing their adverse effects.

Rotterdam Convention

- The Rotterdam Convention was adopted by a Conference of Plenipotentiaries in Rotterdam, Netherlands, in 1998, and went into effect on 24 February 2004.
- It emphasises shared responsibility and coordinated efforts among parties in the worldwide trade of certain hazardous chemicals to safeguard human health and the environment against potential harm.
- The convention includes pesticides and industrial chemicals that have been prohibited or severely restricted for environmental or health grounds.
- It creates legally binding responsibilities for implementing the prior informed constant (PIC) procedure.

Stockholm Convention

- The Stockholm Convention is an international treaty designed to safeguard human health and the environment from persistent organic pollutants (POPs). POPs are compounds that persist in the environment for extended periods, become extensively dispersed geographically, accumulate in the fatty tissue of living animals, and are harmful to humans and wildlife.
- POPs can cause damage wherever they go and circulate globally. Governments will adopt measures to eliminate or decrease the emission of POPs into the environment in the course of implementing the Convention. The Convention comes into force on 17 May 2004 after ratification by over 152 countries.
- The Stockholm Convention focuses on decreasing or eliminating POP emissions. It establishes a method for addressing additional substances deemed unacceptable in terms of their toxicity. Ultimately, the Convention promises to transform our economy's dependence on toxic chemicals and pave the way for a future devoid of hazardous POPs.
- The Stockholm Convention is likely best regarded as having five fundamental objectives for the elimination of hazardous POPs, beginning with the 12 worst known as the Dirty Dozen (aldrin, chlordane, DDT, dieldrin, endrin, heptachlor, hexachlorobenzene, mirex, polychlorinated biphenyls, polychlorinated dibenzo-p-dioxins, polychlorinated dibenzofurans, and toxaphene).
- Encourage the switch to safer options and Intentionally target additional POPs. Remove obsolete POP-containing materials and equipment. Cooperate for a future devoid of POPs.

Laws and Policies for Controlling Water Pollution in India

Water (Prevention and Control of Pollution) Act (1974)

- It instituted Central and State Pollution Control Boards and granted them the authority to prevent and control water pollution as well as to advise governments on the measures to control water pollution.

National River Conservation Plan

National River Conservation Plan was launched along with the Ganga Action Plan in 1985. Its primary aim is to minimise pollution load in rivers by implementing various emission reduction projects, consequently enhancing their water quality.

National Water Monitoring Programme (NWMP)

- Through a network of monitoring stations across the country, the CPCB monitors the water quality of surface and groundwater.
- Namami Ganga programme for effective pollution control, conservation, and revitalisation of the National River Ganga.

The National Water Policy of 2012

- Among its major provisions are the preservation of river corridors and water bodies, as well as the prevention of industrial effluents, fertiliser, chemical wastes, etc., from reaching the water source.
- The CPCB has been directed by the NGT to undertake a nationwide biodiversity programme in order to evaluate the performance of river cleaning programmes.
- Rivers are monitored and indexed. The NGT has also asked MoEFCCC to consider presenting environmental awards to institutions and governments that comply with regulations and facilitate a pollution reduction.

Solved Examples

1. Which of the following can be found as pollutants in the drinking water in some parts of India? **[UPSC 2013]**

I. Arsenic
II. Sorbitol
III. Fluoride
IV. Formaldehyde
V. Uranium

Select the correct answer using the codes given below.

(A) I and III only
(B) II, II, and V only
(C) I, III, and V only
(D) I, II, III, IV, and V

Solution: (C)

2. Which of the following is/are the possible consequence/s of heavy sand mining in riverbeds? **[UPSC 2018]**

I. Decreased salinity in the river
II. Pollution of groundwater
III. Lowering the water-table

Select the correct answer using the codes given below.

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

Solution: (B)

Soil Pollution

Soil pollution is the addition of pollutants that negatively impact the quality or fertility of the soil.

Major Soil Pollutants

Municipal Solid Waste

- Low-density polyethylene (LDPE) plastic bags are essentially indestructible and pose an enormous environmental risk.
- The abandoned bags clog sewers and drains. Cows and dogs who consume leftover food, vegetable trash, etc. may suffocate because of plastic bags.
- The combustion of solid waste produces hazardous gases, including carbon monoxide, phosgene, dioxins, and other harmful chlorinated chemicals.

Agricultural Waste

- India is estimated to produce around 620 million tonnes per annum of agricultural wastes, 43% of which are animal dung and slaughter wastes.
- Agricultural chemicals contaminate the soil. These farms' runoff water may induce eutrophication in aquatic bodies due to the use of fertilisers. Pesticides are highly poisonous substances that have negative effects on humans and other animals, including respiratory issues and cancer.

Industrial Hazardous Waste

Industrial Waste

It consists of metals, chemical residues, flies ash, and nuclear wastes. Numerous carcinogenic industrial chemicals, dyes, etc. enter the soil.

Bio-Medical Waste

As per Assocham, the total quantity of medical waste generated in India is 550 tonnes per day (TPD) and by 2022, it is likely to increase to 775.5 (TPD).

Electronic Waste

India generates about 1.85 million tonnes per annum of e-waste and ranks fifth in the world among the top e-waste-producing countries. It is projected that by 2020 the waste generation in India will be 5.2 million tonnes per annum.

Control of Soil Pollution in India

India does not have a specific law regulating land pollution as likewise air and water pollution. Land pollution is governed by the Environmental Protection Act, which specifies dangerous substances and grants the central government the authority to enact rules and regulations pertaining to them. In accordance with its authority, the Central government has enacted the Solid Waste Management Rules of 2016, the Hazardous and Other Wastes (Management and Transboundary Movement) Rules 2016, the Plastic Waste Management Rules 2016, the E-Waste (Management) Rule 2016, and the Biomedical Waste Management Rules 2018.

Solved Examples

1. With reference to agricultural soils, consider the following statements. **[UPSC 2018]**

I. A high content of organic matter in soil drastically reduces its water holding capacity.

II. Soil does not play any role in the sulphur cycle.
III. Irrigation over a period of time can contribute to the salinisation of some agricultural lands.

Which of the statements given above is/are correct?

(A) I and II only
(B) III only
(C) I and III only
(D) I, II, and III

Solution: (B)

2. Due to improper/indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste? **[UPSC 2013]**

I. Beryllium
II. Cadmium
III. Chromium
IV. Heptachlor
V. Mercury
VI. Lead
VII. Plutonium

Select the correct answer using the codes given below.

(A) I, III, IV, VI and VII only
(B) I, II, III, V and VI only
(C) II, IV, V, and VII only
(D) I, II, III, IV, V, VI and VII

Solution: (B)

Radiation Pollution

- Radioactive pollution refers addition of radiation due to the emission of harmful ionising radiation such as He-nuclei, high energy electrons, emission of a neutron, etc.
- It is the ionised type of radiation that directly breaks down chemical bonds and is likely to be harmful.
- At the time of nuclear fission, high energy generates α, β, and γ particle emissions, which are found in their ionised form and can move from one place to another without a medium. The radiation adversely affects flora and fauna.
- Examples of radioactive elements are thorium, uranium, plutonium, polonium, radon, etc.

Thorium ---→ - Smaller molecules of Thorium + Neutron + Energy

- α, β, and γ particles are in ionic form and have high energy. These ionic contents are harmful as they cause a breakdown of chemical bonds.
- Nuclear radiation can cause a change in the DNA arrangement pattern of gene, which can be life-threatening. It can also cause skin and lung cancers, thyroids, etc.

Preventive Measures of Radiation Pollution

- Extreme care should be exercised in the disposal of waste containing radioactive elements. Fission reactions should be minimised.
- Nuclear devices should be exploded underground. Nuclear medicines and radiation therapies should be applied only when necessary.
- Use of higher chimneys and ventilation at the working place where radioactive contamination is high.
- Production of radioisotopes should be minimised.

Noise Pollution

- Unacceptable noise levels created by human activities (traffic, factories, construction, loud parties, etc.) that degrade the quality of life in the affected region is known as noise pollution.
- Some sources of noise pollution may be transient, while others are long-term. There may be hearing loss, wildlife disturbances, and an overall decline in quality of life as a result.
- A persistently noisy atmosphere can lead to discomfort, impatience, headaches, and sleeplessness, which can have a negative impact on a person's productivity.
- Noise pollution generally consists of three interrelated elements: source, receiver, and path.
- Types of noises based on frequencies:

  Ultrasonic > 20000Hz

Sonic 20000 Hz [it can be heard by a human]
Subsonic < 20 Hz

- The unit of sound is measured in sound pressure level (SPL) and the common unit of measurement is decibel (dB) or dB(A). SPL is weighted to a scale known as the sound level meter. It is denoted by the unit dB(A).

Noise Levels of Some Sources

Source	Noise Level
Whispering	20 to 25 dB
Conversation	60 dB
Busy city traffic	85 dB
Trains	96 dB
Jet aircrafts	135 dB
Fireworks and gunshots	145 dB

Table 3.1

- Noise levels can range from 0 to more than 120 dB, at which point physical discomfort starts.
- The Central Pollution Control Board (CPCB) has recommended zone-wise ambient noise levels as given below

Desirable Level of Sound (As per CPCB)

Zones	Day (6.00–2100 h)	Night (21.00–6.00 h)
Industry	75 dB	70 dB
Commercial	65 dB	55 dB
Residential	55 dB	45 dB
Silent zone	50 dB	50 dB

Table 3.2

Laws Governing Noise Pollution

- The Air (Prevention and Control of Pollution) Act of 1981 includes “Noise” as a pollutant in the air.
- Noise Pollution (Regulation and Control) Rules of the year 2000: It defines and regulates the sources of noise pollution.
- Environment (Protection) Rules, 1986: It specifies noise limits for motor vehicles, air conditioners, refrigerators, and specific types of construction equipment.
- Under the Air (Prevention and Control of Pollution) Act of 1981, State Pollution Control Boards / Pollution Control Committees (SPCBs /PCCS) for states / Union territories regulate industrial noise.

Bio Pollution

- Bio pollution is a term that defines the adverse effect of invasive alien species on the quality of the aquatic and terrestrial environment.
- The remnants of flora and fauna if increased excessively, also cause a lowering down of the quality of life of biota, which is also considered a form of bio pollution.
- The forms of bio pollution can be described in the following ways.
  - At the individual level: Bio pollution caused by parasites or pathogens.
  - At population level (species): Genetic changes, i.e, hybridisation of invasive alien species within native species.
  - At the community level (biocoenosis): Replacement or elimination of native species by the dominance of invasive alien species.
  - At habitat level: Modification of physical and chemical conditions can cause bio pollution. Shifting cultivation has changed physical and chemical components of the ecosystem of north-east India, i.e., responsible for more soil erosion, siltation, and flooding of rivers, which all culminates in the damaging of crops.
  - At the ecosystem level: Bio pollution occurs when energy and organic matter circulation gets altered.

Practice Exercise

1. Consider the following statements regarding biomagnification.

I. Biomagnification is an increase in the concentration of metals in successive tropics levels, which will lead to drastic negative effects.

II. Biomagnification happens with metals with a density of less than 5 gm/cm3, which are medium-density metals.

III. DDT biomagnifies and disturbs calcium metabolism in birds and results in premature breaking of eggs.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

2. Excessive release of the pollutant carbon monoxide (CO) into the air may produce a condition in which the oxygen supply in the human body decreases. What causes this condition? **[UPSC 2011]**

(A) When inhaled into the human body, CO is converted into carbon dioxide.
(B) The inhaled CO has a much higher affinity for hemoglobin compared to oxygen.
(C) The inhaled CO destroys the chemical structure of hemoglobin.
(D) The inhaled CO adversely affects the respiratory centre in the brain.

3. Consider the following effects of smog.

I. It hampers the visibility
II. Heavy smog decreases the ultraviolet radiation
III. Acid rains
IV. Decrease in production of natural vitamin D

Select the correct answer using the codes given below.

(A) I, II, and III only
(B) II, III, and IV only
(C) I, II, and IV only
(D) All of the above

4. Consider the following statements regarding fly ash.

I. Fly ash is one such residue that is produced whenever combustion of the solid material takes place.

II. Coal-based thermal power plants are the major emitters of fly ash.

III. Fly ash is collected by passing hot steam before the flue gases escape into the chimney.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

5. Consider the following statements regarding the effects of a particular air pollutant.

I. Carbon monoxide will result in difficulty in breathing problems, severe headache, unconsciousness, and even death.

II. Ozone will result in nervous disorder.

III. Mercury will result in Minamata disease.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

6. Consider the following regarding chemical oxygen demand.

I. It is the amount of oxygen needed for chemical reactions of inorganic matter in water.

II. It is a slightly better mode of pollution measured in water rather than biological oxygen demand (BOD).

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

7. The term 'marble cancer' is associated with which of the following pollutants?
(A) Nitrogen oxides
(B) Acid rain
(C) Particulate matter
(D) Smog

8. Consider the following statements regarding 'oil zapper'.
I. This is a bioremediation technique developed by TERI.
II. It is a mixture of bacteria that degrades pollutants in oil-contaminated sites by leaving behind no harmful substances.
III. The production of oil zapper is very costly, which makes this process economically unviable.
Select the correct answer using the codes given below.
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

9. Consider the following statements regarding tetraethyl lead (TEL).
I. It works as an antiknock agent and increases the efficiency of a vehicle.
II. This also complements catalytic convertors in vehicles in collecting particulate pollutants.
III. Because of its toxic nature, it is now banned in most countries and replaced with other materials.
Select the correct answer using the codes given below.
(A) I and II only
(B) I and III only
(C) II and III only
(D) All of the above

10. Which of the following are the effects of acid rain?
I. Leaching nutrients may make the soil less fertile
II. Increase in ammonia
III. Increase in decomposition because of increased chemical reactions
Select the correct answer using the codes given below.
(A) I and II only
(B) I and III only
(C) II and III only
(D) All of the above

11. The Central Pollution Control Board's National Air Quality Monitoring Programme (NAMP) monitors, which of the following pollutants at all locations?
I. Carbon dioxide
II. Sulphur dioxide
III. Oxides of nitrogen
IV. Suspended particulate matter
Select the correct answer using the codes given below.
(A) I, II, and III only
(B) I, III, and IV only
(C) II, III, and IV only
(D) I, II, and IV only

12. Regarding eutrophication, consider the following statements.
I. Primary productivity becomes low
II. Oxygen availability decreases
Which of the statements given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

13. Which of the following gases absorbs harmful ultraviolet rays in the stratosphere but acts as a pollutant in the troposphere?
(A) Carbon dioxide
(B) Carbon monoxide
(C) Ozone
(D) Nitrogen dioxide

14. The gas responsible for the Bhopal gas tragedy was:
(A) Methane
(B) Methyl propane
(C) CO
(D) Methyl isocyanate

15. Photochemical smog mainly contains:
(A) CFC
(B) SO_2
(C) H_2O
(D) NO_2

16. Which of the following organism can monitor air pollution?
(A) Fungi
(B) Algae
(C) Lichen
(D) Bacteria

17. Non-point water pollution source is:
(A) Nutrition run-off
(B) Industries
(C) Domestic waste
(D) Oil spills

18. The primary cause of acid rain around the world is:
(A) Carbon dioxide
(B) Sulphur dioxide
(C) Carbon monoxide
(D) Ozone

19. Formation of algal bloom in a water body is due to the:
(A) Enrichment of plant diversity.
(B) Enrichment of animal diversity.
(C) Enrichment of hazardous pollutants.
(D) Enrichment of nutrients.

20. Chemical smog is also known as:
(A) London smog
(B) Britain's smog
(C) Australia's smog
(D) India's smog

21. Which of the following is not a source of mercury pollution?
(A) Automobiles
(B) Coal-fired thermal power plants
(C) Waste incinerators
(D) Supersonic jets

22. Which of the following is a secondary air pollution?
(A) CO_2
(B) SO_2
(C) O_3
(D) NO_2

23. Ashok and neem plants are planted in front of buildings because they:
(A) Absorb harmful gases
(B) Absorb noise
(C) Emit scent
(D) None of the above

24. Los Angeles smog occurs due to:
(A) SO_2+H_2S
(B) O_3+NO_2
(C) CO_2+CO
(D) None

25. Blue baby syndrome is caused due to:
(A) Nitrate
(B) CO_2
(C) Sulphate
(D) Hg

26. Which of the following are particulate matter?
I. Soot
II. Smoke
III. Dust
IV. Asbestos fibres
(A) I, II, and III
(B) I, II, and IV
(C) I, III, and IV
(D) I, II, III, and IV

27. Which of the following is the most dangerous and long-lasting waste?
(A) Biomedical waste
(B) Ash from volcano
(C) Mining waste
(D) Nuclear waste

28. Which of the following is not a primary air pollutant?
(A) Methane
(B) Sulphur dioxide
(C) Ozone
(D) Asbestos

29. In the process of enriching uranium, the common by-products are:
(A) Depleted uranium ores
(B) Contaminated tools
(C) Contaminated clothing
(D) All of the above

30. Fluoride in the atmosphere is contributed by:
(A) Burning coal
(B) Steel industries

(C) Brickmaking
(D) All of the above

31. Common heavy metals that pollute soil and water include:
(A) Cadmium
(B) Lead
(C) Mercury
(D) All of these

32. Consider the following statements regarding Basel Convention.
I. It is a treaty on the control of transboundary movements of hazardous wastes and their disposal.
II. It does not address the movement of radioactive wastes.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

33. Match List I with List II and select the correct sequence of answers using the codes given below the lists.
List I
A. Carbon monoxide
B. Carbon dioxide
C. Chlorofluorocarbon
D. Sulphur dioxide

List II
I. Acid rain
II. Ozone layer depletion
III. Asphyxiation
IV. Green house effect

	A	B	C	D
(A)	III	IV	I	II
(B)	IV	III	I	II
(C)	III	IV	II	I
(D)	II	I	III	IV

34. Worldwide declines in amphibian populations have been attributed to:
(A) UV-B radiation
(B) Pesticide residues
(C) Outbreak of fungal infections
(D) All of the above

35. Consider the following statements regarding catalytic converters.
I. Catalytic converters used in vehicles. They have platinum-palladium and rhodium as a catalyst.
II. Unleaded petrol should be used in vehicles because it will inactivate the catalyst.
Select the correct answer using the codes given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

36. Consider the following. **[UPSC 2011]**
I. Carbon dioxide
II. Oxides of nitrogen
III. Oxides of sulphur
Which of the above is/are the emission/ emissions from coal combustion at thermal power plants?
(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

37. Lead ingested or inhaled, is a health hazard. After the addition of lead to petrol has been banned, what still are the sources of lead poisoning?
I. Smelting units
II. Pens and pencils
III. Paints
IV. Hair oils and cosmetics
Select the correct answer using the codes given below.
(A) I, II, and III only
(B) I and III only
(C) II and IV only
(D) All of the above

38. Which of the following are components of fly ash?
I. Aluminium oxide
II. Silicon dioxide
III. Calcium oxide
IV. Toxic heavy metals such as lead, arsenic, and cobalt

Select the correct answer using the codes given below.
(A) I, II, and III only
(B) I, II, and IV above
(C) II, III, and IV only
(D) All of the above

39. Consider the following statements regarding radioactive pollution.
I. The more half-life period of material, the more potential polluter it is.
II. Gamma rays have two protons and two neutrons, which are the heaviest of all radiations and have a more damaging effect.
Select the correct answer using the codes given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

40. Consider the following statements.
I. Acid rains have a lesser effect in India because our soils are alkaline.
II. In the long run, because of acid rain, microbial action in the soil will shift from fungi bound to bacteria bound, which will increase the rate of decomposition.
Select the correct answer using the codes given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

41. Pollutants can be classified as primary and secondary. Which of these is/are primary pollutants?
I. Volatile organic compounds
II. Ground-level ozone
III. Acid rain
Select the correct answer using the codes given below.
(A) I only
(B) I and II only
(C) I and III only
(D) I, II, and III

42. Which of the following is not correct regarding oil spills?
(A) They make the water deficient in oxygen.
(B) They can cause algal blooms.
(C) They can poison exposed organisms.
(D) Due to oil spills, many aquatic species are vulnerable to hypothermia.

43. Regarding 'fly ash' produced by power plants using coal as a fuel, which of the following statements is/are correct? **[UPSC 2015]**
I. Fly ash can be used in the production of bricks for building construction.
II. Fly ash can be used as a replacement for some of the Portland cement contents of concrete.
III. Fly ash is made up of silicon dioxide and calcium oxide only and does not contain any toxic elements.
Select the correct answer using the codes given below.
(A) I and II only
(B) II only
(C) I and III only
(D) III only

44. There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture of various products. Why? **[UPSC 2014]**
I. They can accumulate in the environment and contaminate water and soil.
II. They can enter the food chains.
III. They can trigger the production of free radicals.
Select the correct answer using the codes given below.
(A) I and II only
(B) III only
(C) I and III only
(D) I, II, and III

45. Consider the following statements regarding Minamata Convention, 2013.
I. The convention has prohibited a myriad of products containing mercury, and their production and trade will be altogether prohibited by 2020.

II. This also is prohibited in vaccines containing mercury and the use of mercury in dental mercury fillings.
III. The biggest mercury release comes from coal-fired power stations and the use of mercury to separate gold from ore-bearing rock.

Select the correct answer using the codes given below.
(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

46. Consider the following statements.
I. Bioremediation is a process of using microorganisms to degrade environmental contaminants into a less toxic form.
II. Biomagnification is a process by which the concentration of toxic chemicals increases in the tissues of organisms at successively higher levels in a food chain.

Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

47. About bio-toilets used by the Indian Railways, consider the following statements. **[UPSC 2015]**
I. The decomposition of human waste in the bio-toilets is initiated with a fungal inoculum.
II. Ammonia and water vapour are the only ends released into the atmosphere.

Which of the statements given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

48. Which one among the following industries is the maximum consumer of water in India?
(A) Engineering
(B) Paper and pulp
(C) Textiles
(D) Thermal power

49. Consider the following statements regarding National Air Quality Index (NAQI).
I. It has six categories of air quality with each category having a distinct colour code.
II. It considers eight pollutants for which National Ambient Air Quality Standards are prescribed.

Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

50. What is the role of ultraviolet (UV) radiation in water purification systems?
I. It inactivates/kills the harmful micro-organisms in water.
II. It removes all the undesirable odours from the water.
III. It quickens the sedimentation of solid particles, removes turbidity, and improves the clarity of the water.

Which of the statements given above is/are correct?
(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

51. Consider which of the following are persistent organic pollutants (POPs).
I. Hexachlorobenzene
II. DDT
III. Hydro bromine compounds
IV. Aldrin

Select the correct answer using the codes given below.
(A) I, II, and III only
(B) I, III, and IV only
(C) I, II, and IV only
(D) All of the above

52. Consider the following statements regarding persistent organic compounds.
I. These are organic compounds that are resistant to degradation.
II. They are the main causes of bioaccumulation.

III. Stockholm Convention, 2001 is signed to eliminate and restrict the production and use of these.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

53. Photochemical smog is a resultant of the reaction among NO_2, O_3, and peroxyacetyl nitrate in the presence of sunlight.

(A) CO, O_2, and peroxyacetyl nitrate in the presence of sunlight.
(B) CO, O_2, and NO_2 at low temperature.
(C) High concentration of NO_2, O_3, and CO in the evening.

54. Consider the following statements.

I. Stockholm Convention on Persistent Organic Pollutants (POPs) was adopted in 2001 and it became effective in 2004.
II. Stockholm treaty is an international environmental treaty aiming to eliminate/ restrict the production and use of POPs.
III. POPs accumulate in the organism by biomagnification and can cause cancer, allergies, hypersensitivity, damage to the nervous system, reproductive disorder, and disruption of the immune system.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

55. What is an oil zapper? **[UPSC 2011]**

(A) An eco-friendly technology for the bioremediation of oil spills.
(B) It is the latest technology developed for undersea oil exploration.
(C) It is a genetically engineered high biofuel yielding maize variety.
(D) It is the latest technology to control the accidentally caused flames from oil wells.

56. Consider the following statements regarding National Air Quality Index (NAQI):

I. It was launched by Union Government in October 2014.
II. NAQI uses a colour-coded table to show air quality and its effect on health.
III. Green colour in the NAQI table indicates moderately polluted air, which may cause breathing discomfort to asthma patients, elders, and children.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

57. The Rotterdam Convention is related to which of the following?

(A) Control of persistent organic pollutants (POPs).
(B) Control of transboundary movements of hazardous wastes and their disposal.
(C) Control of pesticides and industrial chemicals.
(D) Protection of the ozone layer from depletion.

58. Which of the following diseases are caused by the consumption of water contaminated by mercury and nitrates respectively? **[UPSC 2017]**

(A) Osteoporosis and blue baby syndrome
(B) Osteoporosis and Minamata disease
(C) Minamata disease and blue baby syndrome
(D) Minamata disease and osteoporosis

59. The water pollution in an aquatic system is measured by the dissolved amount of:

(A) Carbon dioxide
(B) Chlorine
(C) Hydrogen
(D) Oxygen

60. Consider the following statements regarding London smog.

I. London smog was caused by the use of sulphur-bearing fossil fuels like coal.
II. This type of smog is aggregated by dampness and a high concentration of suspended particulate matter in the air.

Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

61. Consider the following statements regarding the oil spill.
I. It results in hypothermia, which causes fatalities in birds and mammals.
II. It reduces photosynthesis activities in water bodies.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

62. Consider the following statements to Biological Oxygen Demand (BOD).
I. It is expressed in milligrams of oxygen per litre of water.
II. Microbes in sewage treatment plants increase the BOD of the effluent.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

63. The decomposition of human waste in the bio-toilets is carried out by which of the following?
(A) Fungal inoculum
(B) Aerobic bacteria
(C) Virus
(D) Anaerobic bacteria

64. Which of the following conventions/protocols deals with persistent organic pollutants (POPs)?
(A) Stockholm Convention
(B) Nagoya Protocol
(C) Geneva Convention
(D) Vienna Protocol

65. Consider the following statements.
I. Smog is a combination of various pollutant gases with water vapour and dust.
II. Ground-level ozone is formed through reactions, involving hydrocarbons, nitrogen oxides and sunlight.
III. Heavy smog can result in the decrease of natural vitamin 'D' production leading to arise in the case of rickets.
Which of the above statements is/are correct?
(A) I only
(B) I and III only
(C) II and III only
(D) I, II, and III

66. A low biological oxygen demand in a water body refers to:
(A) High-level of microbial pollution
(B) Absence of microbial pollution
(C) Pure water
(D) Low-level of microbial pollution

67. Which of the following is not a secondary pollutant?
(A) Acid rain
(B) Photochemical smog
(C) Ozone
(D) SO_2

68. Consider the following statements.
I. Black carbon is a form of particulate pollutant produced from incomplete combustion.
II. Black carbon is emitted by the high-temperature combustion process whereas brown carbon by low-temperature combustion.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

69. Which of the following is/are the effects of acid rain?
I. It causes the death of the biota of lakes.
II. It corrodes monuments.
III. It degrades the fertility of the soil.
Select the correct answer using the following codes.
(A) I and II only
(B) II only

(C) I and III only
(D) I, II, and III

70. In the context of the environment, the term 'dirty dozen' refers to one of the following?
(A) 12 radioactive elements
(B) 12 most harmful greenhouse gases
(C) 12 persistent organic pollutants
(D) 12 ozone-depleting substances

71. Consider the following statements regarding sulphur dioxide pollution.
I. Automobile exhaust is an important source of sulphur dioxide pollution.
II. High concentration of sulphur dioxide leads to stiffness of flower buds.
Which of the above statements given is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

72. Which of the following constitute primary pollutants?
I. Sulphur dioxide
II. Nitrogen dioxide
III. Particulate matter
Select the correct answer using the codes given below.
(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

73. A river with high Biological oxygen demand (BOD) is:
(A) High productive
(B) Highly clean
(C) Highly polluted
(D) None of these

74. Fluoride pollution mainly affects:
(A) Brain
(B) Heart
(C) Teeth
(D) Kidney

75. Air freshener which contains phthalates have the following adverse effects on human.
I. It suppresses androgen and causes asthma problems.
II. It retards the development of children.
III. It is carcinogenic and also affects central nervous system (CNS).
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

76. Consider the following statements regarding oil zapper.
I. Oil zapper is a mixture of five micro-bacterial strains which are capable to degrade crude oil and oil sludge very fast.
II. After degradation, the oil zapper releases harmless CO, and water.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

77. Consider the following statements regarding SAFAR.
I. SAFAR is an abbreviation of System of Air Quality and Weather Forecasting and Research.
II. It is a computerised system jointly developed by the Indian Institute of Tropical Meteorology, Pune along with the India Meteorological Department, Delhi, and the National Centre for Medium-Range Weather Forecasting.
III. It also provides data on ultraviolet rays.
Which of the above statements are correct?
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

78. Consider the following statements.
I. Smog formed in the area of intense solar radiation is called grey air.

II. Smog formed in the area of lesser solar radiation is called brown air.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

79. Which of the following statements is/are correct regarding noise pollution?

I. The World Health Organization recommends that the indoor sound level should be less than 30 dB.
II. An increase of 10 dB is approximately double the increase in loudness.

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (C)	**2.** (B)	**3.** (C)	**4.** (A)	**5.** (C)	**6.** (B)	**7.** (B)	**8.** (A)	**9.** (B)	**10.** (A)
11. (C)	**12.** (B)	**13.** (C)	**14.** (D)	**15.** (D)	**16.** (C)	**17.** (A)	**18.** (B)	**19.** (D)	**20.** (A)
21. (D)	**22.** (C)	**23.** (A)	**24.** (B)	**25.** (A)	**26.** (D)	**27.** (D)	**28.** (D)	**29.** (D)	**30.** (D)
31. (D)	**32.** (C)	**33.** (C)	**34.** (D)	**35.** (C)	**36.** (D)	**37.** (D)	**38.** (D)	**39.** (A)	**40.** (A)
41. (A)	**42.** (B)	**43.** (A)	**44.** (D)	**45.** (D)	**46.** (C)	**47.** (D)	**48.** (D)	**49.** (C)	**50.** (A)
51. (C)	**52.** (D)	**53.** (A)	**54.** (D)	**55.** (A)	**56.** (A)	**57.** (C)	**58.** (C)	**59.** (D)	**60.** (C)
61. (C)	**62.** (C)	**63.** (D)	**64.** (A)	**65.** (D)	**66.** (D)	**67.** (D)	**68.** (C)	**69.** (D)	**70.** (C)
71. (C)	**72.** (D)	**73.** (C)	**74.** (C)	**75.** (D)	**76.** (C)	**77.** (D)	**78.** (D)	**79.** (C)	

4 Global Warming and Climate Change

Global Warming

- Global warming is defined as an increase in the average temperature of the Earth's atmosphere over a long time.
- Human activities, especially fossil fuel combustion, have increased heat-trapping greenhouse gas levels in the Earth's atmosphere, resulting in global warming which is observed since the pre-industrial era (between 1850 and 1900).

Effects of Global Warming

- The average global temperature has increased by about 1.4 degrees Fahrenheit (0.8°C) over the past 100 years, according to the National Oceanic and Atmospheric Administration (NOAA).
- As the average global temperature warms, weather patterns are changing. An immediate consequence of global warming is extreme weather. Scientists project that extreme weather events, such as heatwaves, droughts, blizzards, and rainstorms will continue to occur more often and with greater intensity due to global warming,
- The atmosphere's moisture carrying capacity will be significantly increased as it warms.
- Because of the change in the pattern of air-mass movement, as the atmosphere warms, extensive alterations in precipitation patterns will occur. Because temperature changes are projected to be most pronounced in the region of middle and higher latitudes, precipitation should increase at higher latitudes in summer and winter.
- The global sea level has risen by about eight inches since reliable record-keeping began in 1880, and it is projected to rise another one to eight feet by 2100. This is the result of added water from melting land ice and the expansion of seawater as it warms. Many small islands are in danger to be submerged in near future.
- Temperature ranges are predicted to alter as a result of global warming, affecting the organism's altitudinal and latitudinal diversity pattern. Vegetation in the temperate zone may move from the equator to the pole.
- Trees are vulnerable to temperature changes, a significant temperature rise could result in widespread tree loss, with scrub vegetation taking their place.
- As the temperature rises, plant diseases erupt, needs explode, and the basal rate of plant respiration rises.

Efforts to Deal with Global Warming

Limiting the dependency on fossil fuels, which will reduce greenhouse gas emissions. The vegetation canopy has to be increased. Reduce N_2O emissions by lowering the usage of nitrogen fertiliser. Chlorofluorocarbons (CFCs) substitutes have to be developed.

Ozone Layer

- British meteorologist Dobson studied ozone properties in depth and developed the Dobson Meter, a simple spectrometer for measuring ozone thickness.
- A Dobson Unit (DU) measures the total amount of O_3 in an overhead column of the atmosphere. Dobson Unit is measured by how thick a layer of ozone would be if it is compressed into 1 layer at 0°C and with a pressure of 1 atmosphere above it.
- Every 0.01 mm thickness of the layer is equal to 1DU.

  1 DU = 0.01 mm
  1mm = 100 DU
- The average amount of ozone in the stratosphere across the globe is about 300 DU (i.e., thickness is only 3 mm).
- In the stratosphere, UV rays from the sun break apart chlorofluorocarbon, releasing free chlorine atoms. Free chlorine atoms then react with ozone molecules, taking one

oxygen atom to form chlorine monoxide and leaving an ordinary oxygen molecule, which finally destroys the ozonosphere.

Stratospheric Ozone Depletion

- $O_3 \rightarrow O_2 + O$
- The absorption of UV radiation by the ozone layer increases exponentially with its thickness. Therefore, the maximum amount of UV radiation passing through the atmosphere reaches the surface in the tropic region (i.e., near the equator).
- The concentration of O_3 in the stratosphere changes with seasons, the concentration is highest during the February-April (Spring season) period and lowest during the July-October period.
- The ozone hole was first discovered in 1974 over Antarctica. It was also confirmed over the Arctic in 1990.
- The traditional measure of stratospheric ozone assumes that if the thickness of ozone is about 300 DU-350 DU, then it is in normal condition. But, if it decreases lower than 200 DU, then it is known as an ozone hole.
- Such ozone loss now occurs every springtime above Antarctica and to a lesser extent above the Arctic region where special meteorological condition and very low air temperature causes the formation of the ozone hole (due to less temperature, oxygen in the poles is less reactive and the poles receive fewer UV rays).

Causes of Formation of Ozone Holes

- Natural causes are UV rays which can be divided into three categories UV-A, UV-B (most destructive for O3 as well as humans), and UV-C.
- Anthropogenic causes are the release of ozone-depleting substances such as CFCs, halons, methyl chloroform, and methyl bromide.
- Substitutes of ozone-depleting elements are hydrofluorocarbons (HFC), perfluorocarbon (PFC), nitrogen trifluoride (NF_3), and sulphur hexafluoride (SF_6).

Policies and Programmes to Phase-Out Ozone Depletion

- The Co-ordinating Committee on Ozone Layer was established by the UN Environment Programme (UNEP).
- In 1985, the Vienna Convention on the Protection of the Ozone Layer was held. The goal of the Convention was to foster international cooperation by exchanging data on the effects of anthropogenic activities on the ozone layer. The makers of the Convention believed that by doing so, policymakers would take steps to mitigate the activities that cause ozone depletion.
- The Montreal Protocol on Substances that Deplete the Ozone Layer is an international environmental agreement that governs the consumption and production of roughly 100 man-made substances that deplete the ozone layer known as ozone-depleting substances (ODS).
- When such compounds are released into the atmosphere, they deplete the stratospheric ozone layer, which protects people and the environment from the sun's harmful UV radiation.
- Since its adoption on 15 September 1987, the Protocol has been approved by all 198 UN Member States, making it the only UN treaty ever to do so.
- The Montreal Protocol gradually reduces ODS consumption and production, with various timelines for developed and developing nations. All parties have specific responsibilities under this convention for the phase-out of certain ODS groups, ODS trade control, annual data reporting, and national licensing systems to manage ODS imports and exports. Both developing and developed countries have equal but differing commitments, yet both sets of countries have enforceable, time-bound, and measurable commitments.
- By the year 2000, developed countries were required to phase out ODS, while developing countries were required to do so by 2010. India passed the Ozone Depletion Substances Act of 2003, which mandated the phase-out of all ODS except for medical uses.

Greenhouse Gases

- In recent times due to domestic and industrial coal burning, the concentration of some gases such as CO_2, CH_4, N_2O, and CFCs are increased in the lower atmosphere. As these gases can absorb long-wave infrared radiation, they are known as greenhouse gases.
- The increased amounts of greenhouse gases in the atmosphere are affecting the global climate and this phenomenon is now recognised as Global Climate Change.
- The sequence of greenhouse gases according to their harmful effect are

  $CO_2 > CH_4 > N_2O >$ CFCs
- The concept of greenhouse gases is derived from the glasshouses, which were constructed in the polar region to ensure the growth and development of greenery in the form of vegetable crops.
- These glasshouses in polar areas are transparent to incoming solar radiations and opaque to outgoing longwave radiations, that is why they trap heat energy, which is used in the growth of vegetables and crops. Since greenhouse gases also work in the same way i.e., they are also transparent to incoming solar radiation and opaque to outgoing longwave terrestrial radiation, hence they raise the temperature of the atmosphere and are responsible for global warming.

Types of Greenhouse Gases

There are two types of greenhouse gases.

Controlled: CO_2, CH_4, nitrous oxide (N_2O), CFC, HFC, and SF_6

Uncontrolled: Water vapour and ozone

The excessive increase in concentrations of these greenhouse gases in the atmosphere would retain more and more of the infrared radiation, resulting in an enhanced greenhouse effect.

Carbon Dioxide (CO_2)

CO_2 levels in the atmosphere have risen from 280 parts per million (ppm) in pre-industrial times to over 368 ppm in 2000. Fossil fuel combustion, deforestation, and changes in land use have all contributed to an increase in CO_2 levels.

Methane (CH_4)

Methane is a by-product of incomplete decomposition and is formed anaerobically by a strain of bacteria known as methanogens. Freshwater wetlands, livestock fermentation, etc are all key sources of methane.

Chlorofluorocarbons (CFCs)

- Chlorofluorocarbons are non-toxic, non-flammable synthetic gaseous carbon and halogen compounds which are highly stable. Air conditioners, refrigerators, evaporation of industrial solvents, and propellants in aerosol cans are all major sources of CFCs.
- CO_2 is expected to account for around 60% of total global warming, whereas CH_4 and CFCs account for 20% and 14%, respectively.

Hydrofluorocarbons (HFCs)

Hydrofluorocarbons (HFCs) are chemicals used in air conditioning and refrigerating systems. HFCs are getting prominence as a viable alternative to CFCs and hydrochlorofluorocarbons (HCFCs). They are non-toxic, non-flammable, recyclable, energy-efficient refrigerants. HFCs are one of the six greenhouse gases included in the Kyoto Protocol.

Nitrous Oxide (N_2O)

Nitrous oxide (N_2O) is produced during the manufacture of nylon, burning of nitrogen-rich fuels, livestock manure, the decomposition of nitrogen-rich fertilisers in the soil.

Sulphur Hexafluoride (SF_6)

Sulphur hexafluoride (SF_6) is a colourless, odourless, non-flammable, inorganic extremely strong greenhouse gas that also serves as an efficient electrical insulator. SF_6 is a tracer gas used to determine building ventilation rates and air movements.

Major Effects of Green House Gases

- The increasing abundance of greenhouse in the atmosphere has the following possible effects CO_2 fertilisation effect, Global warming, and Depletion of the Ozone layer in the stratosphere. As we already discussed Global Warming and the Ozone layer lets us deal with the CO_2 fertilisation effect.
- The response of a plant to elevated concentrations of CO_2 is known as the carbon dioxide fertilisation effect.
- Due to increased CO_2 concentration, the rate of photosynthesis will increase, and stomatal conductance will decrease (due to partial closure of stomata).
- ↑ CO_2 concentration → ↑ Photosynthesis → ↓ Stomatal conductance → ↑ Transpiration rate water use efficiency. Thus, the transpiration rate may be reduced, and concurrently water use efficiency increases.

Global-Warming Potential (GWP)

- Global-warming potential (GWP) is a relative measure of how much energy a greenhouse gas absorbed in the atmosphere.
- It compares the amount of energy absorbed by a certain mass of the gas over a given period relative to the amount of energy absorbed by a similar mass of carbon dioxide (CO_2).
- GWP is expressed as a factor of carbon dioxide (whose GWP is standardised to 1) as CO_2 last thousands of years in the atmosphere. Thus, a high GWP correlates with a large infrared absorption and a long atmospheric lifetime.

Carbon Sequestration

Carbon capture and storage are known as carbon sequestration. It describes the technologies designed to tackle global warming by capturing CO_2 at Source itself or even directly from the air and permanently storing it underground.

Causes of Climate Change

Many factors cause climate change, some of them are mentioned here.

Natural Causes

- Variable sun theory
- Sunspot cycle
- Milankovitch principles of climate change

Variable Sun Theory

According to variable sun theory, when the sun is in its waxing phase, then the insolation (exposure to sun rays) intensity increases 10–20 times, and when it is in the waning phase, then the insolation intensity decreases, which may affect the climatic system of the Earth.

Sunspot Cycle

- The sunspot cycle is a 22-year cycle. It is divided into the first and second 11 years. During the first 11 years, the number of spots on the sun's surface increases and they disappear during the second phase of 11 years.
- When the number of spots increase on the sun's surface, then more magnetic radiation along with solar radiation comes towards the Earth. When it reaches the exterior part of the atmosphere, it gets repelled and, along with magnetic content, cosmic rays also get repelled. That is why, during the first phase of a sunspot cycle, fewer cosmic rays reach the atmosphere of the Earth.
- Since cosmic rays are known for cooling the atmosphere, so less amount of cosmic rays causes less cooling, and finally atmospheric temperature rises than the normal year.

Milankovitch Principles of Climate Change

Milankovitch's principles of climate change describe the collective effects of changes in the movements of the Earth on its climate.

- **Eccentricity:** In each 95000 to 125000 years, the orbital shape of the Earth changes. Currently, it is on an elliptical path. In the future, its orbit will convert to a circular path, which may cause a cooling effect. The cooling effect triggers climate change.
- **Obliquity:** It ranges between 22.1° and 24.5° in every 41000 years, that is, it refers to the inclination axis of the Earth, which can range between 22.1° and 24.5°.

- **Precession:** On each gap of 26000 years, the shape of the Earth changes from oblate spheroid to circular because of the wobbling effect.

Anthropogenic Causes

- CO_2 and greenhouse gas theory
- Ozone depletion concept

CO_2 and Greenhouse Gas Theory

- According to the theory, incoming solar radiation is allowed to enter in Earth's atmosphere by greenhouse gases, but for outgoing long wave terrestrial radiation, greenhouse gases are opaque, which is why they raise the atmospheric temperature of the Earth.
- Primary causes of emission of the greenhouse gases are burning of fuel, industrialisation, vehicular pollution, deforestation, and urbanisation.

Ozone Depletion Concept

- Atmospheric ozone has two effects on the temperature balance of the Earth. It absorbs solar ultraviolet radiation in the lower stratosphere and also absorbs infrared radiation emitted by the Earth's surface, effectively trapping heat in the troposphere.
- The depletion of the ozone layer in the lower stratosphere occurs due to human-produced chlorine and bromine-containing gases having an adverse effect on the environment and climate change.
- Terrestrial Causes are Volcanic eruption, Tectonic Plate, and continental drift.

Carbon Sequestration

Carbon capture and storage are known as carbon sequestration. It describes the technologies designed to tackle global warming by capturing CO_2 at Source itself or even directly from the air and permanently storing it underground.

Conventions on Climate Change

United Nations Framework Convention on Climate Change (UNFCCC)

- United Nations Framework Convention on Climate Change (UNFCCC) is an international environmental treaty negotiated at the United Nations Conference on Environment and Development (UNCED), also called Earth Summit, In 1992, a conference was held in Rio de Janeiro, Brazil, to reconcile worldwide economic development with protection of the environment.
- Using treaties and other documents signed at the conference, most of the world's nations nominally committed themselves to the pursuit of economic development in ways that would protect the Earth's environment and non-renewable resources.
- The Earth Summit resulted in the following documents Rio Declaration on Environment and Development, Agenda 21, Forest Principles.

Rio Declaration on Environment and Development

The Rio Declaration on Environment and Development, often shortened to Rio Declaration, consisted of 27 principles intended to guide countries in future sustainable development. Referring to the "integral and interdependent nature of the Earth, "our home", the Rio Declaration proclaims 27 principles.

Agenda 21

- It outlines actions that governments, international organisations, industries, and the community can take to achieve sustainability. It is divided into four sections
- Social and Economic dimensions include combating poverty, changing consumption patterns, population, and demographic dynamics, promoting health, promoting sustainable settlement patterns, and integrating environment and development into decision making.
- Conservation and Management of Resources for development includes atmospheric protection, combating deforestation, protecting fragile environments, conservation of biological diversity (biodiversity), and control of pollution.
- Strengthening the Role of Major Groups includes the roles of children, youth, women, NGOs', local authorities, businesses, and workers.

- Means of Implementation include science, technology transfer, education, international institutions and mechanisms, and financial mechanisms.

Forest Principles

- The Forest Principles (also Rio Forest Principles, formally the Non-Legally Binding Authoritative Statement of Principles for a Global Consensus on the Management, Conservation and Sustainable Development of All Types of Forests) is a 1992 document produced at the Earth Summit.
- It makes several recommendations for the conservation and sustainable development of forestry.
- The Montréal Process, also known as the Working Group on Criteria and Indicators for the Conservation and Sustainable Management of Temperate and Boreal Forests, was started in 1994 as a result of forest principles

Evolution of the Concept of Conference of Parties (COP)

- Under the broad framework of UNFCCC, parties are supposed to meet annually to discuss and negotiate measures of reduction of GHG to mitigate the adverse effect of climate change. Therefore, the Conference of Parties is held annually, called COP.
- COP-1 was organised in Berlin in 1995 to access the progress of dealing with climate change.
- COP-2 was held in Geneva, Switzerland in July.
- COP-3 was held in Kyoto city of Japan in 1997.
- COP-04 was held in Buenos Aires, Argentina in November 1998.
- COP-05 was held at Bonn, Germany in November 1999.
- COP-06 was held at The Hague, Netherlands in November 2000.
- COP-07 was held at Marrakech, Morocco in November 2001.
- COP-08 was held in New Delhi, India in November 2002.
- COP-09 was held in Milan, Italy in December 2003.
- COP-10 was held in Buenos Aires, Argentina in December 2004.
- COP-11 was held in Montreal, Canada in December 2005.
- COP-12 was held in Nairobi, Kenya in November 2006.
- COP-13 was held in Bali, Indonesia in December 2007.
- COP-14 was held at Poznan, Poland in December 2008.
- COP-15 was held in Copenhagen, Denmark in December 2009.
- COP-16 was held at Cancun in Mexico in 2010. Parties convened in Cancun to set a goal for limiting climate change (Temperature). According to the agreement, by the end of the twenty-first century, the goal was to keep temperature increases to a minimum (the reference time of temperature increment was the Pre-Industrial Revolution).
- COP-17 was held in Durban, South Africa in December 2011.
- COP-18 was held in Doha, Qatar in December 2012.
- COP-19 was held in Warsaw, Poland in November 2013.
- COP-20 was held in Lima, Peru in December 2014.
- COP-21 was held in Paris, France in December 2015.
- COP-22 was held at Marrakech, Morocco in November 2016.
- COP-23 was held at Bonn, Germany in November 2017.
- COP-24 was held at Katowice Poland in December 2018.
- COP-25 was held in Madrid, Spain in December 2019.
- COP-26 was held in Glasgow, Scotland in November 2021.

Kyoto Protocol

- The Kyoto Protocol is the result of COP-3, which was convened in Kyoto (Japan). It is a UNFCCC-sponsored international voluntary agreement. It establishes legally binding climate change targets, requiring industrialised countries (developed countries) to cut GHG emissions by an average of 5.2% below 1990 levels by 2012.

- Commitment Periods of the Kyoto Protocol are the First Commitment Period (2005–2012), Second Commitment Period (2012–2020)
- The Protocol emphasises that countries must achieve their targets through national measures. However, it also offers three market-based mechanisms to meet its targets. They are Emission Trading/Carbon Trading, Clean Development Mechanism, Joint Implementation

Solved Examples

1. Kyoto Protocol operationalises the UN framework convention on?

[UPSC ESE 2022]

(A) Sustainable development
(B) Renewable energy
(C) Climate change
(D) Soil erosion

Solution: (C)

Emission Trading/Carbon Trading

- It is a market-based strategy to control pollution by giving economic benefits and achieving the desired reduction in emissions. Under Kyoto Protocol, each signatory country is allowed to emit a fixed amount of greenhouse gases known as a quota limit or target.
- Suppose two signatories, developed countries such as India and the USA, are given a quota limit of 400 KU/year and 200 KU/year respectively. If India emits 440 KU in a year and the USA only emits 160 KU, i.e., the USA has a spare 40 KU. Now, under the Emission Trading System, India can buy this spare 40 KU from the USA to remain within the limit quota.
- The purchasing and selling of carbon emissions under the quota limit is known as carbon trading. It is also called carbon credits.

Clean Development Mechanism

- It permits a country to have an emission reduction commitment to establish an emission reduction project in other developing nations. This type of project earns Emission Reduction Credits equivalent to 1 tonne of CO_2 or GHG which will be considered as meeting the Kyoto Targets.
- Suppose a developed country such as India is given an emission quota of 400 KU, but it emits 440 KU GHG, so it needs to manage an extra 40 KU, India can finance a solar project (green energy) in any village in Congo (a developing country) and get certificate that the solar plant led to reduction of 40 KU units of GHG. In this way, India remains within its quota limit.
- At the firm level: Suppose the Indian government has passed a law that a particular steel company with an output of 500 tonnes of steel/day must not emit more than 50 units of GHG in a year, but this company wants to produce more steel, then its GHG emission has increased to 54 units (four more units above the quota limit). Therefore, this company can also do the same as the Indian government and set up solar plants (or any green energy) in any developing country and get a certificate that it has led to a reduction of four units of GHG emission to remain under its quota limit.

Joint Implementation Mechanism

- Under a joint implementation mechanism, a developed country that has a higher cost of domestic GHG emission (it has exceeded the quota limit and is being changed for an extra emission limit) can establish a project in another country that can cut emissions.
- The joint implementation mechanism is identical to clean development mechanism (CDM), but in the case of CDM, a developed country can set up a project only in a developing country, while in the joint implementation mechanism, any signatory developed country can set up a project in developed countries (these developed countries may or may not be a signatory) to meet the quota limit.

Durban Summit (COP-17)

The following initiatives were charted out in Durban Summit. They are the Kyoto Protocol

Commitment Period, Green Climate Fund, Technology Mechanism, Reduction of Emissions from Deforestation and Degradation (REDD), an agreement for Climate Change Plan beyond 2020, Carbon Capture, and Storage Mechanism.

Kyoto Protocol Commitment Period

Under Kyoto Protocol measures, a concrete policy was laid down to make the successful implementation of the second commitment period of the Kyoto Protocol which runs from 2012 to 2020. For the first time, developing and developed countries both were brought under a universal legal agreement to cut down greenhouse gases emission.

Green Climate Fund

Under this mechanism, developed countries were supposed to contribute 100 billion USD on yearly basis to strengthen measures to cut down emissions. Green Climate Fund limit will remain at 100 billion USD up to 2020 and beyond 2020, it will be increased.

Technology Mechanism

Through technology mechanisms, developed countries are supposed to invest in Green Technology in developing countries and in that process, greenhouse gas emissions will be cut down through green technology such as the solar power plant, wind power plants, biomass power plants, etc.

Reduction of Emissions from Deforestation and Degradation (REDD+)

- United Nations Framework Convention on Climate Change (UNFCCC) Parties devised the REDD+ mechanism. It gives the carbon stored in forests a monetary value by providing incentives for developing nations to cut emissions from forest areas and invest in low-carbon options for sustainable development. REDD+ encompasses conservation, sustainable forest management, and the enhancement of forest carbon stores, in addition to deforestation.
- REDD+ intends to promote developing nations to contribute to climate change mitigation efforts by slowing, halting, and reversing forest loss and degradation, and increasing the removal of GHGs from the atmosphere through forest conservation, management, and expansion.

Solved Examples

1. Which of the following statements is/are correct?

Proper design and effective implementation of UN-REDD+ Programme can significantly contribute to: **[UPSC 2016]**

I. protection of biodiversity
II. resilience of forest ecosystems
III. poverty reduction

Select the correct answer using the codes given below.

(A) I and II only
(B) III only
(C) I and III only
(D) I, II, and III

Solution: (D)

Agreement for Climate Change Plan beyond 2020

Under this provision, a climate change plan was laid down which will work after the completion of the second commitment period of the Kyoto Protocol, i.e., it will work after 2020. According to this measure, all countries were supposed to reach on consensus by 2014 to have a climate change plan beyond 2020.

Carbon Capture and Storage Mechanism

- Derived from the Clean Development Mechanism, this scheme rewards governments or companies that invest in clean energy projects in developing countries with carbon credits that they can trade and sell for profit.
- The new rules force project developers to put 5% carbon credit earned in a reserve, to be awarded to them only after site monitors have proved that no CO_2 has leaked from the underground store 20 years after the end of crediting period.

COP-21

The COP-21 conference of 196 parties took place in Paris, France 2015. It is a global agreement on climate change control, signed at the meeting. If at least 55 countries sign on, representing at least 55 percent of global carbon emissions, the agreement will become legally enforceable. Such parties must sign the agreement in New York between 22 April 2016 (Earth Day) and 21 April 2017 and incorporate it into their legal systems.

Targets set by Paris Agreement

- Keeping global warming below 2°C compared to pre-industrial levels. During the second part of the twenty-first century, the agreement calls for zero net human greenhouse gas emissions.
- The parties will work to limit the temperature rise to 1.5°C. Between 2030 and 2050, achieving the 1.5°C targets will necessitate zero emissions. Beginning in 2023, the agreement sets a goal of updating and improving them every five years. The Paris Agreement, on the other hand, does not include a detailed schedule or country-specific emission goals.
- The members agreed to reduce their carbon output and to do their best to keep global warming well below 2°C. The Agreement will not become binding on its member states until 55 parties who produce over 55% of the world's greenhouse gas have ratified the agreement.
- Each country that ratifies this agreement will be required to set a target for emission reduction, but the amount will be voluntary. There will be neither a mechanism to force a country to set a target by a specific date nor enforcement measures if a set target is not met. Institutions/companies investors should also finance greenhouse gas reduction.

Practice Exercise

1. What are the possible limitations of India in mitigating global warming at present and in the immediate future? **[UPSC 2010]**

I. Appropriate alternate technologies are not sufficiently available.
II. India cannot invest huge funds in research and development.
III. Many developed countries have already set up their polluting industries in India.

Which of the statements given above is/are correct?

(A) I and II only
(B) II only
(C) I and III only
(D) I, II, and III

2. Consider the following statements. **[UPSC 2012]**

Chlorofluorocarbons, known as ozone-depleting substances, are used:

I. In the production of plastic foams
II. In the production of tubeless tires
III. In cleaning certain electronic components
IV. As pressurising agents in aerosol cans

Which of the statements given above is/are correct?

(A) I, II, and III only
(B) IV only
(C) I, III, and IV only
(D) I, II, III, and IV

3. Which of the following are the sources of methane?

I. Bacteria in wetlands
II. Digestion of livestock
III. Production of natural gas

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

4. Consider the following statements regarding fluorinated gas.

I. Four main categories are hydro-fluorocarbons (HFCs), perfluorocarbons (PFCs), nitrogen trifluoride (NF_3), and sulphur hexafluoride (SF_6).

II. Nowadays they are used as refrigerants, fire retardants, etc.
III. They are eco-friendly as they do not deplete the stratospheric ozone layer and are not greenhouse gases.

Select the correct answer using the codes given below.
(A) I and II only
(B) II only
(C) I and III only
(D) I only

5. Arrange the below-given gases in increasing order of global warming potential.
I. Carbon dioxide
II. Nitrous oxide
III. Methane
IV. Hydrofluorocarbons

Select the correct answer using the codes given below.
(A) I – II – III – IV
(B) I – III – II – IV
(C) I – II – IV – III
(D) II – I – III – IV

6. Consider the following regarding the impacts of climate change on agriculture.
I. Increased yield in temperate regions for a moderate increase in temperature.
II. Increased yield in tropical regions for a high rise in temperature.
III. Increased soil erosion may decrease soil fertility.

Select the correct answer using the codes given below.
(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

7. Consider the following regarding various protocols of environmental protection.
I. Hydrofluorocarbons should be phased out by all countries as per the Montreal Protocol.
II. Kyoto Protocol is a legally binding agreement on all developed countries but not on developing countries.

Select the correct answer using the given codes.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

8. Consider the following statements regarding carbon emission trading.
I. It is a form of emission trading aimed at all greenhouse gases and ozone in the troposphere.
II. Under carbon trading, a country having more emissions of carbon can purchase the right to emit and the country having fewer emissions trades the right to sell carbon credit to other countries.
III. This is implemented under Kyoto Protocol.

Select the correct answer using the codes given below.
(A) I and II only
(B) I and III only
(C) II and III only
(D) All of the above

9. Consider the following statements.
I. 'Global warming potential (GWP)' of a gas is the measure of the total energy that a gas absorbs over a particular period, compared to methane.
II. Among greenhouse gases, Hydrofluorocarbons (HFCs) have maximum global warming potential.

Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

10. Ozone depletion would impact the plant community in several ways. These include:
I. Increase in photosynthesis
II. Decline in water use efficiency
III. Decline in yield of plants

Select the correct answer using the codes given below.
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III

11. The 'Agenda-21', a global action plan for sustainable development was adopted in which one of the following climate change summits?
(A) Rio Summit, 1992
(B) Marrakech Summit, 2001
(C) Cancun Summit, 2010
(D) Durban Summit, 2011

12. Selling of carbon by the developing industries is known as:
(A) Carbon sequestration
(B) Carbon dating
(C) Carbon credit
(D) None of the above

13. Which of the following is not a greenhouse gas?
(A) Oxygen
(B) Methane
(C) CFCs
(D) Carbon dioxide

14. Global atmospheric temperature is likely to increase due to:
(A) Burning of fossil fuel
(B) Water pollution
(C) Soil erosion
(D) None of the above

15. The largest reservoir of nitrogen on our planet is:
(A) Ocean
(B) Atmosphere
(C) Biosphere
(D) Fossil fuel

16. Which of the following gas is used as a reference to calculate Global Warming Potential (GWP)?
(A) Ozone
(B) Methane
(C) Carbon dioxide
(D) Nitrogen dioxide

17. The only gas in the atmosphere that can absorb the sun's dangerous UV is:
(A) Nitrogen
(B) Oxygen
(C) Hydrogen
(D) Ozone

18. Unit of ozone measurement is:
(A) ppm
(B) Domg/1
(C) Dobson
(D) Decibel

19. Which is not a part of the national action plan on climate change?
(A) National Water Mission
(B) National Mission for Sustainable Agriculture
(C) National Mission on Pollution Control
(D) National Mission on Sustainable Habitat

20. What are the primary causes of climate change?
I. Excessive burning of fossil fuels
II. Exploding the number of automobiles
III. Increasing solar flares
IV. Heavy deforestation
Select the correct answer using the codes given below.
(A) III only
(B) I and IV only
(C) I, II, and IV only
(D) I, II, III, and IV

21. Which of these natural events affect the climate?
(A) Thawing permafrost
(B) The path of the Earth around the sun
(C) Volcanic eruptions
(D) All of the above

22. Which of the following activities contributes the most to carbon emission globally?
(A) Agriculture
(B) Transport
(C) Forestry
(D) Energy supply

23. Consider the following.
I. Photosynthesis
II. Respiration
III. Decay of organic matter
IV. Volcanic action
Which of the above adds carbon dioxide to the carbon cycle on Earth? **[UPSC 2014]**
(A) I and IV only
(B) II and III only

(C) II, III, and IV only
(D) I, II, III, and IV

24. Consider the following statements.
I. Photosynthesis process adds carbon dioxide to the carbon cycle.
II. The increasing amount of carbon dioxide in the air is raising the temperature of the atmosphere as it absorbs the outgoing radiation.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

25. Which gas is responsible for ozone depletion?
(A) CFC
(B) SO_2
(C) H_2O
(D) CO

26. Which of these pairs is/are correctly matched?
I. Carbon monoxide: asphyxiation.
II. Sulphur dioxide: the greenhouse effect.
III. Chlorofluorocarbon: ozone layer depletion.
Select the correct answer using the codes given below.
(A) I and II only
(B) III only
(C) I and III only
(D) II and III only

27. Which of the following is the most abundant greenhouse gas in the Earth's atmosphere?
(A) Carbon dioxide
(B) Water vapor
(C) Sulphur dioxide
(D) Tropospheric ozone

28. The difference between carbon dioxide emission allowed and emitted carbon dioxide is known as:
(A) Carbon dioxide equivalent
(B) Carbon footprint
(C) Carbon capture
(D) Carbon credit

29. Consider the following statements regarding carbon credits. **[UPSC 2011]**
I. The carbon credit system was ratified in conjunction with the Kyoto Protocol.
II. Carbon credits are awarded to countries or groups that have reduced greenhouse gas emissions below their emission quota.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

30. Which of the following can help in reducing the carbon footprint? **[UPSC 2013]**
I. Using fly-ash-based cement in building construction.
II. Using LED-based electric lamps instead of incandescent lamps.
Select the correct answer using the codes given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

31. Kigali Amendment is related to which of the following protocols?
(A) Kyoto protocol
(B) Montreal protocol
(C) Nagoya protocol
(D) Geneva protocol

32. Consider the followings sectors of India.
I. Agriculture
II. Industrial processes and product use
III. Energy
IV. Waste
Arrange the above in the increasing order of their contribution to greenhouse gas emissions.
(A) IV – I – II – III
(B) IV – II – I – III
(C) I – IV – III – II
(D) IV – II – III – I

33. Which one of the following protocols is associated with the issue of control and

phasing out of the use of ozone-depleting substances?
(A) Nagoya Protocol
(B) Kyoto Protocol
(C) Montreal Protocol
(D) Bretton Woods Conference

34. Consider the following statements.
I. Carbon sequestration describes the long-term storage of carbon dioxide to mitigate or defer global warming.
II. Carbon offsets are the credits for reductions in greenhouse gas emissions made at locations that create renewable energy.
III. Green carbon refers to coastal, aquatic, and marine carbon sinks held by vegetation, marine organisms, and sediments.
Which of these is not an expected effect of climate change?
(A) I only
(B) I and II only
(C) II and III only
(D) I, II, and III

35. Which of these is not an expected effect of climate change?
(A) Sea-level rising
(B) Flooding in coastal cities
(C) Expanding glaciers
(D) Extreme weather

36. Kyoto Protocol is related to:
(A) Global warming
(B) Ozone depletion
(C) Biodiversity
(D) All of the above

37. The Earth Summit of Rio de Janeiro was held in the year:
(A) 1992
(B) 1993
(C) 1994
(D) 1995

38. Consider the following statements:
I. Kyoto Protocol places a heavier burden on developed countries under the principle of common but differentiated responsibilities.
II. Kyoto Protocol was adopted in 1997 and entered into force in 2005.
Which of the statements given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

39. Consider the following statements.
I. The 13th Annual Conference of Parties of the United Nations Framework Convention on Climate Change (UNFCCC) was held in Bali.
II. Bali action plan of UNFCCC endorsed a mitigation programme called reducing emissions from deforestation and forest degradation (REDD) in developing countries.
Which of the statements given above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

40. The Montreal Protocol on substances that deplete the ozone layer aims to control the production and use of:
I. Dichloromethane
II. CFCs
III. HFCs
Select the correct answer using the codes given below.
(A) I and II only
(B) II only
(C) II and III only
(D) I, II, and III

41. Which of the following statements is not true concerning the Kyoto Protocol?
(A) It commits countries to specified reductions in emissions.
(B) It has not been implemented seriously.
(C) It concerns specified greenhouse gases.
(D) It has a uniform emission reduction target for all.

42. Consider the following statements.
Proper design and effective implementation of the United Nations designated Reducing

Emissions from Deforestation and Forest Degradation (REDD+) programme can significantly contribute to:

I. Protection of biodiversity
II. Resilience of forest ecosystems
III. Poverty reduction

Select the correct answer using the codes given below.

(A) I and II only
(B) III only
(C) II and III only
(D) All of the above

43. Consider the following statements regarding Kyoto Protocol.

I. Under the Kyoto Protocol, the clean development mechanism (CDM) permits a signatory nation to establish emission reduction projects in other developing nations.
II. Each signatory country is allowed to emit a fixed amount of greenhouse gases under the quota limit.
III. The Kyoto Protocol legally binds developed country parties to emission reduction targets.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

44. Consider the following statements regarding Kyoto Protocol.

I. Clean Development Mechanism (CDM) defined in Kyoto Protocol allows signatory countries to set up an emission reduction project in developing countries.
II. The second commitment period of the Kyoto Protocol is from 2008 to 2012.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (A)	**2.** (C)	**3.** (D)	**4.** (A)	**5.** (B)	**6.** (C)	**7.** (D)	**8.** (C)	**9.** (D)	**10.** (C)
11. (A)	**12.** (C)	**13.** (A)	**14.** (A)	**15.** (B)	**16.** (C)	**17.** (D)	**18.** (C)	**19.** (C)	**20.** (C)
21. (D)	**22.** (D)	**23.** (C)	**24.** (B)	**25.** (A)	**26.** (C)	**27.** (B)	**28.** (D)	**29.** (C)	**30.** (C)
31. (B)	**32.** (B)	**33.** (C)	**34.** (B)	**35.** (C)	**36.** (A)	**37.** (A)	**38.** (C)	**39.** (C)	**40.** (C)
41. (D)	**42.** (D)	**43.** (D)	**44.** (A)						

5 Energy and Its Resources

Introduction

- The word 'energy' is derived from the Greek word 'en-ergon', which means 'in-work' or 'work content'. Therefore, energy may be defined as the capacity to do work (physical).
- Energy is the engine for economic growth and when the country moves ahead on the path of development, it is necessary to utilise energy resources available in the country to achieve energy security. It is vital for Human development and urbanisation.
- The wide range of agriculture, industrialisation and household activities has compelled to meet ever-increasing demands, which cannot be fulfilled by conventional sources of energy alone, such as coal, petroleum, gas, etc.
- Hence, there is a need to explore the possibilities of harnessing energy from renewable sources of energy such as solar energy, wind energy, tidal energy, geothermal energy, etc.

Types of Energy Resources

Energy resources can be classified into two types:

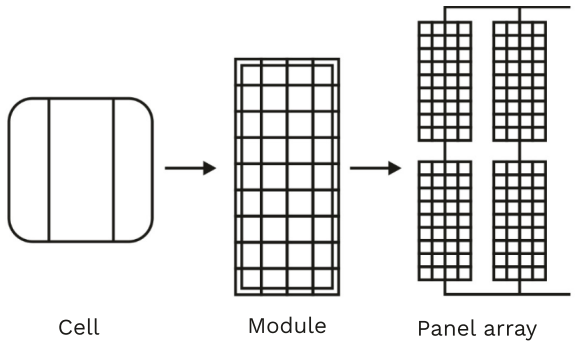
- Renewable energy resources
- Non-renewable energy resources

Renewable Energy Resources

- The energy sources which can be renewed by nature again and again, and their supply is not affected by the rate of their consumption are called renewable energy resources.
- These are environmentally friendly and have the potential to replace non-renewable energy resources. These are also known as inexhaustible sources of energy. For example, solar, wind, ocean, hydro, biomass, geothermal, etc.
- These are available in an unlimited amount in nature and can be renewed over a relatively shorter period. Most of renewable sources of energy are non-polluting and considered as clean.

Solar Energy

- Solar energy is a primary source of energy which is directly obtained from the sun by capturing the solar radiation and converting it into another form of energy to perform various useful activities.
- There are two ways of solar energy utilisation. They are the conversion of solar energy into thermal energy and photovoltaic cells.
- The conversion of solar energy into thermal energy can be done by using solar collectors, whereas in photovoltaic cells direct sunlight is used to generate electricity. Photovoltaic (PV) cells convert sunlight to direct current (DC) electricity. The inverter in the solar power plant converts direct current (DC) into alternating current (AC) electricity. The electrical panel sends power to be consumed. The distribution board supplies electricity to the grid.
- Solar cells have no moving parts, therefore they require very little maintenance. The quantity of free energy a solar panel can generate can result in a substantial return on investment over time.
- The power generation is wholly reliant on the abundance of sunlight, which may be limited by climatic circumstances. The power output of solar power plants cannot compete with that of conventional power plants of equal size, and the infrastructure costs could be exceedingly costly.
- Solar photovoltaic system consists of photovoltaic (PV) cells made up of at least two semiconductor layers, one positively charged and the other negatively charged. As the PV cell is exposed to sunlight, photons are absorbed by the cell. When enough photons are absorbed by the negative layer of the photovoltaic cell, electrons are freed from the negatively charged semiconductor material. These freed electrons migrate to the positive layer creating a voltage difference. When these two layers are connected to an external load, electricity is produced.
- Solar photovoltaic systems are used in space satellites, remote radiocommunication booster stations, marine warning lights, medical refrigeration, and lighting.



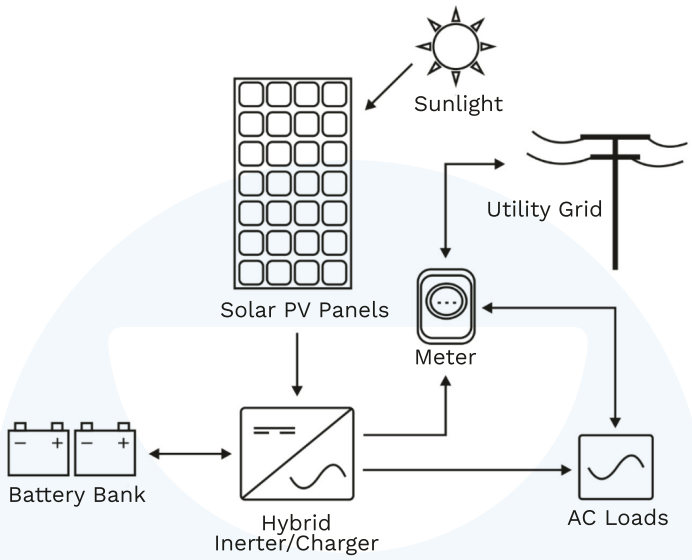


Fig. 5.1

International Solar Alliance (ISA)

- India and France collaborated to establish the ISA in order to encourage efforts to address climate change by deploying solar energy technologies, ensuring energy security, and speeding up to shift to solar energy usage by its member nations.
- The International Solar Alliance (ISA) was launched on 30 November 2015 in Paris, France, by Narendra Modi, India's Hon'ble Prime Minister, and Francois Hollande, France's former Hon'ble President. Former United Nations Secretary-General Ban Ki-moon attended the launch, as did the heads of approximately 120 nations, who pledged their support for the Alliance's efforts to promote solar energy.
- The ISA's mission is to develop and deploy cost-effective and transformative solar energy solutions to assist member countries in pursuing low-carbon growth paths, with a particular emphasis on countries classified as Least Developed Countries (LDCs) and Small Island Developing States (SIDS).
- The ISA is guided by its 'Towards 1000' strategy, which aims to mobilise USD 1000 billion in investments in solar energy solutions by 2030 while providing clean energy access to 1000 million people and resulting in the installation of 1000 GW of solar energy capacity. This would contribute to the annual reduction of 1000 million tonnes of CO_2 from global solar emissions.
- The ISA pursues these objectives through a programmatic approach. Currently, the ISA offers nine comprehensive programmes, each of which is focused on a specific application that could aid in the scale-up of solar energy solutions. The programmes' activities are focused on four priority areas: analytics and advocacy, capacity building, programmatic

support, and readiness and enabling activities. These activities contribute to the establishment of a favourable environment for solar energy investments in the country.

- With the amendment of the ISA's Framework Agreement in 2020, all United Nations member states are now eligible to join. The ISA Framework Agreement currently has 101 signatories, of which 80 have submitted the necessary instruments of ratification to become full members of the ISA.

Suryamitra Skill Development Programme

- The National Institute of Solar Energy (NISE) is a Ministry of New and Renewable Energy-affiliated autonomous institute. The Government of India recently established an Apex National Centre for Research and Technology Development and Related Activities in the Areas of Solar Energy Technologies in the Country by converting its erstwhile Solar Energy Centre.
- As part of the MNRE's skill development initiatives, NISE has been designated as the nodal agency for the 'Suryamitra Skill Development Programme (SSDP).'
- SSDP aims to develop the skills of youth in light of the growing opportunities for employment in the installation, operation, and maintenance of solar energy power plants in India and abroad.
- In addition, the SSDP is designed to prepare candidates to become new entrepreneurs in the solar energy industry.

Wind Energy

- Wind energy is the kinetic energy formed by the movement of large air masses as a result of the sun's differential heating of the atmosphere.
- Wind energy is one of the fastest-growing forms of renewable energy. Wind energy usage is increasing globally, owing to the low cost of harnessing wind energy.
- Wind energy capacity installed onshore and offshore has increased by nearly 75 times in the last two decades, from 7.5 gigawatts (GW) in 1997 to around 564 GW in 2018, according to the International Renewable Energy Agency's (IRENA) latest data.
- It is one of the clean, renewable energy sources that have the potential to meet a substantial portion of energy demand indirect, grid-connected modes as well as remote applications such as water pumping, desalination, and telecommunications.
- The wind resource is more variable and is strongly influenced by terrain or geographic factors.
- Using the kinetic energy generated by flowing air, wind energy is used to generate electricity. Wind turbines and wind energy conversion systems convert this to electricity (WECS). It is a wind-powered machine that directly converts mechanical energy to electrical energy.
- The wind impacts the turbine's blades first, allowing them to rotate and so turning the turbine to which they are attached. This turns kinetic energy into rotational energy by rotating a shaft attached to a generator and producing electrical energy through electromagnetism. This electricity then charges batteries, which can be connected to the building's main power or to the power grid.
- The amount of energy that can be harvested from wind is proportional to the size of the turbine and the blade length. The output is proportional to both the rotor's dimensions and the cube of the wind speed. Wind energy potential increases by a factor of eight when wind speed doubles.

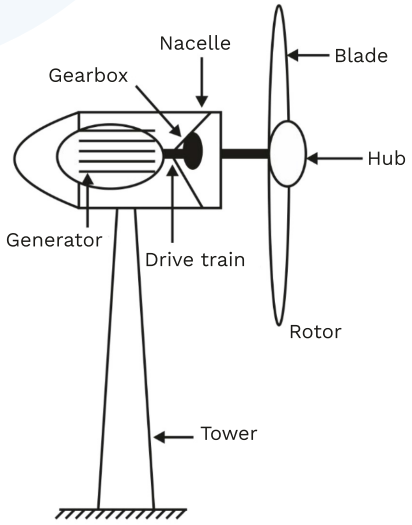


Fig. 5.2

- Wind energy is a non-polluting, environmentally friendly energy source with a low energy density. The cost of electricity generation is lower because there is no shortage of input and recurring expenses are almost non-existent.
- It can be made easily accessible in a variety of offshore, onshore, and remote areas. As a result, it aids in the distribution of electricity to remote and rural areas.

Ocean Energy

- Two types of energy are generated by the ocean: thermal energy (sun) and mechanical energy (tides and waves).
- The temperature differential provides thermal energy, as the sun's heat warms the surface water far more than the deep ocean water. A small portion of the heat stored in the oceans produces sufficient energy to power the whole Earth.
- Wave energy, tidal energy, and ocean thermal energy conversions (OTEC) are the primary oceanic energy sources. The hydrological cycle (energy emitted by solar-heated surface water evaporation) is also a kind of ocean energy.
- OTEC makes use of the temperature difference between the ocean's surface water and its depth. In tropical oceans, where the temperature differential is greatest (28°C at the surface and 5°C at a depth of 1000 m), ocean thermal energy can be harvested.
- The operation of the OTEC plant is based on the thermodynamic principle which states that if a heat source is available at a higher temperature and a heat sink is available at a lower temperature, then the temperature difference can be used to run a prime mover (turbine) that can convert a portion of the heat extracted from the source into mechanical energy. The leftover heat is discharged at a reduced temperature to the sink.
- There are three types of electricity conversion systems: closed-cycle, open-cycle, and hybrid. Using the ocean's warm surface water, closed-cycle systems vaporise a working fluid with a low boiling point, such as ammonia. Vapor expansion drives a turbine. The turbine then activates a generator to produce electricity. Open-cycle systems heat seawater by running under low pressures. This results in the production of steam, which is then fed through a turbine/generator. In addition, hybrid systems incorporate both closed- and open-cycle systems.
- A floating OTEC plant can generate power even in the middle of the ocean and can be used to power offshore mining.

Wave Energy

- The endless motion of the sea surface in the form of wind waves constitutes a source of energy, which is known as wave energy. Waves form as wind blows over the surface of open water in oceans and lakes.
- Waves are caused by tides, which vary depending on the lunar cycles.
- Depending on the lunar cycles, tides, winds, and weather, waves can vary in size and strength. As waves roll through the ocean, they create kinetic energy or movement. This movement can be used to power turbines, which, in turn, create energy that can be converted into electricity and power.
- There are also several ways of harnessing wave energy that utilises the up and down motion of the waves to power pistons/turn generators.
- Ocean waves are a tremendous source of energy. Energy extraction from waves is more efficient than direct wind energy collection, as wave energy is concentrated as a result of the wind's interaction with the free ocean surface.
- The most common method of harnessing this energy is through the use of a turbine coupled with an electrical generator. However, the primary issue with the system is its inability to generate electricity when the airflow velocity exceeds the design value.
- Wave energy has a higher energy density than both wind and solar energy. Wave energy devices, unlike solar and wind energy devices, do not require large landmasses.
- Wave energy is extremely expensive at the current state of technology. Additionally, the relative scarcity of accessible sites of large wave activity is a concern.

- Wave energy extraction equipment must be able to operate in a marine environment and withstand extremely severe stresses associated with storms. The physical presence of wave energy structures could also have an effect on the marine mammal and sea bird populations.

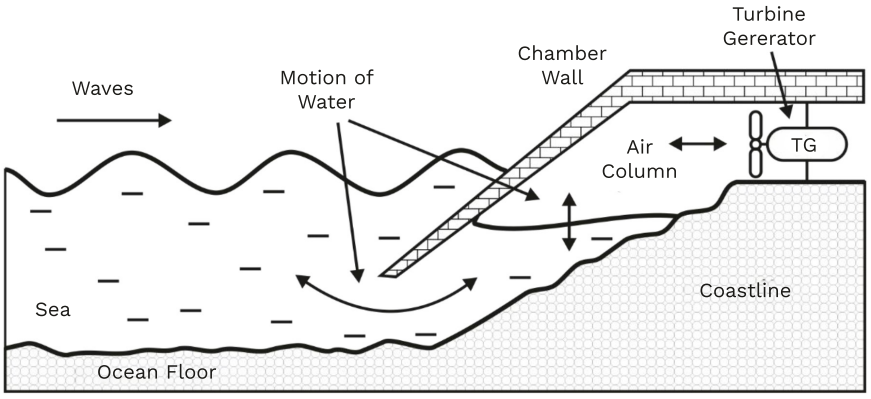


Fig. 5.3

Tidal Energy

- Energy can be generated from the tides formed by the gravitational pull of the sun and moon on Earth. This gravitational pull produces periodic changes in sea level, which can be utilised to generate electricity.
- Mechanical energy and thermal energy in the ocean are significantly unique. Tides (generated by the moon's gravitational pull) and waves (generated by wind) provide inconsistent energy while ocean thermal energy is quite stable.
- A barrage (dam) is used to convert tidal energy into electricity by forcing water through turbines that power a generator. There are three primary technologies for converting wave energy: channel systems that route waves into reservoirs, float systems that drive hydraulic pumps, and oscillating water column systems which compress air within a vessel using the waves. These systems either directly activate a generator or transfer mechanical energy to a working fluid, such as water or air, in order to power a turbine/generator.
- The turbine unit creates electricity when the tide is higher than the tidal basin. At low tide, the tide is less than the depth of the tidal basin, allowing water to flow out and power the turbine unit. A tidal power plant requires a big tidal range and an appropriate bay to construct a dam that will store water and discharge it during low tides to run a turbine.
- Tidal power schemes employ low-head turbines, which are larger and more expensive than comparable high-head turbines. Due to the variability of tidal ranges, turbines must function throughout a broad range of head fluctuations.

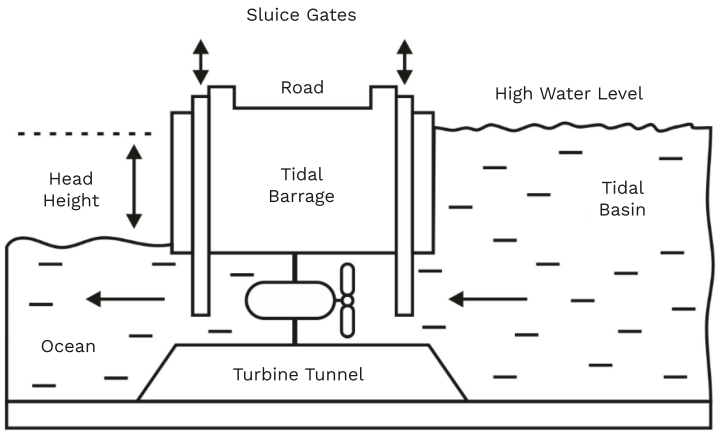


Fig. 5.4

Hydro Energy

- Hydro energy, or hydraulic energy, is the energy stored in moving water that is acquired when a river's gravitational descent is compressed to a single place by a dam or a flume. It is not only a non-polluting and environmentally friendly form of energy but also one of the most significant, second only to thermal energy.
- In a hydroelectric power plant, a turbine turns the kinetic energy of falling water into mechanical energy, which is then converted by a generator to electrical energy.
- Hydroelectric projects are often divided into two groups: small (SHP) and large hydropower projects (LHP).
- SHPs or micro-hydropower projects are hydropower facilities with a capacity of up to 25 MW. These SHPs are particularly helpful in remote areas where standard energy sources are difficult. SHPs can be installed on small rivers or streams with minimum or no damage to natural processes.
- LHP facilities have a capacity of above 30 MW power generation and depend on a dam or reservoir to store river water.
- Hydroelectric power facilities do not produce any primary waste or pollutants. The operation and maintenance costs are low. It has a short gestation period and is rapidly expanding. These power plants can only be constructed if sufficient water is available.
- Hydroelectricity generation plants can cause flooding or submerge substantial forest areas. These facilities are rarely situated close to load centres, mandating the development of lengthy transmission lines.

Biomass Energy

- Biomass energy is the direct combustion or the processing of waste paper, manure, wood, etc. into fuel by specific microbes in the absence of oxygen. Alcohol or methane gas are generated that are combustible and produce energy.
- Biomass energy conversion technology converts renewable biomass fuels to heat and electricity via procedures identical to those used with fossil fuels.
- Currently, the most common method of generating energy from biomass is the combustion, that is, direct firing. and this power plant's combustion systems are similar to those found in the majority of fossil-fuel-fired power plants.
- Biomass energy generated through energy plantation is less expensive than fossil fuel energy generation. Energy stored from biomass is simple and can be easily set up in rural and remote locations.
- The high moisture content of biomass makes its gathering and transportation expensive. Due to the demand for a relatively large land area, this technique cannot be applied in areas with a high land cost.

Biomass Energy Programmes in India

The Ministry of New and Renewable Energy (MNRE) has taken up the following programmes for meeting rural energy needs:

- **The Unnat Chulha Abhiyan** it was launched in 2014 to the promotion of improved biomass cookstove in the country for providing a clean cooking energy solution with a view to reducing the consumption of fuelwood with higher efficiency and low emissions.
- **National Biogas and Manure Management Programme (NBMMP)** promotes the creation of household residential biogas plants to meet the cooking gas needs of rural and semi-urban areas and minimise dependency on firewood.
- **Pradhan Mantri Ujjwala Yojana (PMUY)** it was launched in 2016 by the Ministry of Petroleum and Natural Gas under the slogan 'Swachha Indhan, Behtar Jeevan' (Clean fuel, better life) to protect women and children's health by providing them with clean cooking fuel via LPG. The Scheme provides financial assistance of Rs 1600 per LPG connection to BPL households, as well as an interest-free loan for the purchase of a stove and refilling by Oil Marketing Companies.
- The motto of **Project Urjagram** is to make effective use of locally available renewable energy sources to make rural areas' electricity self-sufficient. this power can be used for domestic purposes, small-scale industries, etc.

Biogas Development Programme

The programme consists of the following.

National Project on Biogas Development (NPBD)

This project was initiated with the following objectives. Efforts to reduce the labour of rural women by providing fuel for cooking purposes, provision of manure composed of organic matter for agriculture purposes, and connecting toilets to biogas plants to recycle human waste so that sanitation is improved.

Biogas Sector Research and Development

This initiative aims to minimise the capital cost of biogas plants and shorten the installation time by developing domestic materials and procedures. It provides research in the fields of microbiology, and biochemistry for improving the biogas field, particularly at low and high temperatures.

Village Energy Security Programme (VESP)

- To the villages lacking grid connectivity, the Ministry of New and Renewable Energy (MNRE) has initiated this Village Energy Security Programme.
- For communities with between 50 and 400 homes, MNRE distributes Central Financial Assistance (CFA) at a rate of $20000 per household.
- This programme is administered by institutions such as forest departments, rural development departments, NGOs, and cooperatives.

Remote Village Electrification Programme

The Remote Village Electrification Programme (RVEP) attempts to provide basic electrification through renewable energy sources to remote communities where grid connectivity is impractical or hugely expensive.

Geothermal Energy

- Geothermal energy is the thermal energy that naturally arises inside rock formations and the fluid enclosed inside these formations. Magma (hot, molten rock in the Earth's inner core) reaches temperatures as high as 4000°C.
- This heat energy is produced by magnetic or radioactive decaying processes, which occur within the Earth's core. In a geothermal power plant, subterranean water gradually absorbs heat from geothermal reservoirs. This hot water is then passed through a steam turbine, which turns thermal energy (heat) into electricity by means of a generator. Cooling the fluid and returning it to the heat source resumes the cycle.
- The injection wells ensure that the water drawn up is returned to the geothermal reservoir, where it regains its thermal energy, completing the cycle.

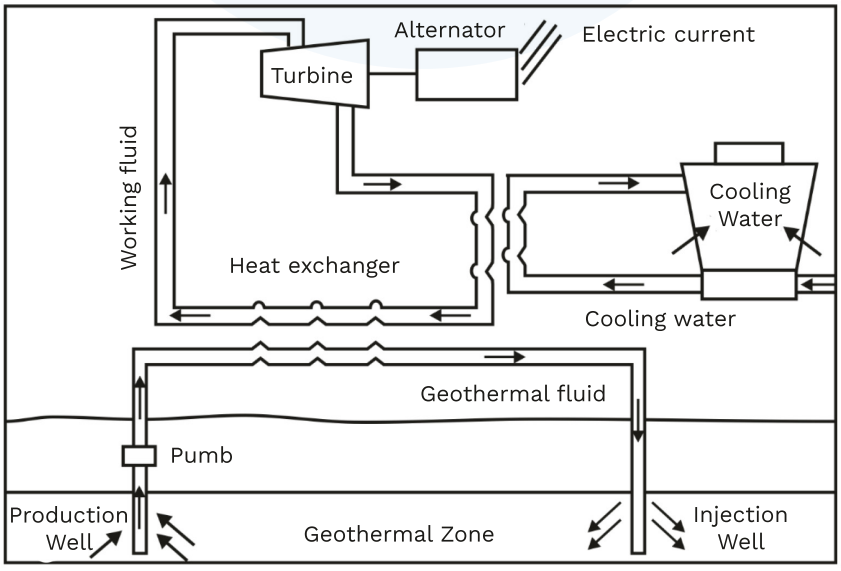


Fig. 5.5

- Geothermal energy is the least polluting and most adaptable form of renewable energy. It provides more net energy than other regular and alternative energy sources.
- In the electric supply system, geothermal sources could provide baseload power with minimal variable costs and at a reasonable price.
- Subsidence may result from the removal of huge quantities of heat. The steam and hot water streaming from the Earth may contain radon gas, CO_2, H_2S, and NH_3. If these gases are released into the atmosphere, severe air pollution will result. Before being released, these gases must be eliminated via chemical action.

Non-Renewable Energy Resources

- Non-renewable energy sources are fossil fuel sources that are found within the top layer of the Earth's crust. They range from volatile materials with a low carbon-hydrogen ratio such as methane to non-volatile materials composed of almost pure carbon, such as anthracite coal, etc.
- It is generally accepted that they are formed from the fossilised remains of dead plants and animals by exposure to heat and pressure in the Earth's crust over hundreds of millions of years. This is known as the biogenic theory.
- Fossil fuels are non-renewable resources because they take millions of years to form, and reserves are depleted much faster than new ones are formed. These are also known as exhaustible sources of energy.
- The production and use of fossil fuels raise environmental concerns. A global movement towards the generation of renewable energy is therefore underway to meet increased energy demands.

Coal

- Since the beginning of industrialisation, coal has been the most common source of energy. It is a solid fossil fuel formed in several stages as buried remains of plants that lived 300 to 400 million years ago and were subjected to intense heat and pressure over many millions of years.
- It is mainly composed of carbon (98%). It also contains hydrogen, oxygen, nitrogen, sulphur, and other elements in small quantities. It has low calorific value.
- value and high transportation cost. Further, it is highly pollutant and when burnt it produces CO_2 and CO.
- The United States and China are the major coal-producing countries and claim about 50% of the world's production of coal.
- India too has vast reserves of coal (7% of the world's coal reserves; 3rd largest in the world after USA and China). According to the rough estimate, the total recoverable coal in India is 75 billion tons.
- The extensive use of coal as a source of energy is likely to disturb the ecological balance of CO_2.

Petroleum and Natural Gas

- Petroleum is a thick liquid consisting of hundreds of combustible hydrocarbons along with a small amount of sulphur, oxygen, and nitrogen impurities.
- It is produced by the decomposition of dead organic matter from plants and animals that were buried under the lake and ocean sediments millions of years ago and subjected to high temperatures and pressures as part of the carbon cycle.
- Petroleum is used as both an energy source and industrial raw material in the petrochemical industry. Almost 40% of the energy needs of the world are met by petroleum.
- Natural gas is a naturally occurring hydrocarbon mixture consisting primarily of methane.

Nuclear Energy

- The energy released by a nuclear reaction, especially by fission or fusion is known as nuclear energy. It has the ability to produce electricity without greenhouse gas emissions.
- Nuclear energy is cleaner than many other forms of energy production. Essentially,

nuclear power would be 'carbon-zero' if the uranium is mined and transported in a more efficient way.

- Nuclear energy is cleaner than many other forms of energy production. Essentially, nuclear power would be 'carbon-zero' if the uranium is mined and transported more efficiently.
- Nuclear reactors make use of uranium as fuel and produce huge amounts of energy from small amounts of it: Moreover, other radioactive fuels such as Thorium are also available for power generation.
- Nuclear energy represents only 15 per cent of the electricity produced worldwide. Though in France, 80 % of electricity production is from nuclear energy and more than one-fourth of electricity in Europe comes from nuclear energy.

Nuclear Fission	Nuclear Fusion
It is the process of breaking a heavy nucleus into two or more light fragments with the liberation of a large amount of energy.	It is the process of fusing two light nuclei into a single nucleus with the liberation of a large amount of energy.
This process results in the emission of radioactive rays such as a, B, y radioactive rays.	This process does not emit any kind of radioactive rays. The end product of this process is harmless helium.
This process takes place at an ordinary temperature.	This process takes place at a very high temperature.
This process gives rise to a chain reaction.	This process does not give rise to a chain reaction.
In this process, neutrons are emitted.	In this process, positrons are emitted.

Table 5.1

- India is the only developing nation to have indigenously developed, demonstrated, and deployed nuclear reactors for electricity generation. This was made possible through several decades of extensive scientific research and technology development primarily in BARC. In this regard, Dr Homi Bhabha was the architect.

India's Three Stage Nuclear Programme

Stage I – Pressurised Heavy Water Reactor [PHWR]

- During the first stage of the programme, natural uranium-fueled pressurised heavy water reactors (PHWRs) generate electricity and plutonium-239 [U-238 → Plutonium-239 + Heat] is produced.
- Enrichment of Uranium to increase the concentration of U-235 is not needed in PHWR. U-238 is a radioactive isotope which can be fed directly into the reactor core. Natural uranium contains only 0.7% of the fissile isotope uranium-235, remaining 99.3 percent is mostly uranium-238, which is not fissile but can be converted to the fissile isotope plutonium-239 in a reactor.
- PHWR uses heavy water (deuterium oxide, D2O) as a moderator and coolant.
- PHWRs were a natural choice for the first phase because they are the most efficient reactor design for uranium utilisation when no enrichment is necessary.
- India accurately estimated that heavy water production facilities (needed for PHWRs) would be simpler to develop than uranium enrichment facilities (required for LWRs).
- Except for the two boiling reactor (BWR) units at Tarapur, the majority of India's nuclear power plants are first-stage PHWRs.

Stage II – Fast Breeder Reactor (FBR)

- In the second stage, FBRs do not require a moderator. It FBRs employ plutonium-239 obtained from spent nuclear fuel from the first stage and natural uranium.
- Plutonium-239 is fissioned to produce energy while uranium-238 in the fuel transmutes into further plutonium-239 in FBRs.
- Both Uranium-235 and Plutonium-239 can sustain chain reactions but Uranium-238 cannot sustain a chain reaction, as a result, it transmutes into Plutonium-239. Therefore, Stage II FBRs are designed to produce more fuel than they consume.
- After storing plutonium-239, thorium can be injected as a blanket material and transmuted into uranium-233 for the third stage.
- The extra plutonium produced by each fast reactor can be utilised to construct additional fast reactors, potentially boosting India's civil nuclear power capability until thorium-fueled third-stage reactors are constructed.

Stage III – Thorium-Based Reactors (TBR)

- A Stage III reactor or Advanced nuclear power system is comprised of a series of self-sustaining thorium-232-uranium-233 fueled reactors.
- This is a thermal breeder reactor which could ideally be refuelled using only naturally present thorium after its first fuel charge.
- According to two independent responses given to the Indian Parliament on 19 August 2010 and 21 March 2012, large-scale thorium deployment cannot be anticipated until three to four decades after the commercialisation of FBRs.
- As there is a long gap until direct thorium use in the three-stage programme, the nation is currently examining reactor designs which permit more direct thorium utilisation in parallel with the sequential three-stage project.
- Accelerator Driven Systems (ADS), Advanced Heavy Water Reactor (AHWR), and Compact High Temperature Reactor are the three solutions being considered.

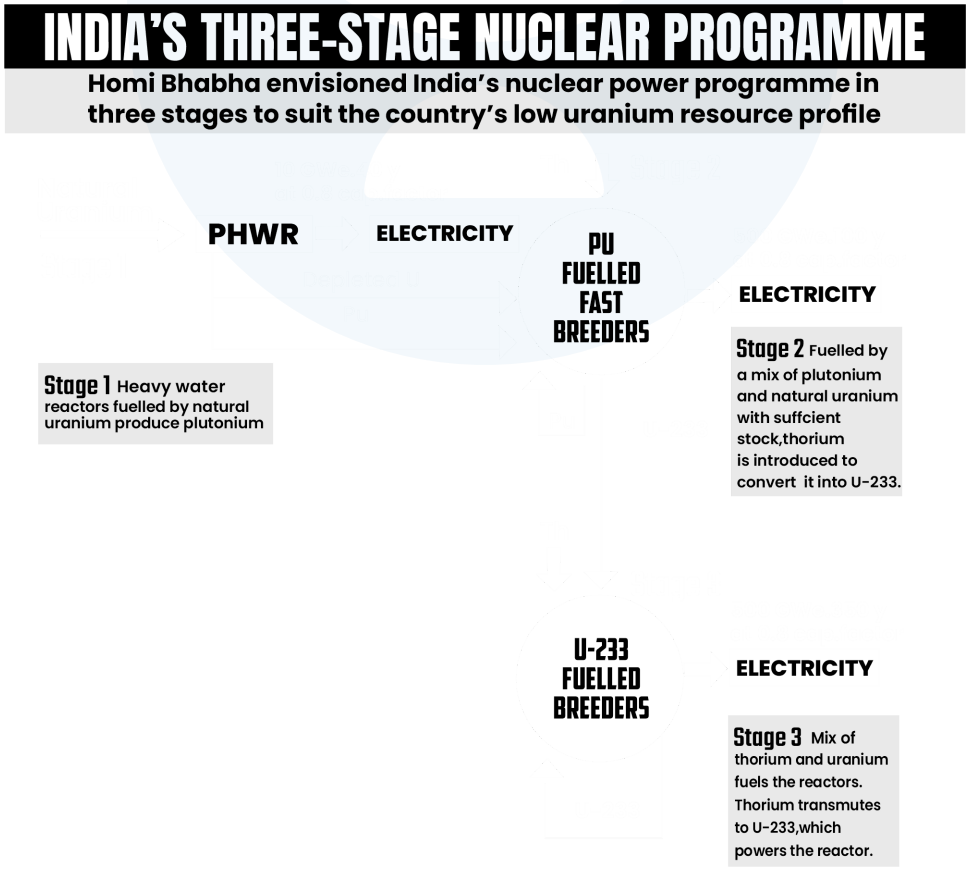


Fig. 5.6

Biofuels

- Biofuels are liquid fuels generated from agricultural crops (food plants and unique energy crops), forests, fishery, garbage, etc.
- Biofuels give significant benefits for sustainable growth and supplement traditional energy sources to meet the fast-expanding demand for petroleum products caused by rapid economic expansion, and the energy needs of the massive population.
- Some of biofuel crops are pongamia, jatropha, karanja, neem, maize, and sugarcane.
- The most common biodiesel blend is B20, which ranges from 6% to 20% biodiesel blended with petroleum diesel. However, B5 (a biodiesel blend of 5% biodiesel, and 95% diesel) is also commonly used in fleet vehicles. B20 and lower-level blends can be used in many diesel vehicles without any engine modification.
- In 2009, the Ministry of New and Renewable Energy established a National Policy on Biofuels to promote biofuels in the country.
- Biofuels have garnered global attention over the last decade, and it is critical to keep up with the rapid advancements in the field. Biofuels are strategically important in India because they align with the government's ongoing initiatives such as Make in India, Swachh Bharat Abhiyan, and skill development. They also present an excellent opportunity to integrate with the ambitious targets of doubling farmer income, reducing imports, creating jobs, and converting waste to wealth.
- The policy categorises biofuels as 'Basic Biofuels,' which include first-generation (1G) bioethanol and biodiesel, and 'Advanced Biofuels,' which include second-generation (2G) ethanol, municipal solid waste, third-generation (3G) biofuels, and bio-CNG. This categorisation permits the provision of effective fiscal and financial incentives for each category.
- The policy expands the range of raw materials available for ethanol production by permitting the use of sugarcane juice, sugar beet, sweet sorghum, corn, and damaged food grains such as broken rice, rotten potatoes, etc., which are unfit for human consumption.
- With the consent of the National Biofuel Coordination Committee, the Policy permits the use of surplus cereal grains to produce ethanol for use in gasoline mixes which helps the farmers to receive fair market price even during the surplus production phase.
- With an emphasis on advanced biofuels, the policy offers a Rs 5000 crore, six-year viability gap funding scheme for 2G ethanol biorefineries, additional tax incentives, and a better purchase price than for 1G biofuels.
- The policy encourages the development of supply chain mechanisms for the manufacture of biodiesel from non-edible oilseeds, spent cooking oil, and quick-growing crops.

Hydrogen

- Hydrogen is emerging as a significant source of energy because it contains no carbon and is a non-polluting fuel, whereas hydrocarbons contain a net carbon content of 75–85%. It has the highest energy content by weight and lowest energy content by volume.
- It is the most abundant element on the planet, but not commercially viable Hydrogen can be produced from hydrocarbons such as natural gas, oil, and coal via processes such as steam methane reforming and coal gasification, as well as from renewable sources such as water, sunlight, and wind via electrolysis and photolysis, along with other thermochemical processes.
- On 15 August 2021, India's Prime Minister announced the launch of the National Hydrogen Mission (NHM) in commemoration of the country's 75th anniversary of independence, with the goal of reducing carbon emissions and increasing the use of renewable energy sources.
- The National Hydrogen Mission aims to reduce carbon emissions and increase the use of renewable energy sources while ensuring that India's efforts are consistent with global best practices in technology, policy, and regulation.
- The Indian government has allocated Rs 25 crore in the Union Budget 2021–22 for hydrogen energy research and development,

with the goal of producing three-fourths of its hydrogen from renewable sources by 2050.

- In 2003, the National Hydrogen Energy Board was established by the erstwhile Ministry of Non-Conventional Energy Sources (MNES) to oversee the implementation of the National Hydrogen Energy Program.
- The National Hydrogen Energy Road Map was drafted in October 2003 and approved in 2006 by a steering committee chaired by Ratan Tata.
- This road map identifies two significant initiatives: the Green Initiative for Future Transportation (GIFT) and the Green Initiative for Power Generation (GIP).
- Hydrogen has been colour-coded according to its source of production, with an emphasis on the use of Green Hydrogen, which contributes to greenhouse gas reductions and increases the share of renewable energy in total energy consumption.

Colour Code	Source or Process Used
Green hydrogen	Produced through water electrolysis process by employing renewable electricity. The reason it is called green is that there is no CO_2 emission during the production process. Water electrolysis is a process which uses electricity to decompose water into hydrogen gas and oxygen.
Blue hydrogen	Sourced from fossil fuels, the CO_2 is captured and stored underground (carbon sequestration). Companies are also trying to utilise the captured carbon called carbon capture, storage, and utilisation (CCSU). Utilisation is not essential to qualify for blue hydrogen. As no CO_2 is emitted, so the blue hydrogen production process is categorised as carbon neutral.
Gray hydrogen	Produced from fossil fuel and commonly uses steam methane reforming (SMR) method. During this process, CO_2 is produced and eventually released into the atmosphere.
Black/brown hydrogen	Produced from coal- the black and brown colours refer to the types of bituminous (black) and lignite (brown) coal. The gasification of coal is a method used to produce hydrogen- it is a very polluting process, and CO_2 and carbon monoxide are produced as by-products and released into the atmosphere.
Turquoise hydrogen	Extracted by using the thermal splitting of methane via methane pyrolysis. The process, though at the experimental stage, remove the carbon in a solid form instead of CO_2 gas.
Purple hydrogen	Made using nuclear power and heat through combined chemo thermal electrolysis splitting of water.
Pink hydrogen	Generated through electrolysis of water by using electricity from a nuclear power plant.
Red hydrogen	Produced through the high-temperature catalytic splitting of water using nuclear power thermal as an energy source.
White hydrogen	Naturally occurring hydrogen.

Table 5.2

Energy Conservation in India

- Petroleum Conservation Research Association (PCRA) is an Indian government body created in 1977 which has been engaged in promoting energy efficiency and conservation in every walk of life. In the recent past, PCRA has done mass media campaigns on television, radio, and print media.

- An impact assessment survey by a third party revealed that due to these mega campaigns by PCRA, the overall awareness level has got up leading to the saving of fossil fuels worth cores of rupees besides reducing pollution.
- The Bureau of Energy Efficiency (BEE) is a statutory body set up in 2002 under the provisions of the Energy Conservation Act, 2001.
- The mission of the BEE is to assist in developing policies and strategies with a thrust on self-regulation and market principles within the overall framework of the Act.
- Energy Conservation Act 2001 requires large energy consumers to adhere to energy-consumption norms; new buildings to follow the Energy Conservation Building Code; and appliances to meet energy-performance consumption.

Ujwal Discom Assurance Yojana (UDAY)

- As a result of providing energy at significantly below-cost rates in India, DISCOMs (Power Distribution Companies) have incurred losses and incurred debt. These financially strained DISCOMs are unable to deliver adequate power at reasonable prices, hence affecting the economic growth and development.
- The government of India introduced the Ujwal Discom Assurance Yojana on 5th November 2015 to make affordable and accessible electricity available to everyone 24/7, to support Power Distribution Companies (DISCOMs) of the states to resolve financial and operational difficulties, and to suggest reforms in the power generation, transmission, coal, and energy efficiency sectors.
- The government launched 'UDAY 2.0' as the next phase of power distribution reforms after evaluating the scheme's impact and potential.

Leadership in Energy and Environmental Design (LEED)

- Leadership in Energy and Environmental Design (LEED) is the most popular green building certification program which is used worldwide. It also issues energy compliance certificates.
- LEED includes a set of rating systems for the design, construction, operation, and maintenance of green buildings, homes, and neighbourhoods that aims to help building owners and operators be environmentally responsible and use resources efficiently. It was developed by the non-profit U.S. Green Building Council (USGBC).
- LEED standards are backed by the Confederation of Indian Industry and have been in use in India since 2001.

Green Rating for Integrated Habitat Assessment (GRIHA)

GRIHA is a building rating system which aids in analysing the performance (energy, environment) of buildings relative to nationally recognised standards, therefore creating detailed criteria which constitute a green building.

GRIHA Variants

GRIHA

- The GRIHA V 2015 rating system consists of 31 criteria organised into divisions such as site planning, construction management, occupant comfort, and well-being, sustainable building materials, performance monitoring, and validation, and innovation.
- It is a 100-point system with a few required core points and the rest being optional. The number of acquired points determines the certification level (one to five stars). The minimum requirements for certification are 25 points.
- All structures in the design phase with a total floor area greater than 2,500 m^2 are eligible for GRIHA certification.

Griha Precertification

- In compliance with the Ministry of Environment, Forest and Climate Change (MoEFCC), the GRIHA Council has developed a methodology that provides faster environmental approval through GRIHA precertification.
- Except for industrial complexes, all structures in the Pre-Design/Design Stage are eligible for GRIHA precertification.

Small Versatile Affordable (SVA) GRIHA

GRIHA and TERI developed SVAGRIHA a green building rating system which is conceptually simple, swift, and more affordable. This rating is specifically for projects with a constructed size of less than 2500 m^2. Individual homes, small offices, and commercial buildings can be designed and rated with assistance from SVAGRIHA.

GRIHA For Affordable Housing (AH)

GRIHA created this variant, Pradhan Mantri Awas Yojana (PMAY) intending to bring sustainability to low-income groups. It helps to cut operational expenses, and greenhouse gas emissions, improve resource efficiency, and measure the dwellings' holistic performance throughout their life cycle. All planned projects with approval from a government agency stating that the project is being developed in accordance with the PMAY are eligible for a GRIHA AH grade.

GRIHA For Existing Buildings

This rating system is a 100-point scale with 12 distinct criteria organised into 7 parts, including site parameters, maintenance and housekeeping, human health and comfort, social aspects, energy, water, and bonus points. This certification is open to all operational buildings with a constructed area of more than 2500 m^2.

GRIHA For Large Developments

This rating system was developed in partnership with TERI for big large developments. It is a standardised approach for evaluating the environmental implications of large-scale developments, such as cities, neighbourhoods, and townships. This ranking applies to all projects with a site area of 50 hectares or more.

GRIHA For Cities

This rating was designed as a configuration for the sustainable development of a city attained by rating the 'greenness' of both existing and planned cities. Quantitative and qualitative standards are incorporated in the rating to cover diverse sustainable development aspects. The ranking specifies the performance benchmarks for energy, water, and waste; and evaluates smart governance, social wellness of the project.

Practice Exercise

1. Hydrogen fuel cell vehicles produce which of the following as exhaust?
(A) NH_3
(B) CH_4
(C) H_2O
(D) H_2O_2

2. Given below are the names of four energy crops. Which one of them can be cultivated for ethanol? **[UPSC 2010]**
(A) Jatropha
(B) Sugarcane
(C) Pongamia
(D) Sunflower

3. In the context of alternating sources of energy, biofuels can be obtained from:
(A) Jatropha
(B) Rice
(C) Bamboo
(D) Wheat

4. With reference to technologies for solar power production, consider the following statements. **[UPSC 2014]**
I. Photovoltaics is a technology that generates electricity by direct conversion of light into electricity.
II. Photovoltaics generates alternating current (AC), while solar thermal generates direct current (DC).

III. India has a manufacturing base for solar thermal technology, but not for photovoltaics.

Which of the statements given above is/are correct?

(A) I only
(B) II and III
(C) I, II, and III
(D) None of these

5. Consider the following statements regarding biogas.

I. This is a mixture of different gases, which is formed after the breakdown of an organic compound.
II. This is produced by the decomposition of fungi and algae in an open system where sufficient oxygen supply is available.
III. This gas can be also compressed same as CNG and used for energy needs.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

6. Which of the following are the components of biogas?

I. Methane
II. Carbon dioxide
III. Nitrous oxide
IV. Hydrogen sulphide

(A) I, II, and IV
(B) I, II, and IV
(C) II, III, and IV
(D) All the above

7. Consider the following statements.

I. Biomass can be converted into liquid biofuels by distillation.
II. Biomass is a renewable form of energy and is abundantly available on the Earth in the form of firewood, agricultural residue, etc.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

8. Consider the following statements regarding ocean energy.

I. Ocean Thermal Energy Conversion (OTEC) uses the temperature difference between the surface of the ocean and depths of up to 1000 m.
II. Tidal power uses the same principle employed in hydroelectric power generation to extract energy from the hydrostatic head created by the rising and a difference falling of tides.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

9. Both power and manure are provided by:

(A) Nuclear plants
(B) Biomass plants
(C) Tidal plants
(D) Hydroelectric plants

10. Consider the following statements regarding Ocean Thermal Energy Conversion (OTEC):

I. OTEC is based on the temperature difference between surface water and deep water in an ocean.
II. It can be harnessed in tropical region oceans where the temperature gradient is high.
III. OTEC power can be used for offshore mining and processing of manganese nodules.

Which of the above statements is/are correct?

(A) I only
(B) II and III only
(C) I and II only
(D) I, II, and III

11. Geothermal power is extracted from:

(A) Heat stored in the Earth
(B) Heat stored in the soil
(C) Heat stored in the water
(D) All of the above

12. Consider the following statements regarding biomass energy.

I. It is obtained through the process of photosynthesis.

II. It is considered another form of solar energy.

III. The cost of obtaining biomass energy is less than the energy obtained from fossil fuels.

Which of the above statements are correct?

(A) I only
(B) I, II, and III
(C) I and II only
(D) I, II, and III

13. Which power plant works based on the gravity of Earth?

(A) Wind forms
(B) Hydropower
(C) Geothermal
(D) Solar

14. Consider the following statements regarding solar panels.

I. It has a high maintenance cost.

II. It has no moving parts.

III. It provides pollution-free, noiseless energy.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

15. Consider the following statements.

I. Wind energy is kinetic energy generated by virtue of movement of large air mass caused by differential heating of the atmosphere by the Sun.

II. Wind resource is continuous and does not influence by terrain or geographical factors.

III. Wind energy has low energy density.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

16. Renewable energy resources:

(A) Regenerate continuously
(B) Regenerate in a short period
(C) Regenerate in a long period
(D) None of these

17. Non-renewable energy sources are:

(A) Fossil fuel sources are found in the top layer of Earth's crust.
(B) Ranging from low carbon-hydrogen ratio volatile materials like methane to non-volatile materials composed of pure carbon like anthracite coal.
(C) Formed from fossilised remains from dead plants and animals by exposure to heat and pressure for hundreds of millions of years. (As per the biogenic theory).
(D) All of the above

18. Consider the following statements regarding the International Solar Alliance (ISA).

I. It is the first treaty-based international organisation, which is headquartered in India.

II. It was launched at the Conference of Parties (COP21) in Paris, France.

III. Most of its members are the countries lying on or within the tropics.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

19. With reference to hydrogen as a fuel, consider the following statements.

I. Hydrogen has the highest energy content of any common fuel by weight.

II. Hydrogen fuel cells produce water as a by-product.

Which of the above statements is?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

20. Consider the following statements regarding geothermal energy.

I. The slow decay of radioactive particles in the Earth's core produces geothermal energy.

II. Geothermal energy is relatively expensive with respect to other sources of renewable energy.

III. The carbon footprint of a geothermal power plant is minimal.

Which of the above statements is/are correct?

(A) I only
(B) II and III only
(C) I and III only
(D) I, II, and III

21. Which one of the following statements is incorrect regarding biofuels?

(A) Biofuels produce fewer greenhouse gases compared to fossil fuels.
(B) Second-generation biofuels are produced from non-food crops like agricultural residues.
(C) Fourth-generation biofuels include electro fuels and photobiological solar fuels.
(D) Biofuels have a higher energy output than fossil fuels.

22. Consider the following statements regarding solar energy systems.

I. Solar energy systems on buildings have minimal impact on the environment.

II. The efficiency at which photovoltaic cells convert sunlight to electricity varies by the type of semiconductor material used.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

23. The main constituent of LPG is:

(A) Methane
(B) Ethane
(C) Hydrogen
(D) Propane + butane

24. The main constituent of CNG is:

(A) Methane
(B) Butane
(C) Ethane
(D) Propane + butane

25. Which of the following is not a renewable source of energy?

(A) Sun
(B) Natural gas
(C) Wind
(D) Ocean tides

26. The radiations absorbed by the ozone layer are:

(A) Infrared
(B) Ultraviolet
(C) Visible
(D) Gamma rays

27. The radiations emitted by a hot furnace are:

(A) Ultraviolet
(B) Infrared
(C) X-rays
(D) Microwaves

28. Consider the following statements regarding small hydropower projects.

I. These projects have a short gestation period.

II. These have low operating and maintenance costs.

III. These are installed away from load centres, thus requiring a long transmission line.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

29. The efficiency of the solar cooker can be increased by placing which of the following?

(A) Plane mirror
(B) Convex mirror
(C) Convex lens
(D) Concave lens

30. Consider the following statements regarding tidal power.

I. Tides are created by the movement of wind on the surface of the sea.

II. Tidal power has traditionally suffered from relatively high costs and the limited availability of sites.

Which of the above statements(s) is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

31. Which among the following is/are adverse impacts due to tidal power generation on the environment?

(A) Interference with spawning and migration of fish.
(B) Pollution and health hazards in estuary due to blockage of the flow of polluted water in the sea.
(C) Navigational hazard.
(D) All of the above

32. With reference to the disadvantages of wave energy, consider the following statements.

I. Energy extraction equipment should be capable of operating in a marine environment and withstanding very severe stress during storms.

II. Marine mammal and sea bird populations can be affected due to the presence of energy extraction equipment.

III. There is a relative scarcity of accessible sites for harnessing wave energy.

Which of the above statements are true?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

33. A solar cell converts:

(A) Heat energy into electrical energy
(B) Solar energy into electrical energy
(C) Heat energy into light energy
(D) Solar energy into light energy

34. Floating generators are used in the sea to harness which of the following?

(A) Tidal energy
(B) Wave energy
(C) Hydel energy
(D) Energy from OTEC power plant

35. How much temperature difference is needed between the upper layers and the deeper layers of the ocean to install an OTEC power plant?

(A) 40°C
(B) 50°C
(C) 20°C
(D) 30°C

36. Which of the following may be classified as third-generation biofuels?

I. Algae-based biofuels

II. Jatropha-based biofuels

III. Corn-based biofuels

Select the correct option using the codes given below.

(A) I only
(B) II and III only
(C) I and II only
(D) I, II, and III

37. Consider the following statements regarding hydropower.

I. An impoundment facility of a hydropower plant generally uses a dam to store water in a reservoir.

II. Run-of-river facility of a hydropower plant captures the kinetic energy in rivers or streams, without a large reservoir, and sometimes without the use of dams.

III. Small hydropower (SHP) plant is defined as any hydropower project, which has an installed capacity of less than 50 MW.

Which of the above statements are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

38. Consider the following statements regarding biodiesel.

I. It is produced by fermentation.
II. It is bio-degradable and reduces hydro-carbon emissions.

Which of the above statement(s) is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

39. The daily differences in the amount of solar energy received by Earth are due to its:

(A) Size
(B) Shape
(C) Atmosphere
(D) Rotation

40. Which of the following areas is preferred for solar power plants?

(A) Coastal areas
(B) Mountain tops
(C) Hot arid zones
(D) High rainfall zones

41. Consider the following statements regarding renewable energy resources.

I. These are the source which can be renewed by nature again and again.
II. These are the source whose supply is not affected by the rate of their consumption.
III. These sources are exhaustible.

(A) I and II only
(B) II and III only
(C) I and III only
(D) All of the above

42. Which one of the following is not a disadvantage of solar energy?

(A) Electricity generation is limited by climate.
(B) High installation cost and low power stations.
(C) Batteries being charged by solar power required replacement and also require large space to occupy.
(D) Clear, a noise-free energy source with little maintenance as having no moving part.

43. Ocean Thermal Energy Conversion (OTEC) works on the principle of:

(A) Energy stored by waves in the ocean.
(B) Tides arising out in the ocean.
(C) Pressure difference at different levels in the ocean.
(D) Temperature difference at different levels in the ocean.

44. Consider the following statements.

I. Shale gas is colourless, odourless, and lighter than air.
II. Shale gas is an unconventional source of methane, just like coal-bed gas and tight gas.
III. Shale gas is used in Europe but not in the United States.

Which of the above statements is/are correct?

(A) I only
(B) I and II only
(C) I and III only
(D) I, II, and III

45. Consider the following statements regarding National Solar Mission.

I. National Solar Mission is one of the eight key national missions, which comprise the National Action Plan on Climate Change.
II. Government has revised the target of grid-connected solar power projects from earlier 20000 MW to 100000 MW by the year 2022.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

46. Consider the following statements.

I. Thorium cannot be used directly to produce nuclear energy.
II. Thorium does not contain any fissile isotope.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (C)	**2.** (D)	**3.** (A)	**4.** (A)	**5.** (C)	**6.** (B)	**7.** (C)	**8.** (C)	**9.** (B)	**10.** (D)
11. (A)	**12.** (D)	**13.** (B)	**14.** (B)	**15.** (C)	**16.** (B)	**17.** (D)	**18.** (D)	**19.** (C)	**20.** (C)
21. (D)	**22.** (C)	**23.** (D)	**24.** (A)	**25.** (B)	**26.** (B)	**27.** (B)	**28.** (D)	**29.** (A)	**30.** (B)
31. (D)	**32.** (D)	**33.** (B)	**34.** (B)	**35.** (C)	**36.** (A)	**37.** (A)	**38.** (B)	**39.** (D)	**40.** (C)
41. (A)	**42.** (D)	**43.** (D)	**44.** (B)	**45.** (C)	**46.** (C)				

6 Environmental Impact Assessment (EIA)

Introduction

- The environmental impact assessment is a tool of the 20th century to strike a balance between development and the environment.
- The term Environmental Impact Assessment (EIA) or Environmental Impact Statements (EIS) are often used interchangeably to analyse environmental impacts.
- The term EIA is used in India and Canada. In the USA, it is called an Environmental Assessment (EA).
- Environment Impact Assessment (EIA) is the study that forecasts the environmental impact of planned activity or project. EIA is a decision-making tool; It examines different possibilities of a project, its positive and negative impacts, and aims to determine the one with the optimal balance of economic viability, environmental costs, and benefits. The purpose of an EIA is to assure that development can be sustained with little environmental impact.
- It is intended to be a methodical procedure that results in Environmental Statement (ES). The environmental statement must cover the direct and indirect consequences of the development on a variety of components, including the biotic, abiotic factors, and archaeology. The ES must also provide a non-technical analysis so that affected parties can comprehend the proposed action and its expected implications.
- The National Environmental Policy Act (NEPA) launched EIA in 1970 in the United States. In India, EIA was initiated between 1976–1977, when the Planning Commission requested the Department of Science and Technology to study river valley projects from an environmental perspective and later extended to include projects requiring Public Investment Board clearance. The Union Ministry of Environment, Forest, and Climate Change (MoEFCC) is responsible for coordinating the EIA.
- EIA has addressed the fundamental aspects such as the location and surrounding area, Occupational health and safety, Water and Air quality, transportation of raw materials, handling, treatment, and disposal of effluents.
- The development projects must prepare an environmental impact statement that includes the following
  - Land impacts (land degradation, collapse)
  - Air, Water, Noise Pollution
  - Loss of biodiversity
  - Socioeconomic impact including human migration, cultural destruction, and health aspects
  - Risk analysis, waste management, and disaster management
- Up to now, the environmental impact assessment of development projects has been conducted in accordance with the 'Executive Order' issuing the provision of the Environment (Protection) Act, 1986, to ensure the execution of the prescribed precautions.

EIA Regulations

- The EIA regulations establish two development schedules.
- An EIA is usually required for Schedule-1 projects. For example,. Major power plants, major road construction projects, etc.
- An EIA is required for Schedule-2 projects if the development is projected to have a major environmental impact due to its nature, magnitude, or location. For example, Quarries, overhead transmission lines, coke ovens, dairy processing, etc.
- A proposed development becomes a Schedule-2 development only if it is above the threshold. EIA is not necessary simply because a project falls under one of the Schedule-2 categories and exceeds the Schedule-2 threshold.
- The first legislative step was through Environment Protection Act, 1986 (EPA, 1986) under which EIA was mandatory for environmental clearance for setting up projects.
- This notification issued by MoEF&CC was later upgraded to EIA, 2006 under EPA, 1986., which included new projects such as thermal power plants, Highways, railways, Dams, etc.

Salient Features of EIA 2006

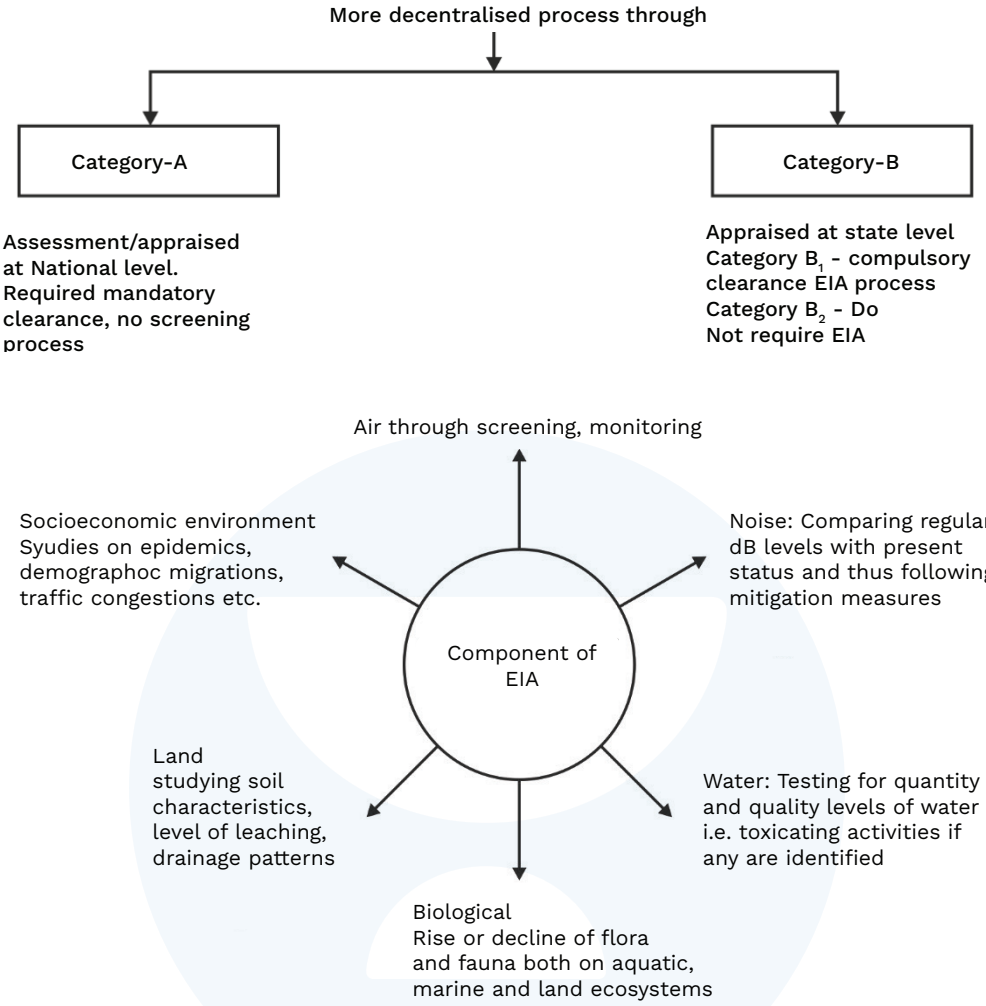


Fig. 6.1

Environmental Impact Assessment Process

The stages of an EIA process will depend upon the requirements of the country. However, most EIA processes have a common structure, and the application of the main stages is a basic standard of good practice.

Steps Involved in the EIA Process

Project Proposal

Before submitting a project proposal, the promoter of a large development project must notify IAA (Impact Assessment Agency) in writing. The project proposal must possess all required information to go to the screening phase. The initiation of the EIA process is notified by the presentation of a project proposal.

Project Evaluation

This requires the use of EIA for projects with the potential for major environmental impacts.

Preliminary Assessment

If a project fails the screening, the developer may well be asked to do a preliminary assessment. This requires appropriate study, examination of existing data, and expert opinion to identify

the key consequences of the project on the surrounding environment, anticipate the extent of those impacts, and concisely assess their significance to decision-makers. It is in the developer's best advantage to do a preliminary assessment because, in practice, this phase can eliminate the need for a comprehensive EIA.

Scoping

This process aims to identify early on the most important environmental issues among the myriad of potential impacts of a project and the viable options.

Evaluation of Options

This is to ensure that the applicant has examined alternate project locations, sizes, processes, layouts, and operational circumstances. Once alternatives have been evaluated, a contingency plan should be drafted for the chosen option and accompanied by an environmental management plan (EMP) to guide the promoter in implementing environmental changes. The EMP is a critical input for monitoring the clearance circumstances; hence, the EMP should include monitoring details.

EIA Report

The promoter creates a comprehensive Project report clearly and coherently based on the information provided in the EIA report regarding various environmental scenarios. The IAA evaluates the procedure according to MoEFCC guidelines.

Public Hearing

- The legislation mandates that the public must be notified and consulted over a proposed development after the EIA report has been completed. Before plans are sent to MOEFCC for environmental clearance, the state pollution control boards will conduct a public hearing.
- The executive summary of the EIA is accessible to everyone who may be impacted by the proposed project. The impacted parties may include local associations, environmental organisations involved in the region, or any person residing at the project site or evacuation sites. They will be permitted to provide the State Pollution Control Board with oral or written opinions.

Decision-Making

The decision-making procedure includes consultation between the project owner (with the aid of a consultant) and the impact assessment authority (assisted by an expert group if needed). The environmental clearance decision is reached through a series of steps, including the examination of the EIA and EMP.

Monitoring the Clearance Conditions

- A project must be monitored during the construction and operation periods. This is done not only to guarantee that promises are met, but also to determine whether or not the EIA reports' projections were accurate.
- When the effects surpass the projected levels, corrective measures must be performed. Monitoring also permits the regulatory body to assess the accuracy of forecasts and the conditions of environmental management plan implementation (EMP).
- The promoter, IAA, and pollution control boards should oversee the implementation of the criteria. Once every six months, the promoter must submit a report demonstrating IAA compliance.

Risk Assessment

- An instrument for estimating the probability of harm occurring from the presence of dangerous conditions or materials at a project site. Risk represents the likelihood and significance of a potential hazard being realised.
- The objective is to evaluate the relative significance of the expected impacts in order to concentrate on the most detrimental ones. A formal definition of significance combines consequence and probability.

  Significance = Consequence × Likelihood

Environmental Appraisal Committee (EAC)

- To elicit multi-disciplinary inputs for appraisal of projects, Union Government has constituted the EAC to ensure multi-

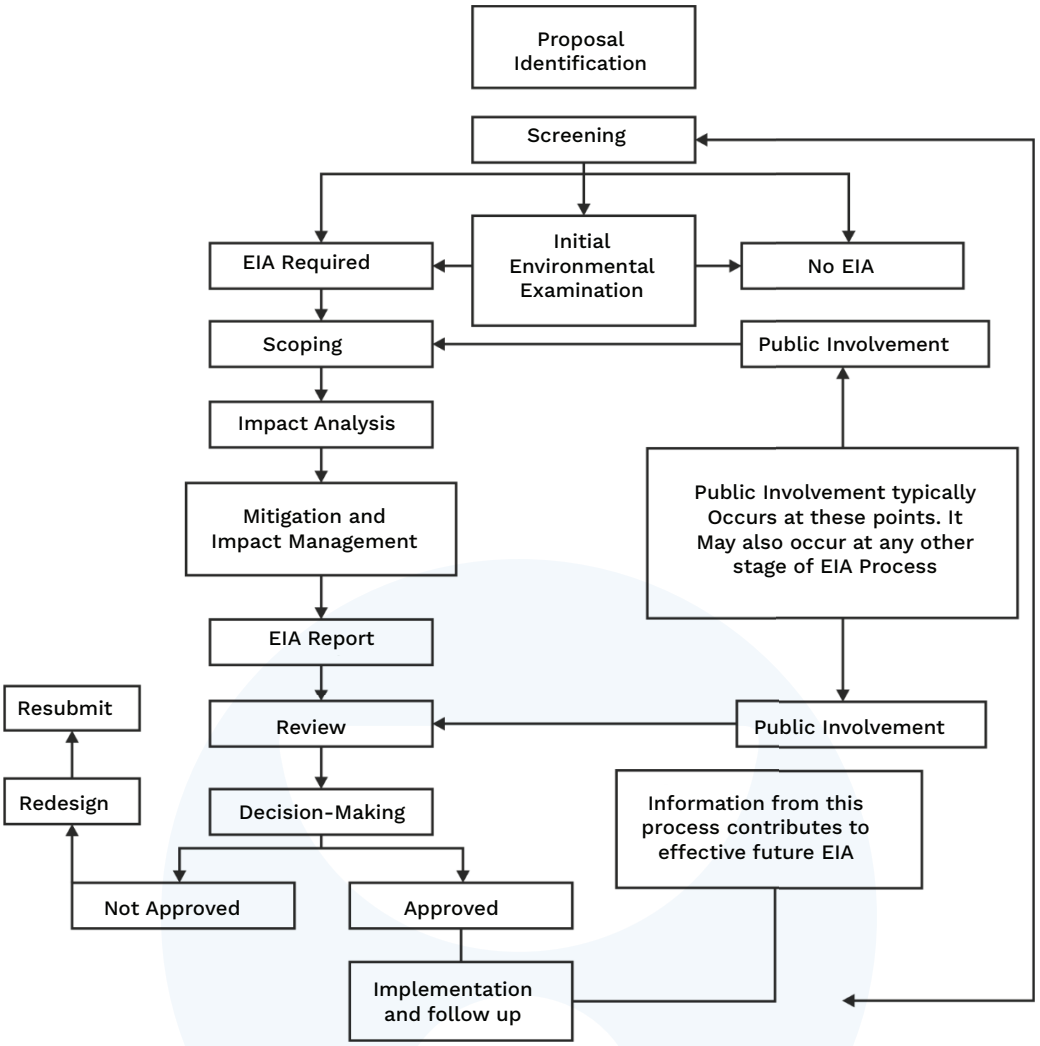


Fig. 6.2

disciplinary input required for environmental appraisal of development projects.

- Mining projects
- Industrial projects
- Thermal power projects
- River valley, multipurpose irrigation projects, hydro-electric plants
- Infrastructure development and miscellaneous projects
- Nuclear power projects

- The project authorities have to furnish the following documents for environmental appraisal of developmental projects: Detailed Project Report (DPR), Filled in questionnaire, and Environmental Impact Statement (EIS) along with the environmental management programme. EIS should provide the possible impact (+ve or -ve) of the project.

Roles of Parties in the EIA Process

- EIA involves numerous partners, which are categorised by their respective roles within the process. The primary responsibilities of various entities are given.
- During the project planning phase, the promoter chooses the project type. Later, the promotermust write the detailed project report/ feasibility report and submit the executive summary, which must include the project's features and the EIA study's conclusions that are made available to the public.
- The promoter must approach the relevant State Pollution Control Board (SPCB) for a No Objection Certificate (NOC) and hold a public hearing. The promoter seeks environmental clearance from the Impact Assessment Agency (IAA) after the public hearing.

- An environmental consultant must be familiar with the current legal and procedural regulations for getting environmental approval for a proposed project. The promoter and environmental consultant are responsible for providing the SPCB and IAA with all environment-related information necessary for both agencies.
- The State PCBs are accountable for managing the proposed development's compatibility with current operating and prescribed requirements. The PCB will subsequently give its NOC if the development is compliant. The PCB host a public hearing and IAA receives the information regarding this public hearing.
- The public plays a significant part in EIA. Through press advertisements, interested parties will be asked to evaluate the information and express their feedback on the proposed development that requires environmental clearance.
- The project's promoter will have the opportunity to present his proposal during this procedure. Based on this evaluation, if a project is accepted, then IAA will develop a set of implementation suggestions and is responsible for environmental monitoring during the project implementation.

Strategic Environment Assessment (SEA)

- Strategic Environment Assessment (SEA) refers to a methodical review of the environmental impacts of proposed strategic policies, plans, actions, and programmes. It is a proactive strategy for integrating these environmental impacts into high-level decision-making.
- This procedure extends the EIA principles upstream in the judgment process, beyond the project level, and when the primary option is still undetermined.
- The major purpose of such a procedure is to ensure the efficacy of proposed measures and, if necessary, to make mid-course adjustments.
- The procedure adopted for monitoring is as follows.
    - Every 6 months, project authorities are expected to report on the status of implementing circumstances indicated in the project's approval.
    - Officers and expert teams from the ministry and/or its regional offices conduct field visits to collect analysed performance data on development projects so that encountered problems can be discussed with the project's promoters in an effort to identify solutions. In the event of significant deviations and inadequate or no reaction, the relevant state government is notified.
    - Modifications in the scope of projects are recognised to determine whether a re-evaluation of a previous decision is necessary.
    - The principal advantages of EIA are reduced project implementation costs and duration. Design improvements that save costs. Enhanced project acceptability Impacts and violations of rules and regulations were avoided. Improved project efficiency. Participation in environmental evaluations offers local communities the following benefits.
    - A more wholesome local ecosystem (forests, water sources, agricultural potential, and clean living in urban areas).
    - The EIA process provides recommendations to decision-makers but not a final determination.
    - The expected bad consequences on the environment may result in the imposition of stringent restrictions to avoid or mitigate these effects, or in the cancellation of a proposal altogether.
    - The EIA necessitates that scientific (technical) and numerous issues be addressed in a single, cumbersome evaluation procedure.
    - The EIA governs the execution of the environmental policy, resulting in an overloaded process. Consultation and engagement of the public are limited.
- The ecological considerations come into light when an ecosystem is a complex system with many parts, both and non-living, if one part of the system is removed, lots of other parts can be affected.

- The Green Public Procurement (GPP) is the integration of environmental considerations in purchasing procedures of the Public Administration, which is the means to be able to choose those products, and services that have a minor or a reduced effect on human health and the environment than other products and services used for the same purpose. It is one of the main measures undertaken to implement strategies for sustainable development.

Recent Case Studies Related to Environment Impact Assessment

Char Dham Highway project

- Char Dham are some of the holiest places of Hinduism and is located in Uttarakhand. Strategically they play an important role as the region borders China. The government of India started building an ambitious network of national highways to connect the holy places and provide mobility to the armed forces of the border region.
- However, in the wake of earlier devastating floods, landslides, and potential ecological impact an NGO based out of Dehradun filed a case in the National green tribunal on grounds that the widening of roads leads to deforestation, which would trigger landslides and potential floods due to increased surface runoff.
- In September 2018 NGT ruled that a fresh environmental clearance was not needed for the project as it comes under strategic roads, which are exempted from EIA. It allowed the project to continue while warning the government that road widening activity might make slopes unstable and trigger landslides, and floods.
- On appeal to the Supreme court, The Honourable Court constituted an expert appraisal committee with representatives from civil societies, environmental experts, and the Ministry of defence to get an idea of the cumulative environmental impact. After a detailed independent study of the environment and evolving threat scenario along the border. The Honourable Supreme Court allowed the construction of the Char Dham Highway project with a reduced shoulder width of the road and identified areas of environmental degradation to take augmentation measures, identify areas of environmental degradation.
- The Bagheera the ecologically sensitive zone shall not be disturbed to prevent damage to the river and its present ecology.

NH 44 Underpass Construction

While often environmental concerns are neglected and are in news because of bureaucratic apathy. The construction of an underpass in Pench Tiger Reserve, Madhya Pradesh is an outlier on National highway 44 from Seoni in Madhya Pradesh to Nagpur a stretch of Highway falls under Pench Tiger Reserve, which is home to a large number of tigers, and other species. To ensure a safe passage for these animals one of the world's largest underpasses has been constructed allowing animals to freely move across the Tiger Reserve. This is an example of how a proper environmental impact assessment and an environmental management plan can help in balancing the environment and developmental concerns to ensure a sustainable balance between both them. It is now being replicated in Delhi to Mumbai Expressway, Delhi to Dehradun Express Way, and many other projects underpasses with noise barriers are being constructed to prevent damage to the environment and wildlife habitat.

Practice Exercise

1. Consider the following statements regarding environmental impact assessment (EIA).

I. This is mandatory under the Biological Diversity Act, 2002.

II. This acts as a tool, which helps in decision-making for government to access the impact of the project.

III. All industries are required to conduct an environmental impact assessment before taking approval from the government.

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) I and II only
(D) II and III only

2. Environmental Impact Assessment (EIA) for some developmental projects is mandatory under which one of the following legislations?

(A) Indian Forest Act, 1927
(B) Air (Prevention and Control of Pollution) Act, 1961
(C) Wildlife Protection Act, 1972
(D) Environment (Protection) Act, 1986

3. Consider the following statements.

I. EIA is a planning tool, which is accepted as an integral component of sound decision-making.

II. Environmental Management Plan (EMP) should assist planners and government authorities in the decision-making process by identifying the key impacts/issues and formulating the mitigation measures.

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

4. Which is not an objective of EIA?

(A) Recycling and reduction of waste
(B) Risk analysis and disaster management
(C) Assessment of international funding
(D) Sustained development with minimal environmental degradation

5. Environmental impacts of development projects can be:

(A) Immediate
(B) Short-term
(C) Long-term
(D) All of the above

6. The important step/steps in the predication of impacts is/are:

(A) Evaluating impacts
(B) Study of EIA
(C) Preparation of environmental management plan
(D) All of the above

7. Consider the following statements.

I. Screening is done to see whether a project requires environmental clearance as per the statutory notifications.

II. Scoping is a process of detailing the terms of reference of EIA.

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

8. Which is not correctly matched?

(A) ISO: International Organisation of Standards
(B) EMS: Environmental Management System
(C) EIA: Environmental Impact Assessment
(D) WTO: Whole Trade Output

9. In EIA, the decision to hold a hearing has to be made within:

(A) 5 days
(B) 10 days
(C) 20 days
(D) 30 days

10. Consider the following statements.

I. The law requires that the public must be informed and consulted on a proposed development after the completion of the EIA report.

II. The decision on environmental clearance arrives through many steps, including evaluation of EIA and EMP.

Select the correct answer using the codes given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (B) **2.** (D) **3.** (C) **4.** (C) **5.** (D) **6.** (D) **7.** (C) **8.** (D) **9.** (D) **10.** (C)